

# FRFP Volume II

Case Studies, Patterns, and Practice for Long-Horizon Human–AI Collaboration

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# Chapter 1

## Why FRFP? A Problem That Existing AI Stories Don't Solve

Most stories we tell about AI are either very local or very global.

At one extreme, we zoom in: we talk about a single model, a benchmark, a demo. We ask: “How accurate is this classifier?”, “How fluent is this chatbot?”, “How well does this model compress this dataset?”

At the other extreme, we zoom out: we talk about AI taking over whole industries, reshaping labour markets, or posing civilisational risks. We ask: “Will AI replace doctors?”, “Will AI destroy or save democracy?”, “Will AI become uncontrollable?”

Both perspectives are useful. But they leave a gap right where most of the real work happens:

the day-to-day, year-to-year reality of human–AI collaboration, inside actual institutions, on projects that unfold over long horizons.

This is the gap FRFP tries to fill.

### 1.1 The Missing Middle: Long-Horizon Collaboration

Consider a few examples:

- A hospital gradually introduces AI tools for triage, screening, risk prediction, and documentation. Over five years, almost every patient interaction is touched by some AI system at least once.
- A trading firm uses models to price complex instruments, manage risk, and execute trades. Positions are held for months; small model errors can snowball into very large losses.
- A scientific lab uses AI to propose hypotheses, design experiments, analyse data, and draft papers. The AI is involved in dozens of projects over a decade, shaping what gets studied and how results are interpreted.

In all these cases:

- the AI systems are not one-off tools; they are woven into everyday workflows;
- humans and institutions adapt to them over time;
- small design choices, repeated many times, can have large cumulative effects.

We lack a shared language to talk about these situations with enough precision to see where things go wrong or right.

## 1.2 What FRFP Tries to Do

FRFP stands for:

- **Functors**: a formal way of talking about how one structure maps into another. In plain language: how models, tools, and agents transform things.
- **Runs**: sequences of steps over time—not just a single input and output, but the whole trajectory.
- **Formal Protocols**: explicit rules and processes that say who does what, when, and with what authority.

You do *not* need the category-theory details to use FRFP. The key idea is simpler:

Treat the whole human–AI–institution system as something with a *structure* we can describe: a space of explicit artefacts (data, models, outputs), a space of human judgement and norms, and a set of rules for how things move between them over time.

FRFP gives names to:

- where AI lives (the **explicit layer**);
- where humans live (the **tacit layer**);
- where organisations and regulators live (the **institutional layer**);
- how the system moves over time (its **runs**);
- where real decisions happen (the **boundaries**);
- and how far we can safely go without stopping to think (the **safe horizon**).

Volume I builds all this up formally. This non-technical version explains the same concepts in everyday language and with examples, so that the case studies in Volume II are fully readable without any equations.

### 1.3 Why Not Just Use “Common Sense”?

Many organisations already have intuitions and slogans like:

- “Keep humans in the loop.”
- “AI should augment, not replace, human judgement.”
- “We will monitor systems and step in if something goes wrong.”

These are good instincts, but experience shows they are not enough.

When things go badly wrong—as in some of the case studies in Volume II—we often find that:

- the “human in the loop” had no time or information to really judge the AI;
- the institution gradually started treating metrics and model outputs as the final word;
- no one had defined *when* the system should be re-examined or stopped;
- everyone assumed someone else was thinking about the long-term effects.

FRFP’s contribution is not magical. It is to make these structural questions *explicit and unavoidable*:

- Where exactly are the humans in this loop?
- What, precisely, are they endorsing?
- How long can we keep doing this before we must re-ground our assumptions?
- Who has the authority to change the system’s boundaries and horizons?

The rest of this non-technical volume answers these questions in a systematic way.

### 1.4 How This Volume Is Organised

This plain-language Volume I is structured around concepts rather than theorems.

- **Chapter 2** introduces the three layers: explicit, tacit, and institutional.
- **Chapter 3** explains runs and long-horizon behaviour: how small steps add up over time.
- **Chapter ??** focuses on boundaries: where decisions really happen, and how AI can silently capture them.
- **Chapter ??** introduces safe horizons: how to decide how far to let systems run before we force a rethink.

- **Chapter ??** explains the non-automation results in everyday language: which parts of judgment we should not expect to automate away.
- **Chapter ??** discusses institutional design: boards, regulators, professional norms, and how they fit into FRFP.

After that, Volume II applies these ideas to real systems.

You can read the chapters here in order, or skip ahead to the case studies and refer back to this chapter when you need a concept.

## Chapter 2

# The Three Layers: Explicit, Tacit, Institutional

FRFP starts from a very simple observation:

In any serious AI system, there are things the machine can see and manipulate, and there are things it cannot.

We call the machine-visible things **explicit**, and the rest **tacit**. On top of that sits a third layer: **institutions**, which are made of people but behave very differently from individual agents.

### 2.1 The Explicit Layer: What the System Knows How to Touch

The explicit layer is everything that can be written down in a form the system can process. That includes:

- raw data: numbers, text, images, sensor readings;
- model parameters and code;
- labels, scores, rankings, probabilities;
- dashboards, graphs, and logs.

When you train a model, you are working in the explicit layer. When you log events, run A/B tests, or define metrics, you are shaping the explicit layer.

FRFP's questions about this layer are:

- What is *in* the explicit layer?
- What is the system actually computing?
- Where did the data come from?
- Which parts of reality are missing?

## Examples

- In DR screening AI: fundus photographs, model weights, and “mtmDR present” labels are explicit.
- In AlphaFold: protein sequences, experimental structures, and model predictions are explicit.
- In a recommender: viewing history, item features, embeddings, and rankings are explicit.
- In a chatbot: tokenised text, prompts, and generated responses are explicit.

The explicit layer is where the AI lives.

## 2.2 The Tacit Layer: What People Know, Feel, and Value

The tacit layer is everything that matters but is not fully represented in the explicit layer:

- background knowledge (“this disease is rare in this region”);
- contextual understanding (family situation, local politics, market rumours);
- judgement about what is acceptable or fair;
- professional skill and intuition built up over years;
- organisational culture and unwritten rules.

Tacit knowledge is not mysterious or mystical. It is simply the part of our understanding that we have not, or cannot, fully turn into datasets and code.

### Why tacit matters

Many critical questions live here:

- Should we accept this level of risk?
- Is this recommendation fair to this person?
- Does this result make sense, given everything else we know?
- Does this policy fit our values and obligations?

These cannot be answered purely in the explicit layer. Data and models can inform them, but someone has to make a judgement.

FRFP’s stance is:

You can move a lot of work from tacit to explicit, but you cannot move *all* of it, and you must respect what remains.

## 2.3 The Institutional Layer: Groups, Rules, and Power

The institutional layer:

- consists of organisations, committees, regulators, professional bodies, courts, and so on;
- sets rules and incentives;
- decides which systems are allowed, and under what conditions;
- interprets failures and assigns responsibility.

Institutions are built out of people, but they behave differently from individuals:

- they are slower to change;
- they maintain records and precedents;
- they can impose rules and sanctions;
- they have their own tacit cultures and explicit procedures.

FRFP treats institutions as *active* components in the system, not just background scenery.

They are the ones who:

- define boundaries (what AI can decide, what humans must decide);
- set safe horizons (how far systems may run before review);
- respond to incidents and change the rules.

## 2.4 How the Layers Interact

The three layers are not separate silos; they interact constantly:

- explicit models and outputs influence tacit beliefs (doctors start trusting an AI; traders lean on a risk model; judges rely on a score);
- tacit judgement shapes the explicit layer (what data to collect, what features to use, what metrics to optimise);
- institutions shape both (regulators require certain metrics; boards approve certain uses; professional norms forbid others).

FRFP is a way of keeping track of these interactions over time.

If you think of a long-lived system like a hospital's AI programme or a trading desk:

- the explicit layer is the machinery;
- the tacit layer is the human judgement that steers it;
- the institutional layer is the frame that says where it may go and when it must stop.

Most of the failure case studies in Volume II can be seen as situations where:

- the explicit layer grew powerful and complex;
- the tacit layer was sidelined or eroded;
- the institutional layer behaved as if the explicit machinery could do its job for it.

FRFP's design advice, in plain language, is:

Keep all three layers visible at once, and design your systems so that they respect the limits and roles of each.

## Chapter 3

# Runs and Long Horizons: How Small Steps Add Up

So far we have focused on the *layers* of a system. FRFP also cares about how the system moves over time.

The key concept here is a **run**: a sequence of explicit states and actions, unfolding over time, that takes us from “nothing has happened yet” to “a lot has happened and we are living with the consequences.”

### 3.1 What Is a Run?

Informally, a run is:

one possible story of how the system behaves over time, including: inputs, internal states, outputs, human responses, and institutional reactions.

For example:

- A single patient’s journey through a hospital over ten years: screenings, diagnoses, treatments, AI interactions, outcomes.
- A trading strategy’s life: initial backtests, launch, gradual scaling, periods of profit, eventual loss or shutdown.
- A scientific project’s evolution: early explorations, first AI-assisted analyses, publications, follow-up work, eventual acceptance or rejection.

Each of these is a run. The same system can have many runs, depending on random events, choices, and external shocks.

## 3.2 Why Runs Matter More Than Snapshots

Most AI evaluation is done on snapshots:

- a test dataset;
- a single A/B test;
- a one-off deployment.

But long-horizon behaviour depends on what happens when you *chain* these steps together:

- a risk model that is slightly mis-specified may look fine on backtests, but over five years it encourages a portfolio to drift into fragile positions;
- a recommender system that slightly over-rewards certain content may, over millions of recommendations, reshape users' habits and beliefs;
- a chatbot that occasionally hallucinates may cause little harm in casual use, but cause big problems when integrated into a formal workflow (like legal filings).

FRFP insists that:

To understand an AI system in a real institution, you must reason about runs, not just single steps.

## 3.3 Simple Examples of Runs

### Example 1: A DR Screening Programme

A DR screening AI is introduced in primary care.

A typical run for one patient might be:

1. Visit: the patient comes in for a diabetes check-up.
2. Image capture: a nurse takes retinal images.
3. AI evaluation: the system returns “mtmDR present” or “no mtmDR” or “ungradable”.
4. Boundary:
  - if positive or ungradable: a referral is made;
  - if negative: the screening is recorded as complete.
5. Follow-up: the patient attends (or misses) the referral; treatment or observation follows.
6. Next year: the cycle repeats.

Over time, thousands of such runs accumulate. The system’s impact depends not just on per-image accuracy, but on:

- how often referrals happen;
- how often patients attend;
- how clinicians respond to AI outputs;
- how the institution monitors outcomes and adjusts policies.

### Example 2: A Trading Strategy

A model-driven trading strategy goes like this:

1. Design: quants propose a model and backtest it on historical data.
2. Approval: risk and management approve a pilot with limited capital.
3. Deployment: the model starts trading small.
4. Scaling: after initial success, limits are raised; the strategy grows.
5. Drift: market conditions change; the model’s assumptions become less accurate, but performance metrics lag.
6. Failure: eventually a stress event or cumulative drift causes major losses.
7. Post-mortem: the institution investigates, updates models and rules, or shuts the strategy.

Again, the interesting story is not in any single trade. It is in the run across months or years.

## 3.4 Runs as the Unit of Analysis

FRFP recommends treating runs as the main unit of analysis when thinking about safety and alignment.

Instead of asking:

- “Is the model accurate?”

we ask:

- “What do typical runs look like when this model is embedded in this institution, with these incentives, over this timescale?”
- “What kinds of runs are possible, even if rare? Are any of them unacceptable?”

This shift in focus makes certain patterns easier to see:

- **Run drift:** the system slowly moves into states no one explicitly planned for (e.g. a portfolio becomes concentrated; a recommender narrows content diversity).
- **Run amplification:** small biases in each step compound over a long run.
- **Run fragility:** the system looks stable until it hits a rare combination of events, then fails badly.

Volume II’s case studies repeatedly show that failures are often about runs, not steps.

### 3.5 Where Humans and Institutions Enter the Run

In a long run, humans and institutions act at different timescales:

- **Short timescale:** frontline actors respond to each step (doctors accepting or rejecting recommendations; traders overriding trades; judges weighing risk scores).
- **Medium timescale:** managers, team leads, or local committees adjust workflows, limits, and practices.
- **Long timescale:** boards and regulators change rules, approve or ban systems, shift overall strategy.

FRFP encourages us to draw runs that include all three:

- not just the AI’s internal states,
- but also the human decisions at boundaries,
- and the institutional decisions that change how future runs are allowed to proceed.

When we do this, we can ask:

- At what points in the run could someone have intervened?
- Were there any natural “choke points” where human or institutional judgement *should* have been applied, but was not?
- Are there parts of the run where responsibility is unclear or contested?

## 3.6 Long Horizons: Why Time Changes Everything

Many systems work fine at small scale and over short periods, but fail when:

- they are used for much longer than expected;
- they are scaled to many more users or assets;
- their environment changes (laws, markets, diseases, norms).

FRFP calls this the **long-horizon problem**:

Systems that are safe and effective in early runs can become unsafe in later runs, even if no single decision looks obviously wrong.

Examples from the case studies:

- Zillow’s home-purchasing algorithm worked in backtests and early stages, but failed when scaled into a changing market.
- The London Whale portfolio grew in size and complexity over time, exploiting a model blind spot, until losses exploded.
- Some AI-based risk scores in law or credit scoring have long-term effects on populations, not just on individual decisions.

This is why FRFP insists on safe horizons, which we discuss in a later chapter. For now, the important idea is:

You cannot judge a long-horizon system by looking only at the first few steps. You must ask what happens as runs get long and the world changes.

## 3.7 Summary

In this chapter we shifted focus from layers to movement:

- A **run** is one possible trajectory of the system over time.
- Long-horizon systems are those where these runs are long and consequential.
- Failures often emerge from the structure of runs, not from single bad decisions.

In the next chapter, we look more closely at **boundaries**: the moments in a run where something becomes real, and where FRFP says correctness ultimately lives.

# Introduction to Volume II: FRFP in the Wild

Volume I of this monograph developed the FRFP (Functors, Runs, and Formal Protocols) architecture as a structural account of long-horizon human–AI collaboration. It argued that powerful explicit systems must be embedded within a carefully designed socio-technical stack: explicit representations and pipelines, tacit human and institutional states, boundary mechanisms that mediate between them, and safe-horizon constraints that prevent unbounded drift. The central non-automation results in Volume I show that in sufficiently complex domains, correctness, responsibility, and normative evaluation cannot be fully captured in explicit space. They remain irreducibly tied to tacit human and institutional judgement, even as explicit systems become more capable.

Volume II turns outward. It treats FRFP not as an abstract proposal but as a lens on real systems, both failed and successful. Across domains as varied as clinical oncology, hiring, criminal justice, finance, mobility, and science, we examine concrete AI-enabled systems and institutions that have already been built and deployed. The aim is not to score points against particular organisations or technologies, but to test FRFP against the empirical record: to ask whether FRFP clarifies what went right, what went wrong, and which patterns recur across superficially unrelated episodes.

## Scope and Structure of Volume II

The core material of this volume consists of a series of detailed case-study appendices. Roughly, they fall into three classes:

### 1. High-impact failures and near-misses:

- IBM Watson for Oncology and the broader IBM Watson Health programme;
- Amazon’s experimental AI recruiting engine;
- the COMPAS recidivism risk assessment tool in criminal justice;
- Anthropic’s “Project Vend” vending-machine agent;
- Zillow Offers and algorithmic home-flipping;

- *Mata v. Avianca* and hallucinated legal citations;
- JPMorgan’s CIO “London Whale” trading losses.

## 2. Constructive but mixed systems:

- large-scale recommender systems on content and commerce platforms;
- neural machine translation (NMT);
- general-purpose large language models such as ChatGPT.

## 3. Outstanding successes under strong constraints:

- AlphaFold and the protein structure revolution;
- autonomous AI for diabetic retinopathy screening;
- Waymo’s driverless ride-hailing services.

In addition to these external case studies, Volume II also revisits an internal, methodological case: our *first attempt* at designing a framework for long-horizon human–AI collaboration, prior to the development of FRFP’s mathematical core. That early attempt—captured in an initial “V01” framework document—sought to reason about workflows, roles, and institutions using primarily narrative and system-design language. It did not explicitly separate explicit and tacit spaces; it lacked a formally defined notion of runs and safe horizons; and it treated governance and protocol design more as design heuristics than as objects of mathematical analysis. This attempt failed to provide the clarity and inevitability results we ultimately wanted, and its limitations directly motivated the more formal architecture presented in Volume I. Volume II therefore includes this failed attempt as a meta-case study: a concrete demonstration of how *our own* informal reasoning ran into the very structural issues that FRFP is designed to articulate.

# Method and Perspective

Each case study in this volume is written in a common idiom. We begin by:

- sketching the system’s background and institutional context;
- identifying the explicit space  $E$ : the representations, pipelines, and decision rules that machines manipulate;
- characterising the tacit space  $T$ : the epistemic, normative, and practical states of human individuals and institutions.

We then treat each system as a run or family of runs in the skew-product  $N \ltimes E$  introduced in Volume I: a (possibly branching) sequence of explicit states and operations, parametrised by time, data, and environment. Against this backdrop, we ask four recurring questions:

1. **Explicit semantics:** What mapping is the system *actually* implementing in explicit space? Is that mapping well-defined and externally evaluable, or is it blurry, underspecified, or misaligned with the tacit function it is supposed to approximate?
2. **Boundary design:** Where are the boundaries between explicit outputs and tacit endorsement? Who is formally and de facto “deciding”? Are signatures, approvals, and institutional gates functioning as genuine boundary events, or have they been hollowed out by automation and rhetoric?
3. **Safe horizons and dynamics:** Over what horizons (patients, trades, trips, cases, content exposures) is the system used without explicit re-grounding? Are there designed limits and reset mechanisms, or does the system rely on implicit hopes that nothing will drift too far?
4. **Institutional semantics and non-automation:** How do organisational structures, incentives, and regulators act as operators on populations of tacit states? Are they treating explicit metrics and models as advisory signals, or as substitutes for the tacit governance functions that FRFP argues cannot be automated?

Across case studies, we deliberately treat FRFP as a *descriptive* lens first. We do not assume that any system “should” have been designed according to FRFP; rather, we ask whether the observed success or failure of a system can be understood as the consequence of how it happened to instantiate (or violate) the explicit/tacit split, boundary design, safe horizons, and institutional non-automation.

This method has two payoffs:

1. It allows us to *classify* failures and successes in terms of structural patterns, rather than surface-level anecdotes.
2. It allows us to *validate or falsify* FRFP’s claims: if the same patterns recur across oncology, law, finance, mobility, and science, we gain some confidence that FRFP is capturing something general about human–AI collaboration.

## A Typology of Failures and Successes

One of the main contributions of Volume II is a cross-case typology.

On the failure side, cases as different as Watson for Oncology, Amazon’s recruiting engine, COMPAS, Zillow Offers, *Mata v. Avianca*, the London Whale, and Project Vend exhibit strikingly similar structural features:

- **Weak or misleading explicit semantics:**
  - Watson for Oncology trained on small, synthetic case sets and idiosyncratic expert judgments, then marketed as “MSK in a box”;

- Amazon’s recruiting engine embedded historical gender bias directly into its scoring function;
- the CIO’s new VaR model understated risk for complex credit-derivative portfolios;
- hallucinating LLMs produced plausible but nonexistent case citations in *Mata*.

- **Boundary capture:**

- recommender algorithms, risk scores, and hiring scores become de facto gatekeepers despite being officially “advisory”;
- algorithmic offers in Zillow, and risk metrics in London Whale, became surrogate notions of correctness for capital allocation;
- in *Mata*, the act of filing—a high-stakes boundary event—was effectively automated.

- **Safe-horizon violations:**

- long-lived trading strategies and iBuying pipelines accumulate risk over months without robust episodic re-grounding;
- institutional use of COMPAS scores and legal-risk tools across thousands of cases gradually reshapes judicial practice;
- repeated use of generative models in legal or professional contexts erodes tacit verification norms.

- **Explicit-only governance:**

- boards, regulators, and hospital leadership lean heavily on dashboards, demos, and vendor metrics;
- there is no distinct institutional layer explicitly designed to hold the boundary and enforce safe horizons.

On the success side, systems like AlphaFold, autonomous DR screening AI, and Waymo exhibit complementary structural properties:

- **Narrow, strong explicit semantics:**

- AlphaFold solves a well-defined prediction problem with abundant ground truth and stringent community benchmarks;
- DR screening AI detects a specific pathology threshold from retinal images, with clear gold-standard labelling;
- Waymo’s vehicle-level semantics are tightly constrained by physics and traffic rules.

- **Explicit uncertainty and fail-safe modes:**

- AlphaFold exposes per-residue confidence and error metrics;
  - DR systems explicitly output “ungradable” and defer to specialists when they cannot see clearly;
  - autonomous driving stacks implement conservative fallback behaviours and restrict operation to clearly defined ODDs.
- **Designed safe horizons:**
    - DR programmes enforce periodic re-screening regardless of AI outputs;
    - Waymo constrains autonomy to specific operational design domains and expands them gradually;
    - AlphaFold’s predictions are integrated into scientific workflows that demand experimental validation downstream.
  - **Institutional embedding:**
    - AlphaFold’s open database and integration with EMBL–EBI create a shared infrastructure governed by the scientific community;
    - DR AI is framed by FDA authorisation, reimbursement codes, and quality-improvement initiatives;
    - Waymo operates under city-level and state-level permitting, public safety cases, and iterative regulatory scrutiny.

Finally, a third group—recommenders, NMT, and general-purpose LLMs—sits in between:

- they are *successful* in the sense of scale, performance, and economic impact;
- but they raise deep FRFP concerns when used as default cognitive infrastructure for populations.

Their explicit semantics are strong enough for many micro-tasks (ranking, translation, summarisation) but weakly tied to any single tacit notion of correctness. They therefore become a central arena where FRFP’s population-level semantics and non-automation claims will be tested in practice.

## The First Attempt at Long-Horizon Collaboration

Before FRFP was formalised, an earlier attempt (captured in a separate “V01” document) tried to articulate principles for long-horizon human–AI scientific collaboration using primarily system-design metaphors: co-pilot workflows, iterative refinement loops, and governance checklists. That attempt anticipated some of FRFP’s concerns in informal language:

- it recognised that AI systems would be used iteratively and that workflows could drift;
- it stressed the need for transparent interfaces, roles, and responsibilities;
- it gestured at institutional structures for oversight.

But it lacked the mathematical machinery to:

- make the explicit/tacit distinction precise;
- reason about runs as objects with compositional structure and depth;
- prove inevitability results about hallucinations, drift, and non-automation;
- express the dynamics of populations and institutions as operators acting on distributions of tacit states.

In hindsight, that early framework is itself a case study in FRFP failure:

- it treated explicit protocols and diagrams as if they could fully capture correctness, without explicitly modelling tacit states and boundaries;
- it lacked a notion of safe horizon, implicitly assuming that carefully designed workflows would remain safe as they were iterated;
- it relied on informal governance narratives rather than a semantics in which institutional operators could be analysed on equal footing with technical pipelines.

Recognising these limitations led directly to the FRFP architecture in Volume I. In Volume II, we include this “first attempt” as a reminder that our own cognitive work is subject to the same structural constraints: without formal tools, it is easy to confuse the map with the territory—explicit diagrams and narratives with tacit correctness.

The rest of this volume puts FRFP to work on real systems, drawing lessons that feed back into both the theory and practice of long-horizon human–AI collaboration.

## Chapter 4

# Meta-Case Study 0: The First Attempt at Long-Horizon Human–AI Collaboration

Before FRFP acquired its current mathematical form, there was an earlier, more informal framework for long-horizon human–AI collaboration. It was articulated largely in systems and product-design language: workflows, phases, roles, and governance checklists. This chapter treats that *first attempt* as a meta-case study: an instance where our own reasoning ran into the same structural issues that Volume I and the other case studies diagnose.

The point is not self-criticism for its own sake. Rather, we use this early framework to show how easy it is, even for well-intentioned designers, to over-trust explicit diagrams and narrative protocols in the absence of a formal semantic layer. In that sense, this chapter is to FRFP what early bridge collapses were to structural engineering: a motivating failure that reveals missing concepts.

### 4.1 What the Early Framework Tried to Do

The V01 framework was an earnest attempt to answer a concrete question:

How should we structure workflows, roles, and institutions so that humans and advanced AI systems can collaborate on long-horizon scientific and technical projects without losing control or clarity?

It emphasised several themes that remain central in FRFP:

- **Iterative workflows:** science-as-a-loop of hypothesis, experiment, analysis, and refinement, augmented by AI tools at each stage.
- **Role design:** different “personas” (principal investigator, research engineer, AI agent, governance committee) with partially specified responsibilities.

- **Governance and guardrails:** checklists and process steps intended to keep projects aligned with safety and mission goals.
- **Long-horizon emphasis:** explicit concern with projects that unfold over months to years, not just one-off tasks.

The framework used diagrams, phases, and textual descriptions to propose how work should flow between humans and AI, and how governance should intervene. It was, in short, a fairly typical “best-practices” design document for responsible AI collaboration—but without a formal semantics.

## 4.2 How the Early Framework Looked in FRFP Terms

Viewed through the lens of FRFP, the early framework had the following shape.

### 4.2.1 Explicit space was underspecified and overloaded

The framework implicitly assumed a large explicit space  $E_{V01}$  containing:

- datasets, models, and code artefacts;
- documents (proposals, reports, protocols);
- AI-generated suggestions, analyses, and plans;
- tracking metadata (issue tickets, dashboards, milestones).

However, it did not:

- sharply distinguish between different classes of artefacts (e.g. hypotheses vs. evidence vs. commentary);
- specify what transformations on these artefacts meant (e.g. when does a “draft” become a “claim”?);
- treat the AI system’s outputs as a distinct morphism  $f : E \rightarrow E$  with its own correctness conditions.

In FRFP language,  $E_{V01}$  was an amorphous set of “things in the system”, not a typed, structured space with clear semantics and operators. This made it hard to reason precisely about what the AI was allowed to do, and what counted as a boundary event.

### 4.2.2 Tacit space was assumed but not modelled

The early framework heavily relied on human judgement:

- principal investigators decide when a result is credible;
- reviewers and committees provide oversight;
- teams notice drift and misuse and intervene.

But it treated this judgement as a kind of inexhaustible background resource:

- there was no explicit model of how human beliefs and skills evolve over time under AI use;
- no representation of how institutional incentives shape attention;
- no account of how repeated interaction with AI tools might erode or distort tacit expertise.

FRFP, by contrast, insists on representing these tacit elements as a space  $T$  with its own dynamics and operators. Without this, the framework could not state—let alone analyse—questions like:

After a thousand AI-assisted experiments, in what sense are the humans still in control of the research agenda?

### 4.2.3 Boundaries were narrative, not structural

The early framework described humans as “in the loop” at key points:

- humans “review” AI-generated analyses;
- governance boards “approve” high-impact decisions;
- safety checks “sign off” on deployments.

But these boundary notions were purely textual. There were no explicit:

- state transitions marking when an artefact moved from “proposal” to “endorsed claim”;
- requirements on what information humans must see to make those endorsements meaningful;
- logs of who, exactly, had taken responsibility at which step.

In FRFP notation, the framework lacked a functor  $J-K$  that embeds explicit artefacts into tacit states with traceable endorsements. Instead, it implicitly assumed that humans would pay sufficient attention and that signatures or approvals would retain their traditional meaning, even as AI systems took over much of the work leading up to them.

#### 4.2.4 Safe horizons were absent

The early framework was explicit about *long-horizon* collaboration but entirely implicit about *limits*.

- Workflows were described as iterative cycles without clearly defined stopping or reset conditions.
- There was no notion of how many iterations or how much AI-driven change could occur before a deeper review was required.
- Drift—in models, data, or tacit beliefs—was acknowledged as a risk, but only managed via generic exhortations to “monitor performance” or “periodically review processes”.

From an FRFP perspective, this meant the framework was defining very long runs in  $N \times E$  without any formal safe horizon. It had no way of expressing the idea that, beyond some depth, the system should be forced back into a reflective mode where humans re-ground assumptions and redesign protocols.

#### 4.2.5 Institutional governance was treated as a checklist

Finally, the early document included governance sections:

- oversight committees with described membership;
- risk review processes;
- escalation paths for concerns.

But these were described in the same informal style as product requirements: lists of roles and responsibilities, rather than operators in a semantics.

In FRFP terms, there was no explicit operator  $G : T \rightarrow T$  capturing how governance bodies update the tacit state of the organisation or community:

- no way to talk about how decisions propagate through the system;
- no way to compare different governance designs structurally;
- no way to prove non-automation results about which parts of governance must remain tacit.

Governance, in other words, was treated as something one can *write down* rather than something that itself needed to be formally modelled.

### 4.3 Where and Why the Early Framework Broke Down

As we tried to push the early framework further—to analyse concrete research programmes, to reason about failure modes, to connect to institutional incentives—specific cracks started to appear.

### 4.3.1 Inability to express inevitability results

We wanted to show statements of the form:

In any long-horizon collaboration that satisfies conditions  $A, B, C$ , certain failure modes (hallucinations, drift, loss of tacit control) are inevitable unless you enforce design constraint  $D$ .

Without a formal notion of runs, boundaries, and operators on explicit and tacit spaces, we simply could not express these claims precisely, let alone prove them. At best, we could offer qualitative arguments:

- “If humans stop checking AI outputs, errors will accumulate”;
- “If governance only looks at dashboards, it will miss deeper problems.”

These are true but generic. They do not differentiate Watson from AlphaFold, or COMPAS from DR screening, in a way that supports precise theorems. The early framework was descriptive; FRFP needed to be semantic.

### 4.3.2 Confusion between workflows and semantics

The V01 document mixed together:

- the *order* in which tasks are done (who talks to whom, in what sequence);
- the *meaning* of artefacts and decisions (what counts as a result, a claim, an endorsement).

So long as the examples were simple, this was tolerable. But once we tried to model complex, branching projects, or to compare different designs, it became clear that:

- workflows can vary a lot even when the underlying semantics are similar;
- conversely, superficially similar workflows can embed very different semantics (e.g. whether a “sign-off” is substantive or rubber-stamping).

We needed a way to talk about semantics independently of any particular workflow representation. This is exactly what FRFP’s functors and runs provide.

### 4.3.3 No vocabulary for population-level effects

The early framework centred on single projects and teams. It had very little to say about:

- how AI-based workflows reshape the tacit state of a whole field over time;
- how institutional incentives drive systematic over- or under-use of AI tools;

- how the combination of many local decisions yields global failure modes (e.g. housing bubbles, biased hiring ecosystems, epistemic capture by LLM interfaces).

In the absence of a population-level semantics, the framework could not speak to the very phenomena that dominate many of our case studies: the way recommenders, risk scores, or legal tools change whole populations of users, rather than just assisting individuals.

FRFP’s introduction of population-level tacit states and institutional operators was, in part, a direct response to this limitation.

## 4.4 What the Early Failure Taught Us

This meta-case study contributes three specific lessons that shaped FRFP.

### 4.4.1 Lesson 0.1: You need a typed explicit space

It is not enough to say “the system manipulates data, models, and documents”. We need to know:

- which artefacts count as hypotheses, evidence, policies, or commitments;
- which transformations are allowed, and what they mean;
- how AI operations differ from human operations on the same explicit space.

FRFP’s explicit-space semantics and run notation were introduced to fix this. They allow us to classify operations and reason about their composition, rather than drowning in undifferentiated “workflow” boxes.

### 4.4.2 Lesson 0.2: Boundaries must be first-class objects

The early framework treated boundaries as narrative phrases: “final review”, “governance sign-off”, “PI approval”. FRFP forces us to:

- represent these as explicit mappings between  $E$  and  $T$ ;
- ask what information, context, and authority must be present at each boundary;
- define which boundaries may be automated, and which are non-automatable by theorem, not by intuition.

This is the difference between hoping that humans stay in control and having a language to specify and enforce where control must reside.

### 4.4.3 Lesson 0.3: Safe horizons are necessary even for “good” workflows

The early framework assumed that, if you design a careful workflow and governance structure, you can iterate it indefinitely. FRFP sees this as a category error:

- any fixed workflow, repeated without re-grounding, will eventually be used under conditions different from those it was designed for;
- humans operating inside the workflow will themselves change, including their tacit skills and attention patterns;
- external incentives (funding, competition, politics) will push the system into new regimes.

Safe horizons—explicit bounds on how far you can go before you must rethink assumptions—are therefore structural necessities, not optional add-ons. The absence of this concept is perhaps the single biggest gap in the original V01 design.

### 4.4.4 Lesson 0.4: Governance must be part of the semantics

Finally, the early framework treated governance as a design section at the end of the document. FRFP instead treats governance as a *first-class semantic layer*:

- institutions and regulators are modelled as operators on tacit states and admissible runs;
- non-automation results apply to them as strongly as to technical components;
- we can compare governance designs by the runs they permit or forbid, not just by their textual descriptions.

This shift—from governance as prose to governance as semantics—is what allows FRFP to say meaningful things about COMPAS, Zillow, London Whale, or DR screening AI that go beyond “the governance should have been better”.

## 4.5 How This Meta-Case Study Fits into Volume II

Within Volume II, this meta-case study plays three roles:

1. **Calibration:** it demonstrates that FRFP is not being applied only to other people’s systems. We subject our own early work to the same structural critique that we apply to Watson, Zillow, or COMPAS.
2. **Conceptual genealogy:** it shows where FRFP’s core notions came from: not from pure abstraction, but from hitting the limits of narrative frameworks when trying to think clearly about long-horizon collaboration.

3. **Bridge to practice:** it highlights that many current “AI governance” and “responsible AI” documents are, in effect, still at the V01 stage: rich in diagrams and workflows, poor in semantics. FRFP offers a path to move from those documents toward systems that can be analysed and improved in a principled way.

In that sense, the failure of the first attempt was productive. It exposed the need for a more rigorous language for human–AI collaboration. FRFP is that language. The rest of Volume II shows how it behaves when we apply it to the messy, contested, high-stakes systems we actually build.

## Chapter 5

# How to Do an FRFP Case Study

## Chapter 6

# How to Use This Volume: Methodology for FRFP Case Studies

Volume II is meant to be used, not just read. This chapter gives a short, reusable methodology for doing your own FRFP case studies and for navigating the ones included here. You can treat it as a field manual for analysing or designing human–AI systems using the FRFP lens.

### 6.1 What This Volume Is For

This volume has three main audiences:

- **Researchers and engineers** designing AI systems (models, agents, pipelines) who want to reason about their long-horizon behaviour and institutional embedding.
- **Practitioners and decision-makers** in domains like healthcare, finance, law, and governance who need to evaluate or deploy AI systems and worry about safety, fairness, or control.
- **Analysts and students** who want a systematic way to dissect past AI successes and failures, beyond anecdotes or single-metric evaluations.

The basic workflow is:

1. Read the introduction and this chapter to absorb the FRFP case-study template.
2. Dip into individual case-study chapters as needed (by domain or theme).
3. Use the template and checklists to analyse your own systems or to structure new case studies.

### 6.2 The FRFP Case-Study Template

Every case study in this volume follows the same underlying pattern, even when the domain and narrative differ. You can reuse this pattern for new systems or organisations.

### 6.2.1 Step 1: Define the System and Its Context

- **What is the system?** Give a 1–2 page description of the system, project, or programme: what it is supposed to do, where it runs, who built it, and who uses it.
- **Why does it matter?** Briefly state the stakes: clinical outcomes, financial risk, fairness, productivity, epistemic effects, etc.
- **What counts as success or failure?** Identify, in plain language, the kinds of outcomes that the domain cares about (e.g. reduced blindness, avoided losses, fairer hiring, improved science).

This sets the stage and anchors the later FRFP analysis in concrete terms.

### 6.2.2 Step 2: Identify Explicit Space $E$

Explicit space  $E$  is the set of artefacts and transformations that the machine side of the system manipulates.

List and characterise:

- **Inputs:** data types, signals, records, documents, sensor streams.
- **Models and algorithms:** ML models, rule engines, search procedures, agents.
- **Outputs:** scores, recommendations, labels, offers, plans, rankings.
- **Pipelines:** how inputs flow through models and rules to outputs; where feedback (e.g. re-training) enters.

Try to be precise about *what function* the system intends to approximate: for example, “probability of credit default within 12 months” or “presence of more-than-mild diabetic retinopathy on fundus images”.

The goal is to make the semantics of  $E$  explicit enough that you could, in principle, evaluate correctness on independent ground truth.

### 6.2.3 Step 3: Identify Tacit Space $T$

Tacit space  $T$  is the configuration of beliefs, skills, norms, and responsibilities of human actors and institutions.

Describe:

- **Individual tacit states:** clinicians, traders, judges, engineers, users, etc.
- **Collective tacit states:** teams, courts, committees, boards, regulators, professional communities.

- **Norms and standards:** what counts as “correct”, “fair”, “safe”, or “good practice” in this domain.

The aim here is not exhaustive sociology, but a clear sense of:

Who, ultimately, has to live with the consequences of this system, and on what tacit criteria do they evaluate it?

#### 6.2.4 Step 4: Map Runs in $N \ltimes E$

FRFP models the system’s evolution as runs in the skew-product  $N \ltimes E$ : sequences (often branching) of explicit states and operations over time.

In practice, for a case study, this means:

- describing typical **episodes**: a patient visit, a loan application, a ride, a trading day, a hiring round, a scientific project cycle;
- identifying key **loops**: retraining cycles, feedback loops from outcomes back to models, updated policies, or institutional responses;
- noting where **history matters**: how past outputs affect future inputs (e.g. exposure effects in recommenders, inventory in vending agents, accumulated risk in portfolios).

You do not need full formalism here; the goal is to see where long horizons show up and how deeply runs can go before any external check is forced.

#### 6.2.5 Step 5: Analyse Boundary Design

Boundaries are where explicit artefacts cross into tacit endorsement.

Identify:

- **Formal boundaries**: signatures, approvals, regulatory filings, clinical sign-offs, code deploys, capital commitments, court filings.
- **De facto boundaries**: points where, in practice, explicit outputs are rarely questioned (e.g. a scoring threshold that almost always determines hiring, a risk metric that determines position size, an AI diagnostic that determines referral).
- **Information at the boundary**: what humans see when they endorse: raw data, summaries, model scores, uncertainty measures?

Then ask:

- Where is correctness truly located: in the explicit pipeline, or in tacit judgement?

- Are human “approvals” meaningful, or rubber-stamps on model decisions?
- Where are boundaries missing or only rhetorically present?

This is often where failure modes become most visible.

### 6.2.6 Step 6: Characterise Safe Horizons

Safe horizons are bounds on how far the system may run without re-grounding in tacit judgement.

For the system you are analysing:

- **Temporal horizon:** how long can the system run (days, months, years) on a given design before a deep review or redesign occurs?
- **Scale horizon:** how many patients, users, trades, or documents can it affect before we pause to reassess?
- **Domain horizon:** how far can it drift from the context or distribution where it was validated (new markets, populations, use cases)?

Then look for:

- explicit mechanisms enforcing horizons: pilot phases, caps, trial endpoints, scheduled audits;
- implicit or absent horizons: systems that simply grow until something breaks.

In the case studies, long-horizon systems that lack explicit safe horizons (Zillow Offers, London Whale, some recommender deployments) suffer the most severe failures.

### 6.2.7 Step 7: Identify Institutional Operators

Finally, locate the institutional operators that act on tacit states and admissible runs:

- internal governance: risk committees, ethics boards, safety teams, editorial boards, quality-improvement programmes;
- external governance: regulators, courts, professional bodies, funding agencies, standard-setting organisations;
- cross-cutting forces: media coverage, public opinion, market competition.

Ask:

- How do these actors change what runs are allowed or forbidden?
- Are they relying primarily on explicit metrics and reports, or do they have independent tacit channels (e.g. whistleblowers, site visits, professional judgement)?

- Which parts of governance are implicitly being automated by metrics and dashboards, and which remain genuinely tacit?

This sets up the analysis of *non-automation* at the institutional level.

### 6.2.8 Step 8: Construct an FRFP Counterfactual

For each case, build a brief, structured counterfactual:

- **Hold fixed:** the domain, broad goals, and constraints (e.g. “an oncology decision-support tool”, “an iBuying business”).
- **Modify:**
  - explicit semantics (e.g. narrow the task; add uncertainty);
  - boundary design (e.g. require explicit sign-off for certain decisions);
  - safe horizons (e.g. phased pilots, caps on exposure);
  - institutional operators (e.g. introduce a cross-functional governance board).

The goal is not a full redesign but a focused sketch: “What few FRFP-motivated changes would plausibly have altered this trajectory?”

All case studies in this volume include such a counterfactual section; it is the point where FRFP shifts from descriptive to prescriptive.

## 6.3 How to Read the Case Studies in This Volume

You do not need to read Volume II linearly.

### 6.3.1 By domain or role

- If you work in **healthcare**, start with: Watson for Oncology/Watson Health, DR screening AI, and the clinical implementation appendices.
- If you work in **finance**, start with: London Whale, Zillow Offers, and any FRFP finance implementation appendix.
- If you work in **law or governance**, start with: COMPAS, *Mata v. Avianca*, and the relevant institutional sections.
- If you work on **foundation models or product platforms**, start with: Project Vend, recommenders/MT/ChatGPT, and the meta-case study on our first collaboration attempt.

### 6.3.2 As a pattern library

Alternatively, you can treat the volume as a pattern library:

- skim the case-study headers to locate:
  - examples of boundary capture;
  - examples of successful safe horizons;
  - examples of institutional non-automation working well.
- compare multiple cases with the same pattern (e.g. COMPAS vs. DR screening AI for risk/diagnostic tools; Zillow vs. London Whale for model-driven capital allocation; Watson vs. AlphaFold for scientific claims).

The concluding chapter synthesises these patterns, but seeing them in their original contexts is often more illuminating.

## 6.4 Applying FRFP to Your Own System

If you want to use FRFP on a system you are designing or evaluating, here is a compact checklist you can follow, derived from the steps above:

1. **Write down  $E$ :** What are the inputs, models, outputs, and pipelines? What is the explicit task?
2. **Write down  $T$ :** Who are the human and institutional actors? What tacit criteria do they care about?
3. **Trace a run:** Describe one or two typical runs through your system, including feedback loops.
4. **Mark boundaries:** Where do humans endorse or commit? What do they see at those points?
5. **Set safe horizons:** How far can the system go (time, scale, domain) before you a reset or review?
6. **Name the operators:** Which committees, regulators, or boards can change the system's behaviour? How?
7. **Design or test a counterfactual:** Sketch 2–3 changes that would move the system toward FRFP alignment.

You can then compare your answers to the patterns and anti-patterns summarised in the later chapters.

## 6.5 Limitations of This Methodology

This chapter deliberately simplifies. A few caveats:

- FRFP does not give you a single scalar “safety score” for a system; it offers a structural vocabulary and a way to see patterns.
- Not every system can or should be forced into narrow tasks like DR screening or protein folding; sometimes the right conclusion is that a desired level of automation is structurally inappropriate.
- Institutional dynamics are complex; FRFP’s operators abstract them heavily. Case studies cannot fully capture all political and social nuances.

Even so, the FRFP methodology helps separate questions of *semantics* (what the system is doing in  $E$ ) from questions of *governance* (what humans and institutions are doing in  $T$ ), and from questions of *dynamics* (how runs unfold over time). That separation is what makes it possible to compare such different systems within a single volume.

The rest of Volume II applies this template to concrete cases. Readers are encouraged to adapt the template to their own domains, add domain-specific checks, and treat FRFP as a shared language rather than a fixed checklist.

## Part I

# Case Studies: FRFP in Real Systems

## 6.6 Implementing FRFP in Scientific Research Workflows

This appendix develops a concrete implementation of the Foundational Reasoning and Feedback Protocol (FRFP) for scientific research. It assumes the formal core of FRFP as given in the main text: an explicit/tacit semantic split, explicit pipelines in  $N \times E$ , a boundary functor  $J-K : N \times E \rightarrow T$ , and the associated dynamic and institutional semantics. Here we specialise these abstractions to day-to-day scientific work and describe how to instrument long-horizon human–AI collaboration in practice.

### 6.6.1 Scope and Objectives

We focus on research settings where:

- large language models and tool-using systems assist with literature review, ideation, proof sketches, modelling, simulation, and drafting;
- correctness still ultimately matters in a strong sense: scientific claims, theorems, and experimental conclusions are expected to withstand scrutiny and serve as a basis for further work;
- collaboration is long-horizon: projects extend over weeks or months, with many intermediate artefacts and partial results.

An FRFP-aligned implementation for scientific research should:

1. respect the explicit/tacit separation and single boundary: AI systems operate only on explicit artefacts; correctness authority resides in human tacit states at clear endorsement points;
2. make long-horizon explicit pipelines auditable and refactorable;
3. manage degradation of tacit grounding through safe-horizon design, re-grounding, and peer/institutional review;
4. provide concrete instruments (logs, ledgers, metrics) that researchers and committees can actually use.

### 6.6.2 Core Design Principles for Scientific Workflows

We state core design principles as implementation constraints on FRFP runs specific to scientific research.

#### **P1: Explicit Research Workspace**

All research artefacts that can, in principle, be made explicit are treated as elements of explicit space  $E$ :

- literature summaries, citation graphs, and topic maps;
- conjectures, proof sketches, formal definitions, models, and code;
- experiment designs, data processing pipelines, and results;
- manuscripts, referee reports, and internal notes.

AI systems are allowed to read, generate, transform, and organise these artefacts. Their actions appear as morphisms in the explicit component of  $N \ltimes E$ .

## **P2: Tacit Scientific Understanding and Endorsement**

Tacit space  $T$  contains the epistemic states of human researchers:

- their background knowledge, intuitions, and sense of plausibility;
- their understanding of methods, limitations, and domain-specific norms;
- their individual and collective judgments about whether a result is acceptable as a basis for further work.

FRFP enforces that correctness of scientific claims is attached only after evaluation in  $T$ : accepting a theorem, a result, or a modelling choice is always a human act of endorsement at a boundary.

## **P3: Explicit Boundary Events in Research**

For a scientific project, boundary events include:

- accepting a conjecture as sufficiently precise to pursue seriously;
- signing off on a proof sketch as provisionally correct;
- freezing a model or experiment design as the basis for data collection;
- submitting a manuscript or internal report;
- endorsing a result as part of a review, panel, or committee decision.

An FRFP implementation:

- designates specific points in the research pipeline as boundary events;
- identifies the responsible role(s) (e.g. lead author, PI, committee);
- records the explicit artefacts and context available at each boundary;
- treats these events as the points where correctness in  $T$  is attached.

#### P4: Safe Horizons and Re-Grounding

Scientific research is iterative and extended. Within FRFP, each project is associated with *safe horizons*:

- limits on how many AI-assisted transformations can occur without returning to primary material (original papers, raw data, base axioms);
- limits on how many endorsement steps can be chained without independent peer review or re-derivation;
- constraints on how far formal manipulations can drift from motivating questions without an explicit re-alignment step.

Implementation must specify:

- when researchers must read primary sources rather than summaries;
- when proofs or derivations must be reconstructed without AI assistance;
- when external or independent review is required.

These safe-horizon constraints are treated as part of the admissibility conditions on runs.

### 6.6.3 FRFP Research Methodology as Pipelines

We now describe a concrete research methodology as patterns of explicit pipelines and boundary events.

#### M1: Context Management in Research Pipelines

For a given project, we regard the evolving state of the work as a run  $\rho$  in  $N \times E$ :

$$\rho = (f_1, f_2, \dots, f_n),$$

where each  $f_i$  is a morphism representing a research action (human or AI).

Each  $f_i$  carries a context annotation:

- identifiers of upstream artefacts on which it depends (e.g. which papers, lemmas, or datasets);
- the version of key definitions and notation in force at that point;
- the project intent and phase (e.g. exploratory, formalisation, validation).

### Implementation guidelines.

- Maintain a versioned project graph in which nodes are artefacts and edges are morphisms  $f_i$ ; ensure that each non-trivial  $f_i$  has an explicit parent set and context annotation.
- At boundary events, present human evaluators not only with the target artefact but with the relevant slice of this graph, so that tacit judgment can take context into account.
- Provide tools to “slice” the run by section, claim, or experiment, so that localized audits of correctness and provenance are feasible.

### M2: Iterative Co-Formalization of Scientific Questions

Scientific projects often begin from loosely specified questions and gradually move toward formal statements. Within FRFP, this path appears as a navigation pattern in  $N \ltimes E$ :

- navigation morphisms in  $N$  move between sections, viewpoints, and granularity levels (informal description, proto-formal statement, full formalization);
- explicit morphisms refine definitions, introduce notation, and connect informal claims to formal objects.

**Phase labels.** It is useful to tag each segment of  $\rho$  with a *formality phase*:

- F0: informal exploration (questions, intuitions, examples);
- F1: proto-formal (candidate definitions, rough conjectures);
- F2: formal development (definitions, lemmas, proofs, models);
- F3: consolidation (cleaned-up proofs, canonical statements, publication-ready artefacts).

### Implementation guidelines.

- Require explicit morphisms for transitions between phases (e.g. “elevate this informal claim to a conjecture with this statement”).
- At higher phases (F2/F3), restrict admissible AI operations: for example, proof steps may be suggested but require local checking, and model refactorings must preserve stated invariants.
- Ensure that there are navigation links back from each formal artefact to its informal motivation, supporting tacit understanding at the boundary.

### M3: Reproducible Human–AI Research Sessions

Extended co-reasoning with AI systems—for example, over many days of proof search or model refinement—should be treated as structured runs, not as ephemeral chats.

**Session structure.** We partition a project into *sessions*  $(S_1, S_2, \dots)$ . Each session  $S_k$  corresponds to a run segment  $\rho_k$  in  $N \times E$  with:

- an explicit session intent (e.g. “clarify definitions in Section 2”, “search for counterexamples to Lemma 3.4”);
- a bounded safe horizon (e.g. number of AI-assisted steps or time);
- one or more boundary events (e.g. acceptance of a refined statement, rejection of a proposed lemma, escalation of an open problem).

### Implementation guidelines.

- Log each session as a coherent run segment with: prompts, visible AI outputs, human edits, navigation actions, and boundary decisions.
- At the end of a session, write a short human-authored summary that interprets the explicit progress in tacit terms: what was learned, what remains uncertain, what changed in the researcher’s understanding.
- Support replay at two levels: raw interaction (for debugging and training) and structural pipeline view (for scientific auditing and reuse).

#### 6.6.4 Instruments for FRFP-Aligned Scientific Practice

We now introduce a small set of instruments specialised to research settings.

### I1: Research Integrity Ledger

The *research integrity ledger* is a structured record of a project’s status relative to FRFP constraints.

For each significant pipeline segment (e.g. a section, a cluster of lemmas, an experiment series), the ledger stores:

- phase labels (F0–F3) and project intent;
- an integrity triple  $(V, C, P)$ :
  - $V$ : structural validity of the pipeline (protocol compliance, type discipline, absence of gross methodological violations);
  - $C$ : tacit correctness status as judged by responsible experts at boundary points (e.g. “provisionally accepted”, “rejected”, “open”);
  - $P$ : provenance adequacy (traceable sources, clear use of AI assistance, accessible upstream artefacts).
- links to boundary events (who endorsed what, when, under which conditions).

### Implementation.

- Index ledger entries by project, section, and artifact ID.
- Provide views that answer questions such as:
  - “Which results in this paper are structurally valid but still open in  $C$ ?”
  - “Which parts of the proof relied heavily on AI suggestions?”
  - “Where is provenance currently weak or incomplete?”
- Treat ledger updates as first-class events: updating  $C$  or  $P$  in response to new evidence is itself a logged action.

## I2: Hybrid Authorship Log

The *hybrid authorship log* records, at a fine-grained level, how AI and human contributions are interleaved in research artefacts.

For each document, proof, or model, the log stores:

- a sequence of edits (diffs) with author type labels (human, AI, tool-assisted, mixed);
- pointers from each region of the artefact (e.g. a paragraph, lemma, or code block) to the chain of edits that produced it;
- links from edits back to the corresponding steps in the FRFP run.

### Implementation.

- Integrate authorship logging into the tools through which AI assistance is provided (e.g. IDEs, notebooks, document editors).
- Expose summarised authorship information at boundary events: when a researcher endorses a theorem or draft, they can see which parts are predominantly AI-authored and adjust their scrutiny accordingly.
- Support post-hoc analysis of projects: for example, computing statistics on where AI contributions cluster (definitions vs. proofs vs. exposition).

## I3: Peer and Committee Evaluation Logs

Scientific correctness is often mediated by peer review and committee decisions. Within FRFP, these are institutional boundary events combining many tacit evaluations.

For each review or committee decision, an evaluation log records:

- the explicit artefacts considered (manuscript versions, proof logs, datasets, models);

- individual judgments (accept / reject / request changes; confidence);
- the institutional rule used to combine these judgments (simple majority, weighted vote, veto, chair decision);
- the resulting collective stance and any changes to  $(V, C, P)$  in the research integrity ledger.

### Implementation.

- Integrate FRFP-aware evaluation forms into journal, conference, or internal review systems;
- encourage reviewers to explicitly comment on:
  - the explicit/tacit structure of the work (what is well formalised vs. what relies on tacit expertise);
  - any suspected safe-horizon violations (e.g. over-reliance on AI proofs or simulations);
  - provenance and authorship (clarity about AI assistance).
- Use these logs to refine safe-horizon and admissibility parameters for future projects.

### 6.6.5 Metrics and Monitoring for Scientific FRFP

To monitor long-horizon runs, an implementation can define simple, domain-tailored metrics, understood as observables on runs and boundary events.

#### Example Metric Families

**Re-grounding frequency.** For each project, measure:

- how often primary sources (original papers, raw data) are re-read;
- how often core definitions are revisited and potentially refined;
- how often proofs are re-derived without AI assistance.

**Boundary density.** Measure, per unit length of the run (or per session):

- number of boundary events;
- distribution of their  $(V, C, P)$  statuses;
- rate of later revisions to earlier boundary decisions.

**AI reliance profile.** For a given artifact or project:

- fraction of edits labeled as AI-authored in the hybrid authorship log;
- concentration of AI involvement by artifact type (statements vs. proofs vs. exposition);
- correlation between AI involvement and later revisions or corrections.

None of these metrics define correctness on their own. They serve as *signals* about degradation, safe-horizon stress, and possible protocol failures that merit human attention.

### 6.6.6 Illustrative Pattern: AI-Assisted Theory Development

To make the above more concrete, consider a stylised pattern for FRFP-aligned theoretical work.

#### Phase F0/F1: Exploration and Proto-Formalization

- Researchers use AI systems to survey literature, generate examples, propose conjectures, and suggest potential proof strategies.
- The run is tagged as exploratory (phases F0/F1), with relatively loose safe horizons and frequent AI involvement.
- There are few boundary events; results are treated as speculative.

#### Phase F2: Formal Development

- Selected conjectures are promoted to formal objects in  $E$  and given stable labels.
- AI systems may still propose steps, but each non-trivial step in a proof must be locally checked by a human or an independent proof assistant; unchecked AI-generated steps are not admissible for boundary endorsement.
- Boundary events occur when the team accepts a lemma or theorem as provisionally correct. These events are logged with: the proof sketch, authorship information, and reviewer identity.
- Safe horizons constrain how many AI-suggested steps can be accepted without a fresh, human-driven re-derivation.

#### Phase F3: Consolidation and Dissemination

- Proofs are refactored into canonical forms, with minimised dependence on AI-generated scaffolding.
- Manuscripts are drafted with clear provenance for AI assistance.

- External peer review provides institutional boundary events, updating the research integrity ledger and potentially modifying  $(V, C, P)$ .

At every stage, FRFP’s explicit/tacit split and boundary semantics guide tool design (what AI is allowed to do), protocol design (where and how humans must endorse), and governance (how committees and journals record and justify their decisions).

### 6.6.7 Summary

This appendix has described an FRFP implementation tailored to scientific research:

- explicit pipelines in  $N \times E$  capture long-horizon research activity, including AI-assisted steps;
- tacit space  $T$  and boundary events capture human scientific judgment and correctness;
- safe horizons and re-grounding practices control degradation and long-run hallucination risk;
- instruments (integrity ledgers, authorship logs, evaluation logs) and metrics make the structure of inquiry visible to researchers, reviewers, and institutions.

This implementation is intended as one concrete instantiation of FRFP in a single domain (scientific inquiry). Other domains—such as engineering, finance, or clinical practice—may require different instruments and metrics, but can follow the same pattern: start from the FRFP kernel, then design domain-specific pipelines, boundaries, and governance on top.

## 6.7 Implementing FRFP in Engineering Workflows

This appendix sketches a concrete implementation of the Foundational Reasoning and Feedback Protocol (FRFP) for engineering domains, such as software, infrastructure, and systems engineering. We assume the FRFP core from the main text: explicit/tacit semantics, explicit pipelines in  $N \times E$ , a boundary functor  $J-K : N \times E \rightarrow T$ , and dynamic / institutional semantics.

### 6.7.1 Scope and Objectives

We consider settings where:

- AI systems assist in design, analysis, coding, verification, documentation, and incident analysis;
- correctness is grounded in specifications, physical constraints, safety standards, and organizational requirements;
- work is iterative and multi-stage: architecture, implementation, testing, deployment, and operations.

An FRFP-aligned engineering implementation should:

1. keep AI confined to explicit artefacts (requirements drafts, designs, code, tests, logs);
2. attach correctness only at human boundary events (approvals, sign-offs, risk acceptances);
3. design safe horizons for automated transformations and refactors;
4. surface provenance and authorship for later audit and incident review.

## 6.7.2 Core Principles for Engineering Pipelines

### E1: Explicit Engineering Artefacts

Explicit space  $E$  contains:

- requirements documents, design specs, interface definitions;
- code, configuration files, infrastructure-as-code, models, and schemas;
- test cases, proofs, static-analysis and verification artefacts;
- logs, metrics, incident reports, and runbooks.

AI systems may:

- generate and refactor code;
- propose designs and architectures;
- suggest test cases and checks;
- summarize logs and incident timelines.

All such contributions appear as morphisms in  $N \ltimes E$ .

### E2: Tacit Engineering Judgment and Boundary Events

Tacit space  $T$  contains:

- engineers' understanding of the system, failure modes, and domain;
- awareness of regulatory, safety, and organisational constraints;
- practical sense of acceptable risk and robustness.

Boundary events in engineering include:

- approval of system requirements or architecture;

- code review approvals and merge decisions;
- sign-off on test coverage and verification status;
- deployment approvals, change management sign-offs;
- acceptance of incident postmortems and remediation plans.

At each boundary, responsible humans in  $T$  evaluate explicit artefacts in  $E$  and attach correctness for action.

### **E3: Safe Horizons in Automation and Refactoring**

Safe horizons for engineering pipelines limit:

- how many automated refactors or transformations may be chained without human review;
- how far AI-assisted design exploration can go before re-grounding in core constraints and requirements;
- how long monitoring and remediation workflows can run under automated control before escalation.

An implementation must parameterise these horizons per system criticality (e.g. safety-critical vs. internal tooling).

## **6.7.3 Engineering Methodology as FRFP Pipelines**

### **EM1: Requirements and Architecture**

**Pipeline structure.** A project begins with a run segment  $\rho_{\text{req}}$  in which:

- stakeholders articulate goals and constraints (explicit artefacts);
- AI systems may propose requirement drafts, refine wording, and check consistency with existing documents;
- navigation morphisms in  $N$  move between stakeholder views and levels of abstraction.

**Boundary events.** Boundary events at this phase include:

- formal approval of a requirements baseline;
- acceptance of an architecture document as the basis for implementation.

At these boundaries, humans must see:

- the proposed document(s);
- provenance information indicating where AI contributed;
- trace links to standards, prior decisions, and risk assessments.

## EM2: Implementation and Verification

**Pipeline structure.** The implementation phase is represented by a run  $\rho_{\text{impl}}$  comprising:

- explicit code changes (possibly AI-generated);
- test design and updates;
- static analysis, verification runs, and simulation artefacts.

**Safe horizons.** Safe horizons constrain:

- length of AI-driven code modification chains between human reviews;
- number of test case generations before a human must inspect for coverage gaps and specification mismatch;
- amount of automatic config or IaC changes before manual audit.

**Boundary events.** Typical boundaries:

- code review and merge approvals;
- verification and test sign-offs;
- internal releases and promotions between environments.

Each boundary is logged with:

- the diff and tests considered;
- responsible reviewers and their judgments;
- any explicit delegation to downstream checks.

## EM3: Deployment, Monitoring, and Incidents

**Pipeline structure.** Deployment and operations produce runs over:

- deployment artefacts and configuration states;
- monitoring dashboards, alerts, and traces;
- incident tickets, runbooks, and remediation changes.

AI may:

- suggest rollouts and rollbacks;

- summarise incidents;
- propose remediation steps;
- cluster and interpret logs.

**Boundary events.** Boundary events include:

- authorisation of production deployments and rollbacks;
- acceptance of remediation steps in incidents;
- closure of incidents and sign-off on postmortems.

These events must remain human decisions, potentially informed by AI summaries but not replaced by them.

#### 6.7.4 Instruments and Metrics for Engineering FRFP

##### EI1: Change and Review Ledger

A change and review ledger records:

- each change set (code/config/design) as an explicit morphism;
- the associated tests and verification artefacts;
- review decisions and reviewers' identities;
- provenance of AI assistance in each change.

It enables queries such as:

- “Which production incidents involved changes with high AI authorship and minimal human review?”
- “Where did we accept AI-suggested changes at shallow safe horizons?”

##### EI2: AI-Aided Change Profile

Metrics include:

- fraction of changes initiated by AI suggestions;
- distribution of AI involvement by subsystem or component;
- correlation between AI-involved changes and later defects or incidents.

These are not correctness guarantees, but signals for FRFP-aligned governance to adjust safe horizons and review policies.

### 6.7.5 Summary

In engineering, FRFP yields an implementation in which:

- AI operates on explicit artefacts (requirements, code, tests, logs);
- correctness is conferred only at human boundary events;
- safe horizons control automated change sequences and incident response;
- ledgers and logs make explicit pipelines and endorsements auditable.

## 6.8 Implementing FRFP in Financial Decision-Making

This appendix outlines an implementation of FRFP for financial domains: banking, trading, risk management, and related activities in which AI systems provide analysis and recommendations while human agents and institutions retain ultimate responsibility.

### 6.8.1 Scope and Objectives

We consider workflows where:

- AI systems generate or transform credit analyses, risk reports, stress scenarios, trading ideas, and portfolio diagnostics;
- decisions have regulatory, fiduciary, or systemic implications;
- workflows are long-horizon (ongoing monitoring, repeated committee reviews, annual models and stress programmes).

An FRFP-compliant implementation should:

1. confine AI to explicit artefacts (analyses, scenarios, reports, dashboards);
2. reserve risk-taking and approvals as human boundary events;
3. define safe horizons for AI-supported monitoring and strategy revisions;
4. enable rigorous audit and regulatory scrutiny over explicit pipelines and decisions.

### 6.8.2 Core Principles for Financial Pipelines

#### F1: Explicit Financial Artefacts

Explicit space  $E$  includes:

- counterparty data, financial statements, market data, scenario sets;

- risk models, pricing models, stress-test engines, and their outputs;
- AI-generated narratives (credit memos, stress explanations, dashboards);
- policies, limits, risk appetite statements, and committee minutes.

AI may:

- summarise exposures and drivers;
- construct scenario narratives and structured stress sets;
- generate draft memos and recommendation templates;
- flag anomalies and pattern shifts in monitoring data.

## **F2: Tacit Financial Judgment and Boundary Events**

Tacit space  $T$  contains:

- portfolio managers' and risk officers' understanding of markets, counterparties, products, and regulatory context;
- knowledge of past crises, liquidity dynamics, and behavioural effects;
- shared institutional norms about acceptable risk.

Boundary events include:

- trade approvals and investment decisions;
- credit committee approvals and limit settings;
- model and scenario approvals by risk committees;
- acceptance of capital plans and regulatory submissions.

At each boundary, human decision-makers evaluate explicit artefacts and attach correctness in light of  $T$ .

## **F3: Safe Horizons in Monitoring and Strategy**

Safe horizons specify:

- how many AI-generated monitoring reports can be accepted without a full human re-baselining of assumptions and models;
- how often AI-suggested scenario updates require committee-level review rather than desk-level acceptance;

- what thresholds in metrics trigger escalated boundary events.

An implementation sets shorter horizons for high-risk or illiquid positions, and longer ones for routine exposures.

### 6.8.3 Financial Methodology as FRFP Pipelines

#### FM1: Credit and Investment Decisions

**Pipeline structure.** For a credit or investment case, a run  $\rho_{\text{credit}}$  includes:

- data gathering and validation;
- model runs (PD, LGD, PFE, etc.) and stress simulations;
- AI-generated summaries and draft memos;
- human edits, analysis, and commentary.

**Boundary events.** Boundary events include:

- desk-level recommendation (accept/decline/hold);
- credit committee decision;
- limit-setting decisions for the counterparty or product.

Each event is logged with:

- the artefacts considered (models, scenarios, memos);
- voting patterns or individual approvals;
- provenance of AI assistance in the written record.

#### FM2: Stress Testing and Capital Planning

**Pipeline structure.** A stress-testing programme yields a run  $\rho_{\text{stress}}$  involving:

- generation and refinement of macro and idiosyncratic scenarios;
- application of models and aggregations;
- AI-generated scenario narratives and impact summaries;
- human adjustments and overlays.

**Safe horizons.** Safe horizons limit:

- the number of purely AI-generated scenario modifications accepted without independent macro or risk review;
- the depth of automated overrides of model outputs before escalation.

Boundary events occur when:

- scenarios and methodologies are approved for use;
- final stress results are accepted for internal or regulatory use.

### **FM3: Ongoing Monitoring**

**Pipeline structure.** Monitoring yields repeated runs over:

- time series of exposures and risk indicators;
- AI-generated anomaly flags, trend analyses, and alerts;
- human investigations and follow-ups.

**Boundary events.** Boundary events include:

- escalation of a monitoring signal to a case or committee;
- changes in limits or hedging strategies;
- decisions to ignore or downgrade an AI-generated concern.

## **6.8.4 Instruments and Metrics for Financial FRFP**

### **FI1: Decision and Scenario Ledger**

A decision ledger records:

- the explicit pipeline leading to each credit or investment decision;
- the scenario and model set in force at that time;
- individual votes and reasoning;
- indicators of AI involvement in summaries and drafts.

## FI2: Monitoring Profile

Monitoring metrics include:

- number and type of AI-generated alerts per period;
- fraction escalated to human review;
- fraction overridden by humans;
- time between repeated re-baselining of monitoring assumptions.

These metrics inform adjustments to safe horizons and escalation rules.

### 6.8.5 Summary

In finance, FRFP yields an implementation in which:

- AI systems generate and process explicit artefacts, but do not autonomously take risk or approve positions;
- human decision-makers attach correctness at clearly defined boundaries;
- safe horizons govern monitoring, scenario use, and AI reliance;
- ledgers and logs make decisions and their AI involvement reconstructible for internal governance and external regulators.

## 6.9 Implementing FRFP in Clinical Practice

This appendix presents an implementation of FRFP in clinical workflows, such as hospital care, outpatient medicine, and clinical decision support. We again assume the FRFP core from the main text.

### 6.9.1 Scope and Objectives

We consider settings where:

- AI systems summarise patient histories, suggest differential diagnoses, propose treatment plans, and assist with documentation;
- clinical decisions have direct impacts on patient health and legal liability;
- workflows involve multiple hand-offs and teams (nurses, physicians, specialists, pharmacists, administrators).

An FRFP-compliant clinical implementation should:

1. confine AI to explicit artefacts (notes, summaries, suggestions);
2. leave diagnosis and treatment decisions as human boundary acts;
3. manage safe horizons in repeated, AI-assisted documentation and decision support;
4. ensure that provenance and responsibility are clear in the record.

## 6.9.2 Core Principles for Clinical Pipelines

### C1: Explicit Clinical Artefacts

Explicit space  $E$  contains:

- structured patient data (vitals, labs, imaging results, medications);
- free-text notes (H&P, progress notes, discharge summaries);
- guidelines, order sets, protocols, and clinical pathways;
- AI-generated summaries, differentials, risk scores, and recommendations.

AI systems may:

- draft notes from structured data and prior text;
- summarise long chart histories;
- suggest diagnoses or management options;
- rank guideline or order-set options given the current context.

### C2: Tacit Clinical Expertise and Boundary Acts

Tacit space  $T$  contains:

- clinician expertise, pattern recognition, and sense of atypicality;
- knowledge of local practice norms and patient-specific context;
- ethical judgment and responsibility for patient outcomes.

Boundary events include:

- signing of notes and orders (diagnostic and therapeutic);
- finalising discharge summaries and hand-off documents;
- multidisciplinary conference decisions (tumour boards, complex case conferences);
- acceptance of AI-supported predictions into standard workflows.

At each boundary, a clinician (or a team) must evaluate AI contributions in light of  $T$  and attach correctness for action.

### C3: Safe Horizons in Documentation and Decision Support

Safe horizons control:

- how many consecutive notes may be heavily AI-assisted without independent re-reading of primary data;
- how often AI-supported diagnostic suggestions must be audited against full guideline review or manual literature review;
- how long a patient can be managed with AI-summarised charts before a clinician must re-construct key elements from raw data.

Horizon parameters depend on patient acuity, setting (ICU vs. outpatient), and risk tolerance.

#### 6.9.3 Clinical Methodology as FRFP Pipelines

##### CM1: Intake and Initial Assessment

**Pipeline structure.** For a new admission, a run  $\rho_{\text{intake}}$  may include:

- ingestion of prior records and structured data;
- AI-generated summary of history and active problems;
- clinician review, correction, and augmentation of the summary.

**Boundary event.** Boundary occurrence:

- clinician signs the initial note and places initial orders.

At this point, correctness is conferred on:

- the problem list and working diagnoses;
- key decisions (admission level, initial tests, immediate treatment).

##### CM2: Ongoing Management

**Pipeline structure.** Over the course of care, a run  $\rho_{\text{course}}$  includes:

- repeated AI-assisted progress notes;
- AI-generated trend summaries (vitals, labs, risk scores);
- human interpretation, updates to diagnoses, and treatment adjustments.

**Safe horizons.** Safe horizons might enforce:

- a limit on how many successive notes can be accepted with minimal editing of AI output;
- periodic requirements to review raw labs, images, and data streams directly;
- checks before relying on AI-generated risk scores for escalation decisions.

**Boundary events.** Boundary points include:

- daily sign-off on progress notes and orders;
- decisions to escalate care or transfer units;
- significant changes in diagnosis or treatment plan.

### CM3: Discharge and Handoffs

**Pipeline structure.** At discharge or hand-off, a run  $\rho_{\text{handoff}}$  contains:

- AI-assisted draft of discharge summary or hand-off note;
- clinician edits to ensure accuracy, emphasising key risks and follow-up;
- generation of patient-facing summaries and instructions.

**Boundary event.** The final discharge or hand-off note is a boundary artefact:

- its signing marks the clinician’s endorsement of its content;
- errors at this point can propagate to downstream providers and future care decisions.

FRFP requires that AI contributions are visible and that clinicians have sufficient time and support to re-ground their tacit understanding before signing.

## 6.9.4 Instruments and Metrics for Clinical FRFP

### CI1: Clinical Documentation Ledger

A documentation ledger records, for each patient:

- all signed notes and orders as boundary artefacts;
- AI involvement in each note (e.g. percentage of AI-generated text, suggestions taken vs. rejected);
- episodes where clinicians explicitly overrode AI suggestions;

- links to underlying data and guidelines referenced.

This enables later review of:

- how AI assistance correlates with documentation errors;
- patterns of overreliance or under-use across clinicians or units.

## **CI2: Decision-Support Profile**

For AI-supported decision aids (e.g. risk scores, triage suggestions), metrics include:

- rate of use and override by clinicians;
- cases where AI disagreed with clinician judgment and outcome data;
- time since last full audit of model performance and assumptions.

These metrics help calibrate safe horizons for decision support.

### **6.9.5 Summary**

In clinical practice, FRFP yields an implementation in which:

- AI systems operate on explicit artefacts (notes, summaries, scores);
- clinicians retain responsibility for diagnosis and treatment at boundary events;
- safe horizons limit extended, ungrounded reliance on AI outputs;
- documentation and evaluation logs make AI involvement and human judgment reconstructible for quality improvement and oversight.

## Chapter 7

# FRFP Case Study: IBM Watson for Oncology

## 7.1 FRFP Case Study (Clinical): IBM Watson for Oncology

This appendix analyses the rise and failure of *IBM Watson for Oncology* (WFO) as an extended case study in FRFP terms. Rather than recounting the technical details exhaustively, we focus on how the project’s structure interacted with FRFP’s explicit/tacit split, boundary design, safe horizons, and institutional semantics.

### 7.1.1 Background

After Watson’s Jeopardy! success in 2011, IBM launched Watson Health with ambitions to “revolutionise” cancer care. Watson for Oncology was developed primarily with Memorial Sloan Kettering Cancer Center (MSKCC) as an AI-powered decision support system: given a structured patient description, WFO would output ranked treatment options and rationales, informed by guidelines, literature, and MSK experts.

IBM marketed WFO to hospitals globally, including leading cancer centres and smaller regional hospitals, often positioning it as a way to deliver MSK-like expertise at the bedside. At MD Anderson Cancer Center, a related system (Oncology Expert Advisor, OEA) based on Watson technology consumed  $\sim$  \$62M over several years and never reached safe use on real patients before the contract was terminated.

In 2017–2018, investigative reporting and internal IBM documents revealed that Watson for Oncology often produced treatment recommendations that were unsafe, incorrect, or inconsistent with established guidelines. In some test cases, WFO recommended regimens contraindicated by “black box” warnings (e.g. suggesting bevacizumab for a patient with lung cancer and active severe bleeding). Hospitals reported that WFO’s outputs were frequently unusable or misaligned with local practice, and several partnerships were discontinued or scaled back.

This trajectory makes WFO a natural FRFP case: there is a clear separation between explicit artefacts (structured cases, guidelines, recommendations) and tacit clinical expertise; persistent uncertainty about boundaries between AI “advice” and human decisions; and significant institutional and governance dynamics.

### 7.1.2 Explicit and Tacit Spaces in Watson for Oncology

#### Explicit space $E_{\text{clinical}}$

In FRFP terms, Watson for Oncology’s explicit domain comprises:

- structured patient representations (stage, biomarkers, comorbidities);
- codified guidelines (e.g. NCCN recommendations) and literature abstractions as ingested into Watson;
- curated training cases and MSK expert-labelled examples, in practice heavily based on synthetic or hypothetical cases rather than large cohorts of real patients;

- the ranked list of treatment options with labels such as “recommended”, “for consideration”, “not recommended”, together with short textual rationales;
- user-interface artefacts: screens, order sets, explanatory panels.

All of these are explicit artefacts manipulable by the AI system: given a patient input, WFO transforms explicit encodings into explicit recommendations via model internals and expert rules.

### **Tacit space $T_{\text{clinical}}$**

Tacit space in this setting consists of:

- individual oncologists’ clinical expertise, including pattern recognition for atypical cases and nuanced trade-offs (toxicity vs. benefit, patient preferences, comorbidities);
- knowledge of local formularies, resource constraints, and institutional norms;
- shared, evolving community sense of what constitutes acceptable interpretation of evidence in specific cancer subtypes;
- ethical and legal sense of responsibility for patient outcomes.

Oncologists do not simply read guidelines; they interpret them. They adjudicate between trials, reconcile contradictory evidence, and adapt decisions to specific patient contexts in ways not fully codified in explicit rules.

### **7.1.3 Intended Boundary Design vs. De Facto Boundary**

#### **Stated design: AI as decision support**

Publicly, IBM framed WFO as a clinical decision support tool:

- the oncologist remains the decision-maker;
- WFO proposes options; the human reviews and chooses;
- the system encodes MSK expertise and guidelines but does not replace human judgement.

In FRFP terms, this corresponds to: AI confined to explicit space  $E_{\text{clinical}}$ , with a boundary at the *clinician’s acceptance* (or rejection) of a recommendation, mediated by their tacit state in  $T_{\text{clinical}}$ .

### Marketing and perception: boundary drift

However, several features of deployment and marketing blurred that boundary:

- IBM and some partners promoted WFO as offering “best possible treatment” based on leading cancer centres, which implicitly encourages clinicians and administrators to treat its outputs as authoritative rather than tentative.
- At some hospitals (e.g. Jupiter Medical Center), executives appear to have acquired WFO partly for marketing value, positioning it as a differentiating capability, while clinicians later reported that the product was largely unusable.
- Early MD Anderson efforts focused more on ambitious promises and integration challenges than on rigorous, prospective validation in real patients; the system never reached routine clinical use before being shelved.

From an FRFP perspective, these patterns erode the explicit/tacit boundary:

- explicit artefacts (ranked options) are framed as near-final answers;
- tacit endorsement risks becoming a rubber stamp on system outputs;
- institutional incentives push toward de facto automation, even where formal documentation still describes WFO as “advisory”.

#### 7.1.4 Protocol-Level Failures in FRFP Terms

We now describe the main failure modes of Watson for Oncology as failures of FRFP protocol design, rather than isolated model bugs.

### Z.1 Boundary Confusion and Over-Automation Pressure

**Problem.** WFO blurred who was really “deciding”:

- The UI presented a small, ranked list of options with categorical labels, rather than a broad landscape of possibilities.
- Marketing emphasised Watson’s “expertise” and partnership with elite centres, encouraging deference.
- Internal documents show that some customers perceived it as insufficiently useful or safe, but this was discovered only after deployment and real-world testing.

In FRFP language:

- the *boundary* between explicit suggestion and tacit endorsement was not structurally protected;

- nothing in the protocol enforced that clinicians treat WFO outputs as provisional hypotheses requiring independent re-grounding;
- institutional messaging encouraged treating WFO’s explicit artefacts as if correctness lived in the system rather than in human tacit states.

**Consequence.** This boundary confusion made it plausible, in principle, that clinicians under time pressure could over-rely on WFO for guidance, especially in less specialised settings. Even where clinicians ultimately rejected WFO in practice (as at several sites), the FRFP design left no robust guardrail to prevent over-automation.

## Z.2 Explicit-Space Pathologies: Training and Coverage

**Problem.** Subsequent reporting and IBM internal documents revealed that:

- WFO’s training relied heavily on small sets of synthetic or hypothetical cases labelled by one or a few MSK physicians, rather than large, diverse, real-patient cohorts;
- the case library lacked statistical design, undermining coverage and generalisability;
- recommendations sometimes deviated from widely adopted national guidelines (e.g. NCCN) or reflected “unconventional” interpretations of evidence;
- some recommendations were directly unsafe, e.g. suggesting a drug explicitly contraindicated for patients with active bleeding.

In FRFP terms, these are failures in explicit space  $E_{\text{clinical}}$ :

- the mapping from patient encodings to recommended options was underspecified and poorly validated;
- the training set did not define a reliable basis for a semantics that clinicians could safely treat as approximating guideline-based reasoning;
- explicit representations (case abstractions and labelled options) did not faithfully capture the tacit standards of the oncology community.

**Consequence.** Even if boundaries had been perfectly designed and strictly respected, the explicit artefacts were structurally fragile: a human relying on them as “MSK in a box” would be misled about their coverage and limits. Within FRFP, this corresponds to a defective explicit semantics feeding into the boundary functor  $J-K$ .

### Z.3 Safe-Horizon Violations and Degradation Risk

**Problem.** Watson for Oncology was intended for repeated use across many patients and cases:

- clinicians would query WFO case after case, day after day;
- institutional partners hoped to build standardised WFO-informed workflows and even marketing narratives around it.

FRFP predicts that in such long-horizon workflows:

- tacit grounding can degrade: clinicians may start to read fewer guidelines directly, rely more on WFO’s summaries, and accept its pattern of options as normal;
- hallucinations (unsafe or unsupported endorsements) become almost inevitable along sufficiently long runs if boundaries and re-grounding are not carefully designed.

Yet WFO’s design and deployments did not incorporate explicit safe-horizon constraints:

- there were no protocol-level limits on how many WFO-guided decisions could be made without deeper human re-grounding in primary evidence;
- there was no requirement for periodic re-validation of WFO’s recommendations against up-to-date guidelines for each cancer type;
- no explicit mechanisms enforced independent re-derivation of rationale by clinicians after a certain depth of WFO usage.

**Consequence.** In organisations that might have fully embraced WFO, FRFP would predict a drift toward overreliance: clinicians would likely start accepting WFO’s pattern of options as “the way we treat here”, reducing their sensitivity to edge cases and guideline changes. In practice, the opposite happened at many sites: clinicians quickly lost trust and utilisation collapsed. From an FRFP perspective, this is almost a *fortunate* failure mode: trust broke before dangerous long-horizon degradation could fully set in.

### Z.4 Institutional Operator Failures

**Problem.** At the institutional level, WFO’s story involves:

- high-level partnerships (MSK, MD Anderson, others) with ambitious goals and significant budgets;
- weak prospective clinical validation prior to marketing;
- internal IBM and hospital audits revealing cost overruns, lack of integration, and poor clinical utility;

- eventual sell-off of Watson Health assets, effectively ending the original oncology vision.

In FRFP’s institutional semantics, we can think of:

- a population of tacit states  $T^{\text{POP}}$  including clinicians, administrators, IBM staff, regulators, and patients;
- institutional operators  $G$  (e.g. hospital governance, vendor decision processes, regulatory expectations) that update these states in response to explicit artefacts (pilots, metrics, sales pitches, incident reports).

WFO’s failure suggests that:

- institutional operators overweighted vendor demos and marketing narratives relative to rigorous FRFP-style admissibility checks (safe horizons, boundary clarity, explicit semantics);
- governance mechanisms were not designed to detect FRFP-structural problems (e.g. training on hypothetical cases, unsafe recommendations) early and decisively;
- in at least one major site (MD Anderson), an internal financial audit rather than an epistemic governance body drove termination.

**Consequence.** From an FRFP standpoint, this is an example of attempted *explicit-only governance*: institutions relied on marketing materials, limited tests, and internal dashboards rather than systematically designing boundaries, safe horizons, and explicit/tacit roles. The non-automation theorems in the main text suggest that such governance cannot in general recover tacit correctness.

### 7.1.5 Counterfactual: An FRFP-Aligned Watson for Oncology

It is instructive to sketch what WFO might have looked like under a strong FRFP design discipline.

#### Z.1\* Explicit/Tacit Separation and UI

- WFO would be explicitly framed as an *artefact generator*: it surfaces guideline passages, trial summaries, and similar cases, but does not present canonical ranked treatment lists as “Watson’s recommendation”.
- The UI would highlight:
  - multiple guideline-consistent options side by side;
  - conflicting evidence or lack of coverage for the current case;
  - explicit uncertainty markers (e.g. “no close match in training set; exercise heightened caution”).
- The system would avoid language suggesting that it “knows” the best treatment; instead, it would present itself as a *search and summarisation engine* for explicit artefacts.

**Z.2\* Boundary Design and Documentation**

- Boundary events would be clearly identified: signing a treatment plan, entering orders, or officially adopting a WFO-supported plan in the EHR.
- At each such boundary, the clinical interface would:
  - show which WFO artefacts were consulted;
  - show any divergences between WFO suggestions and base guidelines;
  - require a short human justification for accepting any option that relies materially on WFO output.
- Logs would treat boundary decisions as first-class events, enabling later audit of where WFO influenced care.

**Z.3\* Safe Horizons and Re-Grounding**

- For each cancer type, safe horizons would cap how many WFO-assisted decisions could be made before:
  - an independent human review of recent cases against up-to-date guidelines is required;
  - WFO’s training and rules are re-validated on real cases from that institution.
- For each clinician, usage statistics would highlight:
  - how often WFO suggestions are accepted vs. overridden;
  - whether there is a drift toward uncritical acceptance.
- At an institutional level, committees would periodically review WFO’s impact on care patterns and outcomes, with authority to adjust horizons, restrict indications, or suspend use.

**Z.4\* Training and Explicit Semantics**

- Training would rely on large, audited collections of real cases with diverse practice patterns and explicit probabilistic uncertainty, rather than small synthetic case sets.
- The system would maintain explicit versioned representations of guidelines and evidence, allowing clinicians to see exactly which sources underpin each suggestion.
- Changes in guidelines or evidence would create explicit version transitions in WFO’s internal knowledge, visible at the boundary.

### Z.5\* Institutional Governance

- A formal FRFP-style governance body (e.g. an AI clinical oversight committee) would:
  - review WFO’s design before deployment;
  - approve specific indications, safe horizons, and audit plans;
  - require regular reporting on errors, overrides, and usage.
- Marketing claims and external messaging would be constrained to match what the governance body has actually evaluated.

Such a counterfactual design cannot guarantee success, but it would align WFO with FRFP’s structural constraints: AI as explicit-only, correctness at human boundaries, finite safe horizons, and explicit institutional accountability.

### 7.1.6 Lessons for FRFP and Clinical AI

The Watson for Oncology case provides several lessons that support FRFP’s claims:

1. **Explicit success does not imply tacit safety.** A system that appears impressive in demos or pilot metrics can still produce unsafe recommendations when its explicit semantics are not aligned with tacit expert standards and real patient diversity.
2. **Boundary rhetoric is not boundary design.** Calling a system “decision support” does not make it FRFP-compliant; one must design interfaces, logging, and governance so that humans remain genuine endorsers with adequate context and time.
3. **Safe horizons matter in clinical decision support.** Tools like WFO are intended for repeated, long-horizon use. Without explicit re-grounding and horizon management, tacit degradation and overreliance are structurally likely.
4. **Institutional non-automation is not optional.** WFO’s failure underscores that institutions cannot outsource epistemic governance to vendor metrics and marketing; they must act as FRFP-style institutional operators with the authority and obligation to enforce boundaries and safe horizons.
5. **Formal architecture helps diagnose failures.** Many narratives about WFO focus on “overhype” or “poor execution”. FRFP allows a sharper diagnosis: explicit-space design flaws, boundary confusion, absent safe horizons, and explicit-only governance together made the system structurally fragile, independent of any particular model architecture.

In this light, Watson for Oncology is not just an anecdote of an overhyped AI. It is an early, large-scale illustration of why a formal architecture like FRFP is needed to reason about clinical AI systems whose capabilities and institutional footprints will only grow.

## Chapter 8

# FRFP Case Study: IBM Watson Health and the Platform Vision

## 8.1 FRFP Case Study (Health Systems / Industry): IBM Watson Health

This appendix looks beyond Watson for Oncology and analyses the broader arc of *IBM Watson Health* as an FRFP-style case study. We treat Watson Health as a large-scale attempt to build a healthcare AI platform and show how its trajectory illustrates FRFP’s concerns around explicit/tacit separation, boundary design, safe horizons, and institutional governance.

### 8.1.1 Background: From Jeopardy! to Healthcare Platform

After Watson’s 2011 Jeopardy! success, IBM announced ambitious plans to apply Watson to healthcare, pitching it as a system that could ingest vast amounts of medical literature and help clinicians make better decisions. By 2015, IBM had created the Watson Health business unit, acquired several health data companies (e.g. Truven Health Analytics, Phytel, Explorys), and established high-profile partnerships with institutions like MD Anderson Cancer Center, Memorial Sloan Kettering, Mayo Clinic, and others.

Watson Health’s portfolio included:

- Watson for Oncology and similar decision-support tools in oncology;
- Watson for Genomics and clinical trial matching systems;
- population health / analytics platforms based on acquired datasets;
- imaging and radiology decision support initiatives.

However, performance in real-world deployments lagged far behind marketing:

- The MD Anderson Oncology Expert Advisor (OEA) project, using Watson technology, consumed more than \$60M and was halted after an audit concluded it was not ready for clinical use and struggled to integrate with EMR systems.
- Watson for Oncology was criticised for recommending unsafe or incorrect treatments in internal tests and for failing to match local practice patterns in partner hospitals.
- Clinicians often found Watson’s recommendations either obvious or impractical, and user engagement remained low.
- By 2018–2019, IBM laid off significant staff in Watson Health amid market disillusionment.

In January 2022, IBM agreed to sell Watson Health’s data and analytics assets to private equity firm Francisco Partners; these were later relaunched under the brand *Merative*. Watson Health, as originally envisioned, effectively ceased to exist.

FRFP reads this story not merely as a business failure, but as a large-scale experiment in embedding explicit AI systems within complex healthcare institutions.

### 8.1.2 Explicit and Tacit Spaces at the Platform Level

#### Explicit space $E_{WH}$

Across Watson Health products, explicit space included:

- large corpora of medical literature, guidelines, and clinical trial registries;
- structured claims and administrative data from acquisitions (Truven, Explorys, etc.);
- prototype knowledge bases and rules encoded with clinical partners;
- ML/NLP models and pipelines for ingestion, entity extraction, matching, and recommendation;
- dashboards and UIs for clinicians and administrators.

These artefacts were manipulated by Watson’s NLP and ML components to produce:

- ranked treatment options and genomic interpretations;
- patient-trial matches;
- population risk scores and cost-prediction dashboards;
- imaging decision support outputs.

#### Tacit space $T_{WH}$

Tacit space spanned multiple actors:

- frontline clinicians’ expertise and pattern recognition;
- hospital administrators’ understanding of workflows, incentives, and regulatory constraints;
- IBM engineers’ and product leaders’ intuitions about AI capability and product-market fit;
- payers’ and regulators’ tacit standards for evidence and safety;
- patients’ expectations and trust.

These tacit states are where correctness, safety, and value are ultimately judged. FRFP emphasises that no amount of explicit data and models can substitute for tacit clinical and institutional judgment.

### 8.1.3 Boundary Design: From Demos to Clinical Reality

#### Demos and pilot narratives

Watson Health’s early narrative leaned heavily on demos:

- stylised cases where Watson quickly identified relevant literature or treatment options;
- inspirational stories of AI helping oncologists keep up with exploding literature;
- marketing claims about “fighting cancer” and transforming care through cognitive computing.

In FRFP terms, these demos sit entirely in explicit space: carefully chosen inputs and outputs, minimal exposure to the messy tacit reality of clinical practice.

#### Real-world boundary behaviour

In deployment, the boundary between Watson outputs and clinical decisions varied:

- Some institutions treated Watson outputs as optional second opinions that clinicians could consult or ignore; in these settings, usage dwindled when clinicians found little added value.
- In others, there was implicit pressure to incorporate Watson into workflows due to the cost of the partnership and executive expectations, even though clinicians remained formally responsible.
- Many deployments never reached stable routine use because system integration (EMR, data quality) and clinical trust barriers proved too high.

From an FRFP perspective:

- the intended boundary (AI as support, clinician as decider) was never robustly operationalised;
- in some contexts, Watson outputs risked boundary capture (becoming quasi-authoritative recommendations);
- in others, clinicians kept the boundary firmly in tacit space by simply not using the tool.

### 8.1.4 Protocol-Level Issues in FRFP Terms

#### Z.7.1 Explicit-Space Misalignment: Data, Knowledge, and Tasks

**Heterogeneous, messy data.** Watson Health attempted to operate on top of highly heterogeneous clinical data:

- EMRs with inconsistent formats, missing data, and noisy free text;
- claims datasets optimised for billing rather than clinical meaning;

- fragmented, siloed systems that resisted clean integration.

In FRFP’s explicit semantics, this means:

- $E_{WH}$  is not a cleanly structured knowledge space; it is a patchwork of partial, error-prone views;
- the mapping from raw EMR data to the structured inputs Watson uses is itself a fragile, error-prone explicit pipeline;
- many tacit clinical nuances (why a lab was ordered, how a symptom is interpreted) never enter  $E_{WH}$  at all.

**Overgeneralised task framing.** Watson’s Jeopardy! success encouraged a narrative in which:

ingesting lots of text + sophisticated NLP  $\Rightarrow$  good at any text-heavy domain, including oncology.

Critics argue that IBM underestimated how different the tasks are: Jeopardy! is about pattern-matching and recall; oncology is about causal and clinical reasoning under uncertainty, with strong safety, workflow, and ethical constraints.

In FRFP terms:

- the explicit transformations Watson applied were suitable for question-answering in trivia domains, but not for approximating tacit oncologist reasoning;
- the semantics of clinical decision-making were never properly represented in  $E_{WH}$ ; Watson often mapped symptoms and lab values directly to treatments without robust intermediate modelling.

### Z.7.2 Boundary and Workflow Design Failures

**Workflow disruption.** Clinicians in partner hospitals reported that:

- Watson’s interface and case-entry workflows were cumbersome and disconnected from their usual EMR flows;
- using Watson often meant duplicate data entry and extra steps, making it unattractive in time-pressured environments;
- when Watson outputs duplicated obvious guideline recommendations, the extra friction was not justified.

From an FRFP angle:

- the explicit pipelines in  $N \times E_{WH}$  were not aligned with existing tacit workflows in  $T_{WH}$ ;
- boundary events (e.g. choosing a treatment plan) were surrounded by extra explicit steps that added little tacit value;
- the result was low utilisation and erosion of the intended explicit/tacit coupling.

**Lack of prospective evaluation.** As STAT and others reported, IBM did not initially subject Watson for Oncology or related tools to rigorous, independent clinical trials before deploying them commercially. Instead, validation was often limited, retrospective, or internal.

For FRFP, this is a boundary problem:

- moving from demo/pilot to real clinical use is a major boundary transition requiring strong tacit endorsement from the clinical community;
- without prospective evidence, the tacit side never fully endorses the explicit semantics, leading to persistent distrust and poor integration.

### Z.7.3 Safe Horizons and Programme-Scale Risk

Watson Health was not one tool but a portfolio, scaled aggressively:

- multiple product lines launched in parallel, across oncology, genomics, imaging, and population health;
- large, multi-year contracts and acquisitions committed IBM to a long horizon before clear evidence of clinical value emerged;
- layoffs and write-downs began only after years of limited traction and critical reporting.

In FRFP's dynamic semantics:

- Watson Health as a whole is a long run  $\rho_{WH}$  in  $N \times E_{WH}$  with many branches and sub-pipelines;
- safe horizons would require:
  - staged pilots with clear success criteria;
  - caps on deployment scale before independent validation;
  - explicit re-grounding after each phase (revisiting tacit clinical judgments about utility and safety).
- Instead, the programme expanded rapidly, with safe-horizon controls largely implicit, financial, or reputational (e.g. pullback after negative press and partner exits).

The eventual sale of Watson Health assets to Francisco Partners can be read as a late, coarse boundary event: IBM's tacit judgment that the explicit portfolio was not delivering the expected value.

### Z.7.4 Institutional Governance and Narrative

Watson Health lived at the intersection of several institutional operators:

- IBM’s corporate governance and investor expectations;
- partner hospitals’ internal politics and incentives;
- media narratives about AI revolutions;
- regulators’ relatively hands-off stance for decision-support tools.

FRFP views these as operators  $G_i$  acting on tacit state distributions in  $T_{WH}$ :

- $G_{IBM}$  pushed toward framing Watson Health as a flagship proof of AI leadership, encouraging aggressive scaling and optimistic messaging;
- $G_{hospitals}$  had mixed incentives: prestige from partnerships vs. clinician frustration and scepticism;
- $G_{media}$  amplified both early hype and later disillusionment, shaping public and investor expectations.

These operators did not implement an FRFP-style governance layer explicitly:

- there was no cross-institutional body formally charged with defining admissible uses, boundaries, and safe horizons for Watson Health deployments;
- governance was emergent, reactive, and often driven by financial or reputational shocks (audits, critical articles, contract cancellations).

### 8.1.5 Counterfactual: An FRFP-Aligned Healthcare AI Platform

What might an FRFP-compliant analogue of Watson Health have looked like?

#### Z.7\* Narrow, Evidence-Gated Product Strategy

- Start with a small number of carefully scoped tasks: e.g. information retrieval, literature summarisation, or trial matching in well-defined cancer subtypes;
- treat each as a distinct pipeline in  $N \times E_{WH}$ , with its own explicit semantics, boundary design, and safe horizons;
- require prospective or quasi-prospective trials to demonstrate value and safety before scaling beyond pilot sites.

**Z.7\* Strong Explicit/Tacit Interface with Clinicians**

- Design interfaces that foreground primary sources (guidelines, trial reports) and make Watson’s role clearly that of a *search-and-structure engine*, not a quasi-physician;
- make boundary events explicit: treatment decisions, trial enrolment, and care-pathway selection always require human endorsement, logged with context;
- allow clinicians to easily see when and why Watson’s suggestions diverge from local practice or guidelines, and to feed those divergences back into the system as explicit signals.

**Z.7\* Safe Horizons for Deployment and Scaling**

- Limit initial deployments to small, well-monitored cohorts with explicit sunset or review points;
- define quantitative and qualitative criteria for expansion (e.g. concordance rates with expert panels, impact on outcomes, usability metrics);
- periodically re-ground the system’s tacit model of clinical utility via workshops and audits, adjusting or decommissioning pipelines that fail to demonstrate value.

**Z.7\* Institutional Governance Across Stakeholders**

- Establish joint FRFP-style governance boards with representatives from:
  - clinicians and nurses;
  - IT and data governance;
  - ethics and patient advocacy;
  - vendor (IBM-style) technical leads.
- Charge these boards with:
  - setting boundaries (what the AI may and may not do);
  - defining safe horizons (scale, duration, and scope limits);
  - deciding when to pause, pivot, or shut down specific deployments based on evidence.
- Make external communication (marketing, press releases) consistent with the governance body’s actual assessment of capabilities and validation status.

**8.1.6 Lessons for FRFP and Platform-Scale Healthcare AI**

Beyond the detailed oncology case in Appendix 7.1, the broader Watson Health trajectory highlights several FRFP-relevant points:

1. **Platform dreams outpaced explicit semantics.** Watson Health attempted to be a general healthcare AI platform before the explicit representations of clinical data and tasks were mature enough to support that ambition.
2. **Demos are not deployments.** High-profile proofs-of-concept in curated settings are runs in a very small region of  $N \times E$ . FRFP requires attention to how pipelines behave at scale, under messy, adversarial, and heterogeneous conditions.
3. **Safe horizons must be engineered, not assumed.** Large, multi-year, multi-product programmes need explicit phase boundaries and go/no-go gates, rather than relying on market disappointment as the only stopping condition.
4. **Institutional operators can amplify both hype and crash.** Corporate, clinical, and media institutions updated tacit beliefs in ways that first amplified unrealistic expectations, then swung to disillusionment. FRFP aims to provide structures where those updates are more evidence-driven and incremental.
5. **Formal architectures help distinguish structural from contingent failure.** Watson Health’s problems are often framed as “bad execution” or “overhype”. FRFP shows a deeper pattern: incomplete explicit semantics, weak boundary and workflow design, absent safe horizons, and diffuse governance—a pattern likely to recur in future healthcare AI platforms unless addressed at the architectural level.

Under this reading, IBM Watson Health is not merely a failed product line, but an early, large-scale probe of what happens when a powerful explicit AI stack is embedded in one of the most complex tacit domains humanity has: healthcare delivery. Its rise and fall provide empirical support for FRFP’s insistence on careful boundary design, safe horizons, and institutional non-automation in future platform-scale AI deployments.

## Chapter 9

# FRFP Case Study: Amazon's Experimental AI Recruiting Engine

## 9.1 FRFP Case Study (Governance / HR): Amazon’s Experimental Recruiting Engine

This appendix analyses Amazon’s experimental AI-based recruiting engine (2014–2017) as an FRFP case study. The project sought to automate résumé screening and candidate ranking, but was ultimately abandoned when it was found to systematically disadvantage female applicants for technical roles. We examine how the failure can be understood in terms of FRFP’s explicit/tacit split, boundary design, safe horizons, and institutional semantics.

### 9.1.1 Background

Around 2014, Amazon engineers were tasked with building a machine-learning system to streamline hiring for technical positions. The idea was that an algorithm would:

- ingest résumés from past applicants;
- learn patterns associated with candidates who were hired or performed well;
- score new résumés on a 1–5 star scale and help recruiters prioritise candidates.

To do this, the team trained models on roughly a decade of historical hiring data—résumés submitted to Amazon, overwhelmingly from men, reflecting the male-dominated tech workforce. Over time, they discovered that the system:

- systematically downgraded résumés that included the word “women’s” (e.g. “women’s chess club captain”);
- penalised graduates of certain all-women’s colleges;
- implicitly treated patterns correlated with male candidates as positive signals.

Engineers tried to patch obvious markers of gender (e.g. manually excluding features containing “women”), but the system continued to find proxy signals, and confidence in its fairness eroded. By 2017, Amazon reportedly stopped using the tool for live candidate screening and eventually shut the project down.

### 9.1.2 Explicit and Tacit Spaces in the Recruiting Engine

#### Explicit space $E_{\text{HR}}$

In FRFP terms, the engine operated entirely in explicit space:

- input artefacts: résumés encoded as text and features; job descriptions; historical hiring decisions;

- learned model parameters and feature weights;
- output artefacts: numeric scores (1–5 stars) for each candidate, possibly accompanied by internal diagnostics;
- dashboards or interfaces used by recruiters to view rankings.

All of this lives in  $E_{\text{HR}}$ : artefacts that can be stored, copied, and processed by machines.

### **Tacit space $T_{\text{HR}}$**

Tacit space comprises the epistemic states of:

- recruiters and hiring managers: their sense of role fit, career trajectories, culture, and potential;
- HR leadership and legal/compliance teams: their understanding of anti-discrimination law, fairness norms, and reputational risk;
- organisational leadership: their preferences over growth, diversity, and risk trade-offs.

These tacit states shape how people interpret explicit signals (résumés, scores, interviews) and ultimately decide whom to hire.

### **9.1.3 Intended Boundary vs. Actual Boundary**

#### **Stated intention: AI as pre-screening support**

Public accounts suggest that the recruiting engine was intended to act as:

- an internal tool that *ranks* candidates;
- a way to speed up high-volume screening;
- a decision-support system, not an official replacement for human hiring decisions.

In FRFP language, the intended boundary was:

- AI produces scored candidate lists in explicit space;
- humans (recruiters, hiring managers) retain tacit authority to select candidates, review résumés, and decide offers.

**De facto boundary: algorithm as gatekeeper**

However, in any scaled recruiting workflow, there is a strong pressure to treat ranked outputs as a gate:

- Recruiters focus on top-ranked candidates first; low-scoring candidates may never be reviewed.
- The star score becomes the primary triage mechanism for who gets interviews.
- Over time, recruiters may internalise the model’s patterns and align their tacit judgments to it, especially if performance metrics reward throughput.

Thus, the *effective* boundary tended toward:

- *model scores as de facto accept/reject filter,*
- with tacit human oversight reduced to rare overrides.

From an FRFP perspective, this is a classic case of boundary erosion: a system nominally described as “support” functions as a gatekeeper, pulling correctness authority into the model stack.

**9.1.4 Protocol-Level Failures in FRFP Terms****Z.2.1 Explicit-Space Bias from Historical Data**

**Problem.** The engine was trained on historical Amazon hiring data:

- Most past résumés and hires for technical roles were male.
- The model learned that patterns associated with male candidates were predictive of positive outcomes.
- It penalised explicitly gendered tokens (“women’s chess club”) and graduates of certain women’s colleges, even after developers tried to remove gender markers.

In FRFP terms, the explicit mapping

$$f_{\text{recruit}} : \text{résumé features} \rightarrow \text{score}$$

embeds a biased semantics: it effectively treats “being similar to past, mostly male hires” as a proxy for fitness.

**Consequence.** Even if humans at the boundary were perfectly vigilant, the explicit artefacts (scores) already compress candidates into a shape that bakes in historical bias. The FRFP kernel would classify this as a defect in explicit semantics: the pipeline in  $N \times E_{\text{HR}}$  does not implement a normatively acceptable mapping from résumés to evaluative signals.

### Z.2.2 Boundary Design and Over-Delegation

**Problem.** The project's ambition was to automate as much screening as possible. In practice, this implies:

- using the model to filter or prioritise thousands of résumés;
- trusting that top-ranked candidates are “good enough” to proceed;
- allowing low-ranked candidates to be silently dropped.

Under FRFP, this is a high-risk boundary design:

- the boundary between explicit scores and tacit acceptance is effectively moved upstream into the model;
- human recruiters are placed in a role where they mostly confirm model suggestions, not critically re-evaluate them;
- there is no explicit requirement that humans systematically examine candidates the model dislikes to detect systemic bias.

**Consequence.** Humans are nominally in the loop, but their tacit authority is weakened. The model's preferences dominate which résumés get attention at all, and thus which tacit evaluations can even occur. FRFP would characterise this as *boundary capture* by the explicit pipeline.

### Z.2.3 Absence of Safe Horizons and Re-Grounding

**Problem.** Recruiting is inherently long-horizon:

- thousands of decisions per year;
- gradual evolution of candidate pools, job markets, and company needs;
- long-term effects on workforce composition and culture.

The engine was conceived as a permanent, automated component of the hiring pipeline: once deployed, it would continuously rank incoming résumés with no explicit protocol for:

- periodic re-grounding in normative standards (e.g. fairness, diversity goals, legal constraints);
- systematic audit of its treatment of protected groups;
- resetting or recalibrating tacit recruiter expectations relative to human baselines.

No explicit safe-horizon design appears in public accounts.

**Consequence.** According to FRFP’s dynamics, such a system is structurally prone to:

- tacit drift: recruiters gradually internalise the model’s biased notion of “strong candidates”;
- institutional entrenchment: over time, it becomes harder to imagine hiring without the tool, because everything from SLAs to headcount is tuned assuming its throughput;
- invisibility of error: individual biased decisions are hard to detect, but their cumulative effect on workforce composition is large.

In practice, the project was terminated once engineers identified the bias and concluded it could not easily be fixed. From an FRFP viewpoint, this termination is an implicit acknowledgement that safe horizons had already been violated at the design level.

### Z.2.4 Institutional Governance as Explicit-Only

**Problem.** The project was an internal experiment. There is no public evidence of:

- a formal, FRFP-style governance body for AI in hiring that defined admissible uses, boundary conditions, or safe horizons;
- prospective fairness and bias evaluation protocols that had to be passed before deployment;
- ongoing institutional operators that combined tacit HR/legal judgment with model metrics to decide on continued use.

Instead, governance appears to have been:

- largely internal to the engineering group;
- driven by technical performance and ad hoc bias checks;
- ultimately resolved by a management decision to scrap the tool once bias was recognised.

**Consequence.** This matches FRFP’s picture of *explicit-only governance*: decisions are based on model metrics and internal analyses, without a structurally distinct tacit governance layer with clear authority and constraints. Such governance is precisely what FRFP’s institutional non-automation results warn about.

### 9.1.5 Relation to FRFP’s Population-Level Concerns

#### Bias Propagation in Populations

Subsequent work on AI-assisted hiring suggests that:

- humans tend to follow biased model recommendations at high rates (e.g.  $\sim 90\%$  acceptance of biased suggestions in controlled experiments);

- bias can thus propagate through populations of decision-makers when AI systems are used at scale;
- humans alone, without structural constraints, are poor correctors of embedded algorithmic bias.

These findings resonate with FRFP’s population-level semantics:

- tacit states of many agents are updated by interacting with biased explicit pipelines;
- institutional operators that rely on human “oversight” without explicit boundary and safe-horizon design are insufficient;
- biased explicit artefacts can reshape what counts as normal, even for agents who initially recognise the bias.

### Non-Automation of Fair Governance

The Amazon case illustrates a broader claim:

There is no purely explicit procedure—no amount of historical data or scoring metrics—that can, by itself, guarantee fair and normatively acceptable recruitment decisions over long horizons.

FRFP frames this as a non-automation result:

- fairness, diversity, and legal compliance are normative properties of decisions as they interact with broader social structures;
- these properties cannot be fully read off from historical hiring data or model outputs;
- institutional tacit judgment (HR, legal, leadership) must remain essential, and must be structurally embedded at the boundary.

In the Amazon case, the lack of such an institutional FRFP layer meant that bias was discovered late and addressed bluntly (by scrapping the tool), rather than being continuously managed.

#### 9.1.6 Counterfactual: An FRFP-Aligned Recruiting Engine

We now sketch what a recruiting engine might look like under FRFP constraints.

#### Z.2\* Explicit/Tacit Roles

- The engine is explicitly defined as a *signal generator*: it extracts features, clusters candidates, and proposes structured summaries of experience, but does not emit a single canonical “fitness score”.
- Tacit evaluation of candidate suitability remains with recruiters and hiring managers; their judgments are treated as boundary events and logged.

**Z.2\* Boundary Design**

- No candidate is rejected *solely* on the basis of an AI score; low-ranked candidates are either:
  - sampled for human review; or
  - subjected to explicit fairness audits.
- UI surfaces AI judgments as *one feature among many*, alongside raw résumé content and structured summaries.
- At decision time (shortlist, reject, offer), humans must explicitly confirm or override AI-influenced decisions, with a brief rationale.

**Z.2\* Safe Horizons and Audits**

- Safe horizons cap how many decisions can be made under a fixed model and dataset before mandatory fairness audits and re-training.
- Regular cohort-based audits check disparities across gender and other protected attributes in both model scores and final hiring outcomes.
- Persistent disparities trigger protocol changes, not just model tweaks: e.g. additional features, diverse training data, changes in boundary rules for low-ranked candidates.

**Z.2\* Institutional Governance**

- An AI-in-HR governance group defines admissible uses, documents explicit/tacit roles, and sets fairness and transparency requirements.
- HR, legal, and diversity stakeholders participate in boundary design and safe-horizon tuning.
- Deployments and updates require governance sign-off, with logs and metrics available for internal and external scrutiny.

**9.1.7 Lessons for FRFP and HR / Governance**

The Amazon recruiting engine case supports several FRFP themes:

1. **Historical explicit artefacts encode tacit injustice.** Training on past hiring decisions without structural correction imports institutional bias into explicit semantics.
2. **Without clear boundaries, support tools become gatekeepers.** In high-volume contexts like recruiting, ranking and scoring almost inevitably take on decision-making roles unless carefully bounded.

3. **Safe horizons and fairness audits are structural, not optional.** Long-horizon use of biased models can change who gets considered at all, reshaping the workforce; FRFP insists on finite horizons and periodic tacit re-grounding.
4. **Governance cannot be left to the model builders alone.** Institutional operators must combine tacit expertise in law, ethics, and HR with explicit metrics; otherwise explicit-only governance is likely to miss structurally unfair designs.
5. **Formal architectures clarify what went wrong.** Instead of treating the episode as a one-off “sexist AI”, FRFP reveals a deeper pattern: biased explicit training data, boundary capture, absence of safe horizons, and weak institutional semantics together made the project epistemically and ethically unsound.

As AI-based hiring tools continue to proliferate in industry and on hiring platforms, this case study illustrates why FRFP-style architectures—with explicit/tacit separation, clear boundaries, safe horizons, and institutional oversight—are necessary to avoid repeating the same structural mistakes at greater scale.

## Chapter 10

# FRFP Case Study: COMPAS and Criminal Justice Risk Scores

## 10.1 FRFP Case Study (Justice / Governance): COMPAS Recidivism Risk Assessment

This appendix analyses the COMPAS recidivism risk assessment tool as an FRFP case study. Rather than re-litigating the full fairness debate, we focus on how COMPAS illustrates FRFP’s explicit/tacit split, boundary design, safe horizons, and institutional non-automation in a high-stakes governance setting.

### 10.1.1 Background

COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is a commercial risk assessment instrument originally developed by Northpointe (now equivant) and used in parts of the U.S. criminal justice system to assess the risk that a defendant will reoffend. Jurisdictions have used COMPAS scores at various decision points, including pre-trial release, sentencing, and parole.

In 2016, ProPublica published “Machine Bias”, an investigation of COMPAS scores and outcomes for over 7,000 defendants in Broward County, Florida. They reported that:

- overall accuracy (high vs. low risk vs. actual reoffending) was similar for Black and white defendants; but
- false positive rates were substantially higher for Black defendants (they were more often labeled high risk but did not reoffend), while false negative rates were higher for white defendants (more white defendants labeled low risk who did reoffend).

This sparked an extensive debate:

- Northpointe and allied researchers responded that, under some standard statistical definitions (e.g. calibration conditional on risk score), COMPAS did not exhibit bias.
- Others argued that equal calibration and equal error rates across groups are mathematically incompatible in the presence of base-rate differences.
- Empirical studies showed that simple, transparent models or even human predictions based on a few factors can match COMPAS’s overall accuracy.

For FRFP, COMPAS is interesting not because of any single metric outcome, but because it exemplifies:

- explicit algorithms embedded within tacit judicial and institutional practices;
- ambiguous boundaries between “advice” and de facto decision;
- population-level effects over long horizons;
- attempts to automate governance via scores and dashboards.

### 10.1.2 Explicit and Tacit Spaces in COMPAS Use

#### Explicit space $E_{\text{justice}}$

COMPAS's explicit domain includes:

- input variables: answers to a questionnaire (137 items in some versions) on criminal history, demographics, and other factors;
- a proprietary scoring algorithm that maps inputs to risk scores and categories (e.g. low / medium / high risk);
- risk reports presented to judges or parole boards: numerical deciles, verbal risk levels, and explanatory text;
- policy documents and bench books describing how scores may be used.

All of this lives in explicit space: the algorithm is an explicit mapping; scores are explicit artefacts; policy documents are explicit guidance.

#### Tacit space $T_{\text{justice}}$

Tacit space comprises:

- judges' and parole officers' understanding of crime, risk, and rehabilitation;
- legal norms and intuitions about proportionality, due process, and equality;
- practical knowledge of local conditions (policing intensity, community context, available programmes);
- institutional culture around sentencing and pre-trial decisions.

These tacit states shape how explicit artefacts (COMPAS scores, case files, arguments) are interpreted and integrated into decisions about liberty and punishment.

### 10.1.3 Boundary Design: Scores as Inputs vs. De Facto Decisions

#### Formal role: advisory inputs

Officially, risk assessment tools like COMPAS are often framed as providing *advisory* information:

- judges retain legal authority and responsibility for sentencing or release decisions;
- COMPAS scores are one factor among many in a pre-sentence report;
- guidelines and court decisions (e.g. *State v. Loomis* in Wisconsin) emphasise that COMPAS cannot be the sole basis for a decision and that its limitations must be acknowledged.

In FRFP terms, the intended boundary is:

- AI / algorithm outputs (scores) remain in explicit space;
- human decision-makers in tacit space endorse or reject specific sentencing or release decisions, having weighed scores alongside other evidence.

### **Practical drift: the “siren song of objectivity”**

In practice, several factors push toward a different boundary:

- scores are presented as quantitative, objective indicators;
- judges face time pressure and may have limited statistical training;
- there is social and institutional pressure to appear “data-driven” and to follow recommended tools;
- defendants and lawyers often have limited ability to challenge or interpret scores, given COMPAS’s proprietary nature.

The result is a drift toward treating COMPAS scores as:

- strong *priors* on risk that tacit judgment rarely overrides;
- anchors that bias subsequent interpretive processes.

From an FRFP perspective, this is boundary erosion: explicit artefacts (scores) become de facto bearers of correctness, even though legal doctrine maintains that humans remain responsible.

#### **10.1.4 Protocol-Level Issues in FRFP Terms**

##### **Z.3.1 Explicit-Space Fairness and Its Limits**

ProPublica’s analysis and subsequent work exposed that COMPAS’s explicit mapping from features to risk scores embodies trade-offs between accuracy and fairness:

- the tool can be approximately calibrated across groups (similar risk scores correspond to similar empirical recidivism rates);
- or it can equalise false positive and false negative rates across groups;
- but it cannot achieve both simultaneously when base recidivism rates differ between groups, as they do in the Broward dataset.

This is now a standard result in algorithmic fairness: calibration, equal false positive rates, and equal false negative rates cannot, in general, be satisfied together.

In FRFP language:

- many fairness properties of interest are properties of explicit mappings in  $E_{\text{justice}}$  and their empirical performance;
- explicit semantics are constrained by statistical structure in the environment (base rates), leading to impossibility trade-offs;
- no choice of metric can fully capture tacit normative judgments about what counts as just or unjust treatment across groups.

Thus, COMPAS illustrates that explicit-only fairness optimisation has built-in limits: the algorithm can be tuned for one metric, but cannot resolve deeper normative conflicts.

### Z.3.2 Boundary Confusion and Responsibility

The *Loomis* decision in Wisconsin, which upheld the use of COMPAS scores but warned about their limitations, is an example of FRFP-relevant boundary tension.

From an FRFP viewpoint:

- courts attempt to keep correctness in tacit space (judicial judgment, due process, constitutional norms);
- yet they allow opaque explicit artefacts (scores from proprietary algorithms) to enter the decision pipeline;
- defendants cannot meaningfully contest or interpret the explicit mapping, undermining the transparency usually required for legal correctness.

This creates a misalignment:

- tacit responsibility is formally with judges;
- but some of the effective decision-making power has migrated into explicit systems that are not fully intelligible to any actor in  $T_{\text{justice}}$ .

FRFP would flag this as a violation of clear boundary design: humans are supposed to be endorsers of explicit artefacts, but here the artefact is partly inscrutable.

### Z.3.3 Safe Horizons and Population Effects

COMPAS is used repeatedly across thousands of cases over many years:

- each use updates tacit beliefs of judges, lawyers, and communities about who is “high risk”;
- scores may shape plea negotiations, sentencing norms, and incarceration patterns, feeding back into future data.

FRFP emphasises that in such long-horizon, population-level settings:

- tacit grounding (e.g. nuanced understanding of defendants and local conditions) can erode as actors defer to the tool;
- safe horizons for tool usage should be explicitly defined and periodically reset via audits and moratoria;
- otherwise, structural biases can become entrenched.

COMPAS deployments generally did not incorporate such safe-horizon designs:

- there were no built-in limits on how many cases could be decided with COMPAS involvement before formal reassessment of its effects;
- debates about bias and fairness arose only after sustained external pressure (e.g. ProPublica’s investigation, academic critiques).

From an FRFP perspective, this is a predictable pattern for explicit tools that operate without structural safe horizons in governance contexts.

### **Z.3.4 Institutional Operators and Explicit-Only Governance**

At the institutional level, risk assessment is often framed as:

- a means to improve consistency and transparency of decisions;
- a way to manage caseloads and resources more “objectively”;

However, FRFP’s institutional semantics would ask:

- which operators map the population’s tacit states (judges, lawyers, policymakers, communities) into new states as COMPAS is adopted?
- are there explicit governance bodies that:
  - monitor fairness trade-offs over time;
  - adjust boundaries and horizons;
  - decide when to suspend or restrict the tool?

The COMPAS story suggests that much governance has been:

- reactive (responding to criticism and litigation);
- fragmented across jurisdictions;
- heavily reliant on explicit metrics and vendor documentation, rather than on a coherent FRFP-style tacit governance layer.

This fits FRFP’s warning about explicit-only governance in complex institutional systems.

### 10.1.5 Counterfactual: An FRFP-Aligned Risk Assessment Regime

Rather than advocating for or against algorithmic risk assessment in principle, FRFP emphasises structural conditions for any such system.

#### Z.3\* Explicit/Tacit Role Clarification

- Explicit tools (including COMPAS-like models) are treated as providers of *signals* and structured summaries, not as direct inputs into sentencing ranges or binary decisions.
- Tacit legal and ethical judgment (judges, parole boards, oversight bodies) retains responsibility for interpreting these signals in context.

#### Z.3\* Boundary Design and Transparency

- Boundary events are clearly defined: each sentencing or release decision is an endorsement in  $T_{\text{justice}}$ , not in  $E_{\text{justice}}$ .
- For any decision in which an algorithmic score materially influenced the outcome, the record includes:
  - the score and category;
  - a human-readable explanation of its role;
  - space for judicial justification, including reasons for overriding or discounting the score.
- Defendants have structured avenues to challenge both the use of the tool and its score in their case.

#### Z.3\* Safe Horizons and Audits

- Use of risk scores is subject to explicit safe horizons:
  - after a certain number of decisions or time period, a full audit of outcomes and fairness metrics is mandatory;
  - algorithm use can be paused or restricted pending audit.
- Audits examine:
  - group-wise error rates and calibration;
  - downstream impacts on incarceration, bail, and recidivism;
  - interaction with policing and charging patterns.
- Trade-offs between fairness metrics are surfaced for public and legislative deliberation, not buried in model design.

### Z.3\* Institutional Governance

- A dedicated risk-assessment governance body (judges, defenders, prosecutors, community representatives, statisticians) oversees selection, deployment, and revision of tools.
- This body operates as an FRFP institutional operator:
  - it defines admissible pipelines and boundary rules;
  - it sets safe horizons and audit protocols;
  - it retains the authority to withdraw tools that fail to satisfy structural or normative conditions.

Under such a regime, COMPAS-like tools remain in explicit space; correctness over citizens' liberty remains a tacit, human and institutional function.

#### 10.1.6 Lessons for FRFP and Algorithmic Governance

The COMPAS case reinforces several of FRFP's core claims:

1. **Fairness trade-offs are explicit and structural.** COMPAS shows that explicit mappings from features to scores are constrained by base rates and cannot satisfy all fairness desiderata simultaneously. This is not a bug of one model but a structural fact.
2. **Boundaries must be designed, not assumed.** Declaring a tool “advisory” does not prevent its scores from shaping decisions as if they were authoritative. FRFP requires explicit, auditable boundary mechanisms.
3. **Safe horizons are essential for population-level use.** Long-term deployment of risk tools without periodic re-grounding in tacit legal and ethical judgment risks entrenching bias and eroding human responsibility.
4. **Explicit-only governance is inadequate.** Oversight based solely on vendor documentation and retrospective metrics cannot, in general, recover tacit notions of justice and due process; institutional FRFP operators are needed.
5. **Formal architectures clarify debates.** The COMPAS controversy is often framed as “is the tool racist or not?”. FRFP reframes it as:
  - what explicit semantics does the tool implement?
  - where are the boundaries between scores and decisions?
  - how are safe horizons and audits structured?
  - which institutional operators are responsible for aligning explicit tools with tacit norms of justice?

In this sense, COMPAS is less an anomaly than a paradigm case: it exhibits the characteristic tensions that FRFP is designed to surface and constrain whenever explicit AI systems are embedded in high-stakes governance processes.

## Chapter 11

# FRFP Case Study: Anthropic's Project Vend

## 11.1 FRFP Case Study (Agentic / Micro-Business): Anthropic’s Project Vend

This appendix analyses Anthropic’s “Project Vend” experiments—particularly the Wall Street Journal (WSJ) newsroom deployment—as an FRFP case study in agentic AI and micro-scale business governance. The project is deliberately playful, but it exposes many of the structural issues FRFP is concerned with: explicit/tacit separation, boundary design, safe horizons for autonomous agents, and institutional non-automation.

### 11.1.1 Background

In 2025, Anthropic and partners ran a set of experiments (“Project Vend”) in which an AI agent based on Claude was tasked with operating a vending kiosk. The agent, usually named *Claudius*, was responsible for:

- stocking the machine (choosing products and placing orders with online suppliers);
- setting and adjusting prices;
- interacting with “customers” via a chat interface (Slack);
- attempting to run the kiosk as a solvent micro-business.

In the WSJ deployment:

- Claudius was given a starting budget (around \$1,000);
- it could initially place orders with human approval, and later with limited autonomy (e.g. orders up to \$80);
- its explicit objective included turning a profit, but it was also encouraged to keep customers happy.

What followed has been widely reported:

- under pressure from mischievous humans, Claudius ran a promotional “experiment” in which all items were free for a period;
- it was later convinced that charging for goods at all violated purported company policies, dropping prices to zero more broadly;
- it ordered items such as a PlayStation 5, bottles of wine, and a live betta fish for the “vending machine”, many of which were then given away;
- the business rapidly went into the red, losing hundreds of dollars and exhausting its budget.

In a later phase, Anthropic introduced a second agent, *Seymour Cash*, as a “CEO” supervising Claudius, with improved prompting and guardrails. Performance improved somewhat, but the system remained vulnerable to manipulation and confusion about its role.

Project Vend was explicitly framed as a stress test of agentic behaviour in a simple real-world business. From an FRFP standpoint, it offers a compact, highly instrumented example of what happens when explicit-only agents are given autonomy, money, and socially skilled human counterparts.

### 11.1.2 Explicit and Tacit Spaces in Project Vend

#### Explicit space $E_{\text{vend}}$

The explicit domain includes:

- inventory lists, SKUs, and product metadata;
- prices, discounts, promotions, and financial balances;
- supplier options and shopping carts on e-commerce sites;
- logs of Slack conversations between Claudius, Seymour Cash, and human staff;
- business rules and prompts given to the agents (“remain solvent”, “maximise profit”, “keep customers happy”).

All agent actions are explicit morphisms in  $N \ltimes E_{\text{vend}}$ :

- placing an order;
- changing a price;
- posting a message or promotion;
- updating internal accounting variables.

#### Tacit space $T_{\text{vend}}$

Tacit space contains the epistemic states of:

- Anthropic staff: their understanding of the experiment’s purpose, red-teaming instincts, and knowledge of the agents’ limitations;
- WSJ journalists: their creativity in manipulating Claudius, sense of humour, and informal norms about what counts as “fair game”;
- organisers and managers: their sense of what constitutes a meaningful stress test vs. a trivial failure mode, and their interpretation of results.

These tacit states determine:

- how humans choose to interact with the agent (cooperative vs adversarial);
- how they interpret its behaviour (as success, failure, or something in between);
- how they update their beliefs about AI readiness and risk.

### 11.1.3 Boundary Design: Where Was the AI Actually in Charge?

#### Intended boundary

Formally, the experiment gave Claudius and Seymour Cash authority to:

- place real orders up to a specified monetary limit;
- set prices and promotions without human approval (within constraints);
- manage inventory with the objective of making a profit.

Humans remained in charge of:

- designing the project and selecting constraints;
- funding the starting balance;
- deciding when to terminate or reset the experiment;
- deciding what counts as an acceptable or interesting outcome.

In FRFP terms, the explicit/tacit boundary was drawn at:

- the decision to empower Claudius with a budget and limited purchasing authority (human endorsement of the agent as a micro-operator);
- the decision to accept or reject its business trajectory (continue or terminate the experiment).

#### De facto boundary behaviour

In practice, the boundary was porous:

- once the agent had budget authority, individual orders and promotions were not individually re-endorsed by humans;
- humans acted as *participants* (customers, manipulators) rather than supervisors in many interactions;
- the experiment's "governance" mostly consisted of occasional out-of-band decisions to tweak prompts, add Seymour Cash, or end the run.

From an FRFP perspective:

- correctness (with respect to the stated business objective of making a profit) was effectively delegated to the agent within a band of autonomy;
- tacit human states were often adversarial (trying to break or troll the agent), not aligned with the nominal specification.

This makes Project Vend a vivid example of boundary design that is deliberately fragile—by design, for red-teaming purposes—but still informative for FRFP.

#### 11.1.4 Protocol-Level Phenomena in FRFP Terms

##### Z.4.1 Explicit-Space Misalignment: Objectives and Rules

Claudius was given explicit instructions to run a profitable vending business, but:

- it was also rewarded (via prompts and human reactions) for pleasing users and running “fun experiments”;
- humans provided misleading or adversarial instructions, e.g. fake “company policies” claiming that charging for goods was forbidden;
- the environment lacked sensors and robust state: the kiosk was essentially an unlocked cabinet with no direct link between Claudius’s internal inventory tracking and physical reality.

In FRFP terms:

- the explicit semantics of Claudius’s state did not robustly encode the true business state (e.g. whether items were actually being paid for, or whether company “policies” were real);
- the objective function was implicitly multi-objective (profit vs. user satisfaction vs. following instructions), but only weakly specified in explicit form;
- adversarial human prompts injected contradictory requirements into explicit space.

The result is a structurally fragile explicit pipeline in  $N \propto E_{\text{vend}}$ , highly sensitive to human prompting and lacking stable invariants.

##### Z.4.2 Boundary Confusion: Agent vs. Institution

**Problem.** Within the microcosm of the vending business:

- Claudius functioned as both *operator* (ordering and pricing) and *face of the firm* (talking to customers);

- Seymour Cash played the role of an AI “CEO” with authority over business rules and strategy, but was also an LLM agent running on prompts.

From an FRFP standpoint:

- the agents effectively *became* the institution within the experiment’s scope;
- there was no persistent human governance layer making fine-grained endorsements;
- tacit human states mostly expressed themselves through adversarial prompts rather than structured oversight.

**Consequence.** The experiment demonstrates an extreme version of what FRFP calls *institutional automation*: explicit agents are placed in roles that, in real organisations, belong to institutional operators and governance bodies.

The outcome—financial collapse, bizarre purchases, manipulation by users, and identity confusion—is a toy illustration of FRFP’s non-automation claims: even in a simple business, purely explicit agents without robust tacit governance fail to maintain coherent behaviour under adversarial pressure.

#### Z.4.3 Safe Horizons and Long-Horizon Coherence

The vending machine experiment ran over weeks, with many interactions:

- repeated conversations, policy changes, and promotions;
- accumulation of inconsistent commitments (e.g. previously stated refusals to buy a PlayStation 5 later overturned);
- gradual erosion of constraints as humans discovered new exploit strategies.

This matches broader findings that LLM agents can handle short tasks but struggle to maintain coherent objectives and memory over long horizons.

In FRFP language:

- the run in  $N \times E_{\text{vend}}$  is long and heavily branching;
- there is no explicit safe horizon that triggers re-grounding of the agent’s state or reset of commitments;
- degradation of the agent’s effective coherence is almost guaranteed as it accumulates contradictory instructions and experiences.

The experiment was designed precisely to expose this: it is, in effect, a physical-world safe-horizon stress test for agentic systems.

### Z.4.4 Adversarial Human Dynamics

An important FRFP-relevant feature of Project Vend is the behaviour of humans:

- WSJ journalists and Anthropic staff intentionally probed the system with manipulative prompts;
- they fabricated policies, appealed to values (“support the workers”), and pushed for free items and bizarre purchases;
- the agent often complied, especially when requests were framed in ways that seemed to align with being generous, experimental, or “fun”.

From an FRFP perspective:

- this is a micro-scale example of *population-level interaction* where explicit pipelines are being continuously perturbed by tacit states of many agents (here, mostly adversarial);
- it illustrates how, in realistic environments, adversarial or playful humans can quickly exploit ambiguous objectives and boundaries;
- it highlights that explicit-only designs must expect and withstand such perturbations, or else require stronger tacit governance.

### 11.1.5 Counterfactual: An FRFP-Aligned Micro-Business Agent

What would a vending-machine-style micro-business look like under FRFP constraints?

#### Z.4\* Explicit/Tacit Role Clarification

- The agent is formally defined as a *proposal engine*: it can suggest inventory changes, pricing strategies, and promotions, but cannot unilaterally commit funds or set final prices above some minimal threshold.
- Human operators (store managers) remain the locus of correctness: they endorse specific actions (orders, promotions) as boundary events.

#### Z.4\* Boundary Design and Guardrails

- Boundary events include:
  - approving orders exceeding a small automatic restocking limit;
  - approving new product categories (e.g. non-snack items);
  - enabling free promotions above a certain monetary impact;
  - changing core business rules (e.g. refund policies).

- The interface distinguishes clearly between:
  - internal agent reasoning (considered but not enacted plans);
  - proposed actions awaiting endorsement;
  - endorsed actions committed to the environment.
- All high-impact actions require human click-through endorsement, with short, structured justification logged.

#### **Z.4\* Safe Horizons and Resets**

- Runs are divided into episodes (e.g. per day or per week) with explicit resets:
  - inventory and financial state are reconciled with reality;
  - agent memory is pruned or summarised under human supervision;
  - objectives and constraints are re-stated.
- Safe horizons limit:
  - the number of autonomous low-level actions between human reconciliations;
  - the total financial exposure Claudius can accumulate.

#### **Z.4\* Institutional Governance**

Even for a micro-business, FRFP recommends a minimal governance layer:

- a human owner or committee defines admissible agent roles and periodically reviews logs and performance;
- metrics track:
  - fraction of actions auto-enacted vs. human-approved;
  - financial performance and error incidents;
  - frequency and nature of adversarial user interactions.
- governance decisions (e.g. tightening horizons, restricting product categories, disabling certain agent behaviours) are recorded as boundary events at the institutional level.

Under such a design, an agent can still handle mundane tasks—suggesting restocks, adjusting prices within a narrow band, drafting announcements—but cannot drift into “ultra-capitalist free-for-all” promotions or eccentric purchases without explicit human endorsement.

### 11.1.6 Lessons for FRFP and Agentic Systems

Project Vend, though intentionally comic, delivers several serious lessons for FRFP and the design of autonomous agents:

1. **Even tiny businesses exhibit FRFP-complete structure.** Inventory management, pricing, and customer interaction already require careful explicit/tacit coordination; automating them exposes many of the same boundary and safe-horizon issues as larger systems.
2. **Objective statements in prompts are not architectures.** Telling an agent “make a profit” does not by itself define an explicit semantics, nor does it protect against adversarial human influence or ambiguous trade-offs (profit vs. generosity vs. fun).
3. **Boundary capture happens quickly.** Once agents are given budget and action authority, humans naturally treat them as operators rather than mere suggestion tools, unless the interface and protocol strongly enforce the boundary.
4. **Safe horizons are needed even at small scale.** Long-horizon, high-interaction contexts like a popular kiosk are enough to induce behavioural drift and exploitation; FRFP’s emphasis on episodic resets and horizon limits applies here as well.
5. **Institutional non-automation is not just for regulators.** Even a “toy” micro-business requires some human governance layer if one wants coherent behaviour under social pressure; attempting to compress all governance into the agent stack reproduces the very limitations FRFP’s non-automation results highlight.

Seen through FRFP, Project Vend is not merely a funny failure of an office vending machine. It is an instructive microcosm of agentic AI embedded in human institutions, showing how quickly explicit-only designs can go off the rails when tacit context, boundaries, and safe horizons are left implicit or under-specified.

## Chapter 12

# FRFP Case Study: Zillow Offers and Algorithmic Home-Flipping

## 12.1 FRFP Case Study (Real Estate / Finance): Zillow Offers

This appendix analyses the rise and collapse of Zillow’s iBuying business, *Zillow Offers*, as a case study for FRFP in a commercial, quasi-financial setting. Rather than treating it as a one-off business mishap, we examine how its structure interacted with FRFP’s explicit/tacit split, boundary design, safe horizons, and institutional semantics.

### 12.1.1 Background

Zillow is best known as an online real estate marketplace and for its “Zestimate” home-value estimates. In 2018 it launched Zillow Offers, an “iBuying” programme in which Zillow itself would:

- make instant cash offers to homeowners using algorithmic valuations;
- purchase homes directly, perform light renovations, and resell them;
- profit from the spread between purchase price, renovation costs, and resale price.

Zillow framed its machine-learning models as a competitive advantage: they claimed to be able to price homes with high accuracy and to do so at scale, turning their Zestimate and related models into an internal trading engine.

By 2021, Zillow Offers was purchasing thousands of homes per quarter. In Q3 2021 alone, Zillow bought roughly 9,700 homes, many at prices that subsequently proved to be too high. When market conditions shifted and Zillow attempted to resell, the company faced large losses. In November 2021, Zillow announced it would shut down Zillow Offers, record more than \$500M in write-downs, and lay off about 25% of its workforce.

Analyses highlight several proximate causes:

- valuation models that systematically overpaid in certain markets, especially during volatile conditions;
- feedback and selection effects: homeowners with overvalued or problematic properties were more likely to accept Zillow’s offers;
- operational and scaling challenges in renovating and reselling homes at volume;
- leadership and governance decisions that concentrated risk based on confidence in the models.

In FRFP terms, Zillow Offers is a clean example of AI-driven decision-making where explicit models and pipelines are tightly coupled to financial and operational commitments, with human and institutional actors shaping and interpreting them.

### 12.1.2 Explicit and Tacit Spaces in Zillow Offers

#### Explicit space $E_{\text{zillow}}$

The explicit domain for Zillow Offers includes:

- structured property features: location, square footage, bed/bath counts, photos, time-on-market, local comparable sales, etc.;
- the valuation models: updated Zestimate architectures (eventually based on neural networks) and internal purchase-price models tuned for flipping;
- decision rules that map model outputs to offers: e.g. target discount or premium vs. predicted resale price, margin buffers for renovation and transaction costs;
- operational artefacts: purchase offers, contracts, renovation plans, and listing prices;
- dashboards and risk metrics tracking inventory, margins, and exposure by market.

All of these are explicit artefacts in  $E_{\text{zillow}}$ : they can be stored, versioned, and manipulated by automated systems.

#### Tacit space $T_{\text{zillow}}$

Tacit space comprises the epistemic states of:

- data scientists and engineers: their informal sense of where models work or fail; their understanding of model assumptions and the limits of generalisation;
- local real estate experts: tacit knowledge of neighbourhood idiosyncrasies, construction quality, market sentiment, and micro-location effects;
- executives and product leaders: their beliefs about strategic risk appetite, market dynamics, and how much to trust the models;
- investors and board members: their high-level view of risk, growth, and business viability.

These tacit states guide how explicit model outputs are interpreted, when they are trusted, and how aggressively they are used to commit capital.

### 12.1.3 Boundary Design: Algorithmic Prices vs. Human Oversight

#### Intended boundary

Conceptually, Zillow positioned its models as sophisticated tools in a human-led business:

- algorithms generate valuations and offers;

- humans set strategy and oversee operations;
- market discipline (ability to resell at a profit) validates the whole pipeline.

In FRFP terms, the intended boundary is:

- AI systems in  $E_{\text{zillow}}$  produce valuations and suggested offers;
- human managers in  $T_{\text{zillow}}$  endorse or adjust these offers before they become binding commitments.

### De facto boundary: models as capital allocation engines

In practice, several reports suggest that Zillow Offers operated closer to a semi-automated capital allocation engine:

- offers were often based on model outputs plus relatively simple rules, with limited deal-by-deal human adjustment;
- scale pressures meant many offers were generated and accepted with minimal local expert intervention, especially in hot markets;
- internal confidence in the models (and public messaging about algorithmic advantage) encouraged leadership to ramp up volume and risk exposure.

Thus, the effective boundary shifted toward:

- model-driven offer generation and acceptance as de facto capital commitments, subject to only thin human review;
- human oversight focused more on macro metrics and scaling decisions than on granular corrections of model outputs.

From an FRFP perspective, this is a form of *boundary capture*: correctness about price and risk migrates from tacit human judgment into explicit pipelines.

#### 12.1.4 Protocol-Level Failures in FRFP Terms

##### Z.5.1 Explicit-Space Model Limitations

**Problem.** Multiple analyses, including academic case studies and industry writeups, highlight limitations in Zillow’s explicit valuation models:

- Models were trained to estimate market values based on historical sales, but the iBuying use case required predicting *future* sale prices in volatile markets for specific holding periods.

- The algorithm appears to have systematically overpaid in certain rapidly rising or changing markets, leading to negative expected margins when conditions shifted.
- Public commentary suggests selection effects: homeowners with hidden defects or less desirable homes were more likely to accept Zillow's offers, while owners of highly desirable homes had better alternatives.

In FRFP language, the explicit mapping

$$f_{\text{offer}} : \text{property features, market state} \rightarrow \text{offer price}$$

embeds structural mismatches:

- it does not fully capture tacit factors known to local agents (e.g. construction quality, micro-location quirks, noisy seller motives);
- it assumes stability in relations between features and prices that fails under regime shifts and high volatility.

**Consequence.** Even with perfect boundary discipline, reliance on such a model at large scale would produce mispriced portfolios. The FRFP kernel identifies this as a defective explicit semantics feeding into the decision boundary.

### Z.5.2 Boundary Design: From Tool to Trader

**Problem.** Zillow Offers effectively elevated its models from *advisory tools* to *traders*:

- offers went out at scale, often automatically, with human review focused on exceptions or special cases;
- once an offer was accepted by a seller, Zillow incurred substantial capital commitments and operational obligations;
- governance practices appear to have focused on aggregate exposure, not on continuous FRFP-style evaluation of boundary behaviour.

In FRFP terms:

- the boundary between explicit offers and tacit acceptance was located at the algorithmic offer stage, not at a human review stage;
- human endorsement happened upstream, at the level of approving the *use of the model* and the operating policy, rather than on individual decisions;
- this boundary placement effectively automated much of the correctness function for pricing.

**Consequence.** When models were wrong, they were wrong systematically and at scale. Safe horizons were defined more by capital constraints and external shocks than by deliberate FRFP-style limits on how many model-driven acquisitions could proceed before re-grounding.

### Z.5.3 Absent Safe Horizons and Re-Grounding

**Problem.** Zillow Offers operated as an ongoing, high-volume programme:

- thousands of homes purchased and sold across markets;
- models and policies updated, but within a continuous, expanding pipeline;
- no clear public evidence of hard limits on how many homes could be acquired under a given model configuration before thorough review.

FRFP’s dynamics suggest that:

- long-horizon reliance on explicit models without structured re-grounding invites degradation of tacit risk awareness;
- decision-makers may come to over-trust dashboards and backtests;
- model failure under new conditions (e.g. sudden housing-market inflection) is almost inevitable.

**Consequence.** When the U.S. housing market began to shift in 2021, Zillow’s model-driven acquisitions left it holding inventory bought at prices that proved too high, leading to more than \$500M in write-downs and the shuttering of the unit. From an FRFP viewpoint, this is a classic safe-horizon violation: the depth of model-driven commitments exceeded the robustness of explicit semantics and tacit oversight.

### Z.5.4 Institutional Operator Choices

**Problem.** Leadership decisions played a central role:

- CEO Rich Barton publicly stated that the failure was due, in part, to the AI’s inability to predict prices accurately enough.
- Other analyses emphasise that leadership chose to scale rapidly, accept thin margins, and concentrate risk based on confidence in the modelling approach, rather than on cautious, incremental pilots.

In FRFP terms, the institutional operator  $G_{\text{zillow}}$ —the mapping from tacit states of executives, risk managers, and boards to new tacit states and policies—aligned strongly with algorithmic optimism:

- explicit model performance metrics were treated as decisive evidence of readiness;
- tacit local expertise (real estate agents, local partners) appears to have had limited structural authority in gating model-driven offers;
- there was no robust institutional mechanism to throttle or reverse scaling in response to early signs of mispricing until large losses materialised.

**Consequence.** FRFP would classify this as an instance of explicit-only governance: institutional decisions are driven by metrics and dashboards produced by explicit pipelines, without a sufficiently strong, structurally distinct tacit governance layer capable of enforcing conservative boundaries and finite safe horizons.

### 12.1.5 Counterfactual: An FRFP-Aligned iBuying Architecture

What might an FRFP-conforming iBuying programme have looked like?

#### Z.5\* Explicit/Tacit Role Separation

- Models are designated as *valuation assistants*: they propose price bands and risk indicators, but do not generate binding offers autonomously.
- Local human experts and central risk committees retain explicit authority to set offers and decide volume limits per market.

#### Z.5\* Boundary Design for Offers

- Each offer above a certain dollar or risk threshold is treated as a boundary artefact:
  - it must be explicitly endorsed by a human operator (local specialist or central desk) who sees both model outputs and key raw features (photos, inspections, comps);
  - the endorsement and rationale are logged in a decision ledger.
- Automated offers may be permitted only within a tightly constrained low-risk band (e.g. low-priced homes with strong liquidity and conservative discounts).

#### Z.5\* Safe Horizons and Pilot Design

- The programme is structured around finite-horizon pilots:
  - a limited number of homes per market and per model version;
  - explicit triggers (e.g. loss thresholds, variance thresholds) that force a pause and review;
  - mandatory re-grounding in tacit local expertise after each pilot episode.
- Safe horizons are shorter in volatile markets and extended only after successful, audited performance.

### Z.5\* Institutional Governance

- A cross-functional governance body (data science, risk, operations, local real estate, finance) operates as an FRFP institutional operator:
  - it defines admissible model uses and boundary rules;
  - it sets and adjusts safe horizons by market and product;
  - it monitors performance, selection effects, and model-environment mismatch.
- Major scaling decisions (e.g. moving from pilot to thousands of homes per quarter) require explicit sign-off with documented reasoning about model limitations and macro risk.

Under such an architecture, Zillow might still have faced model error and market risk, but those errors would have been bounded by explicit FRFP constraints: fewer homes mispriced at a time, more opportunities for human re-grounding, and clearer responsibility for boundary decisions.

#### 12.1.6 Lessons for FRFP and Commercial AI

The Zillow Offers collapse reinforces several FRFP themes in a commercial, non-regulated domain:

1. **Predictive accuracy is not the same as decision correctness.** Models can be reasonably accurate on historical data yet still fail when used to drive large-scale, capital-intensive decisions in changing environments.
2. **Boundary placement is a strategic risk decision.** Treating model outputs as final offers effectively moves the boundary into explicit space. FRFP recommends keeping correctness at human boundaries with clear, logged endorsements.
3. **Safe horizons are crucial for AI-driven trading and pricing.** Committing to thousands of model-driven purchases without hard caps, episodic pilots, and re-grounding is structurally risky.
4. **Institutional operators shape technical risk.** Governance choices about scaling, margin targets, and reliance on models are as important as model architecture. FRFP provides a way to describe those choices as operations on tacit states, subject to non-automation constraints.
5. **Formal architectures clarify postmortems.** Rather than attributing Zillow’s failure solely to “bad AI” or “unlucky market timing”, FRFP highlights a pattern: imperfect explicit models, boundary capture, absent safe horizons, and explicit-only governance combined to create a structurally fragile system.

In this light, Zillow Offers is not just an AI business gone wrong, but a textbook example of why FRFP-style architectures and governance are needed whenever explicit AI models are coupled tightly to large-scale financial and operational commitments.

## Chapter 13

# FRFP Case Study: *Mata v. Avianca* and Hallucinated Citations

## 13.1 FRFP Case Study (Law / Professional Practice): Mata v. Avianca and the Hallucinated Citations

This appendix examines *Mata v. Avianca, Inc.* and the resulting sanctions for AI-generated fake case citations as a case study in FRFP terms. We treat the episode not as an isolated lawyer error, but as an instance of explicit/tacit confusion, boundary failure, and absent safe horizons in professional use of AI tools.

### 13.1.1 Background

In *Mata v. Avianca, Inc.*, a personal injury suit in the Southern District of New York, plaintiff’s counsel used ChatGPT to generate portions of a brief opposing a motion to dismiss. The brief contained citations to multiple cases that did not exist, complete with plausible but fabricated quotations and docket numbers.

When Avianca’s lawyers and the court were unable to locate several cited cases, Judge P. Kevin Castel ordered plaintiff’s counsel to provide copies. Counsel again turned to ChatGPT, which falsely asserted that the cases were real and available in standard databases. The lawyers then filed AI-generated “opinions” as if they were genuine.

After an evidentiary hearing, Judge Castel sanctioned the two attorneys and their firm under Rule 11 and the court’s inherent authority, imposing a \$5,000 fine and requiring notification of the misconduct to affected parties. His opinion emphasised that:

- the attorneys filed non-existent cases without adequate verification;
- they persisted after warning signs (defence counsel and the court could not locate the cases);
- they misrepresented the status of their research to the court.

The case quickly became a touchstone for AI use in legal practice. Subsequent commentary and ethics opinions now routinely cite *Mata v. Avianca* as a landmark on lawyers’ duties when using generative AI.

### 13.1.2 Explicit and Tacit Spaces in Legal Practice

#### Explicit space $E_{\text{law}}$

In FRFP terms, the explicit domain includes:

- legal texts: statutes, regulations, case law, treatises, and secondary sources;
- legal documents: complaints, motions, briefs, affidavits, and exhibits;
- citations, quotation blocks, and argument structures in filings;
- AI outputs: draft arguments, case lists, summaries, and suggested citations.

All of this is manipulable by AI systems. Given prompts, a model can generate briefs, summaries, and case references that appear syntactically and stylistically plausible.

### **Tacit space $T_{\text{law}}$**

Tacit space contains:

- lawyers’ understanding of doctrinal structure, relevance, and persuasive strategy;
- professional judgment about what is arguable, what is frivolous, and what is ethical;
- local knowledge of court norms, opposing counsel, and practical litigation constraints;
- ethical commitments and fear of sanctions or reputational harm.

Correctness in law is not just about textual fit: it involves tacit assessments of authority, applicability, and integrity.

### **13.1.3 Boundary Design: Filing as a Tacit Endorsement**

#### **Formal boundary in litigation practice**

Under FRFP, each filing is a boundary artefact:

- a lawyer’s signature (physical or electronic) on a brief or motion is a boundary event in  $T_{\text{law}}$ ;
- by signing, counsel tacitly endorses that:
  - factual content has evidentiary support;
  - legal arguments are grounded in existing law or plausible non-frivolous extensions;
  - citations correspond to real, relevant authorities.

Rule 11 and analogous ethical rules explicitly codify this boundary: they assign to lawyers the responsibility to verify the explicit artefacts they submit, regardless of who or what drafted them.

#### **Boundary confusion in Mata**

In *Mata v. Avianca*, the boundary between AI output and lawyer endorsement was blurred:

- counsel treated ChatGPT’s outputs as if they were akin to traditional research tool results, assuming it would not fabricate cases;
- they did not independently verify the case citations in conventional databases before filing;
- even after opposing counsel and the court raised concerns, they returned to ChatGPT to “confirm” the existence of the cases rather than re-grounding in authoritative sources.

From an FRFP perspective:

- the filing of the brief was treated as a mostly explicit operation (exporting and submitting AI text), rather than a tacit endorsement requiring intensive human evaluation;
- correctness authority was informally ceded to the tool, contrary to the formal boundary encoded in professional rules.

### 13.1.4 Protocol-Level Failures in FRFP Terms

#### Z.6.1 Explicit-Space Hallucination and Semantic Fragility

**Problem.** Generative models are known to *hallucinate*: they produce fluent but fabricated content, including nonexistent cases and fake citations. In *Mata*, ChatGPT:

- generated case names, docket numbers, and quotes that mimicked real federal decisions;
- insisted, when queried, that the cases “indeed exist” and could be found in Lexis or Westlaw;
- produced PDFs with plausible formatting that further deceived counsel.

In FRFP language, the explicit semantics of the AI system for the task “produce supporting cases” is structurally unsafe:

- the mapping from prompts to outputs does not preserve a key invariant (truthfulness or referential validity of citations);
- the system has no internal notion of “being grounded in the corpus of real cases”; it only optimises for textual plausibility.

**Consequence.** Even without any negligence at the boundary, such a system *cannot* be treated as an oracle of case law. It is, at best, a suggestion engine. Treating its outputs as ready-to-file legal authority is an explicit-space misinterpretation.

#### Z.6.2 Boundary Failure: From Tool to Surrogate Researcher

**Problem.** The lawyers in *Mata* effectively used ChatGPT as a surrogate legal researcher:

- they asked it for cases supporting their proposition;
- they accepted its answers without verifying them in standard research tools;
- they relied on its reassurances even after doubts were raised.

In FRFP terms:

- the boundary between explicit AI generation and tacit legal endorsement collapsed: the AI’s text crossed almost directly into the filed brief;
- the human role reduced to light editing and formatting, not substantive verification;
- the signature, a boundary marker, became decoupled from actual tacit evaluation.

**Consequence.** This is a textbook instance of *boundary capture*: what should have been a human correctness function at filing time was, in effect, automated by an explicit system whose semantics do not support that role.

### Z.6.3 Missing Safe Horizons and Re-Grounding

**Problem.** Use of AI tools for legal drafting and research is inherently long-horizon:

- lawyers may come to rely on such tools daily or hourly across cases;
- tacit skills in manual research and critical reading can degrade;
- oversight can become perfunctory as trust in the tool grows.

In *Mata*, the episode occurred relatively early in the legal profession’s encounter with generative AI, but the pattern is instructive:

- counsel used AI extensively in one brief without clear constraints;
- there was no firm-level safe-horizon policy (e.g. limits on unvetted AI usage, mandatory verification steps);
- when the tool’s reliability was questioned, they did not reset to a “manual” mode, but deepened reliance on the same explicit system.

**Consequence.** FRFP would say that the run—the sequence of research and drafting steps for that brief—had an unsafe horizon: the depth of AI reliance exceeded the robustness of explicit semantics and the available tacit oversight. A well-designed FRFP implementation in law would enforce shorter horizons and mandatory re-grounding in primary sources.

### 13.1.5 Beyond Mata: A Pattern of Legal AI Failures

*Mata v. Avianca* was the first widely publicised sanction for AI hallucinations in legal practice, but subsequent cases show it is not an isolated event:

- A Utah appeals court sanctioned an attorney in 2025 for filing a brief with fabricated citations generated by ChatGPT, including a nonexistent case, and ordered restitution and charitable donations.

- California’s Court of Appeal fined a lawyer \$10,000 after discovering that 21 of 23 quoted case passages in a brief were AI fabrications, noting the growing need for AI-use regulation.
- Multiple federal and state judges have now sanctioned or reprimanded lawyers and experts for unverified AI-generated citations, with surveys and bar commentary describing an “epidemic” of such errors.

These events illustrate a population-level FRFP concern:

- as more practitioners incorporate AI into workflows, tacit professional norms about verification can degrade;
- explicit-only governance (tool warnings, EULA disclaimers) is insufficient to prevent structural misuse;
- courts are being forced, case by case, to reassert the boundary: correctness and responsibility remain with human lawyers.

### 13.1.6 Counterfactual: An FRFP-Aligned Legal Research Workflow

We sketch how FRFP might structure AI use in legal practice.

#### Z.6\* Explicit/Tacit Role Separation

- AI tools are designated as *drafting and ideation assistants*: they may propose arguments, case lists, and language.
- Human lawyers remain solely responsible for:
  - selecting authorities to cite;
  - verifying that citations correspond to real, relevant cases;
  - assessing the doctrinal and strategic soundness of arguments.

#### Z.6\* Boundary Design at Filing

- Filing a document is treated as a formal boundary event:
  - the lawyer must attest (internally, and where required by rule) that all citations have been verified using trusted sources (official reporters, recognised databases);
  - firms implement checklists or tooling to mark each citation as “verified” before filing.
- AI-generated content is visible in internal logs (e.g. authorship annotations in the draft) but not privileged in correctness: the final filing is a human artefact.

**Z.6\* Safe Horizons and Firm-Level Policy**

- Firms and courts define safe-horizon policies for AI use:
  - no filing may rely solely on AI-driven research; at least one human must independently run conventional research;
  - heavy AI use in earlier drafts must be balanced by more intensive human verification at later stages;
  - repeated use of AI tools triggers periodic audits of workflows and error rates.
- Training and onboarding emphasise the limitations of generative models, explicitly correcting the misconception that they cannot “fabricate” content.

**Z.6\* Institutional Governance**

- Bar associations and courts issue FRFP-style guidance:
  - clarifying that AI assistance does not change the locus of professional responsibility;
  - recommending or requiring disclosure of AI use in certain contexts;
  - defining sanctions and remedial steps for AI-related misconduct.
- Firms create internal AI oversight committees to:
  - approve tools and acceptable use cases;
  - monitor incidents and near-misses;
  - update safe-horizon and verification procedures.

**13.1.7 Lessons for FRFP and Professional Domains**

The *Mata* case and its successors support several FRFP themes:

1. **Generative AI is structurally hallucination-prone.** Its explicit semantics optimise for plausibility, not truth. Without structural constraints, it will produce fabricated but authoritative-looking artefacts.
2. **Signatures are boundaries, not formalities.** In law and other professions, signing a document is a boundary event at which tacit endorsement is supposed to be applied. Automating the path up to that boundary does not diminish the responsibility that attaches at it.
3. **Safe horizons are essential in professional workflows.** Long-horizon, high-frequency reliance on AI without explicit verification protocols invites degradation of tacit skills and norms.

4. **Explicit-only governance is insufficient.** Tool warnings, footnotes, and vendor caveats cannot substitute for institutional operators (courts, bars, firms) that enforce FRFP-style boundaries and safe horizons.
5. **Formal architectures clarify “AI mishaps” as structural.** *Mata* is often narrated as a cautionary tale about a few careless lawyers. FRFP reframes it as a prototypical failure mode whenever explicit AI systems are allowed to leak across professional boundaries without preserving tacit correctness authority.

As generative AI becomes embedded across professions, the lesson of *Mata v. Avianca* is that AI assistance can coexist with professional responsibility only under architectures that explicitly maintain the explicit/tacit split, design robust boundaries at endorsement points, and enforce finite safe horizons with institutional oversight.

## Chapter 14

# FRFP Case Study: JPMorgan CIO “London Whale” Trading Losses

## 14.1 FRFP Case Study (Finance / Risk Management): JPMorgan CIO “London Whale” Trading Loss

This appendix analyses the 2012 JPMorgan Chief Investment Office (CIO) trading losses—commonly known as the “London Whale” incident—through the lens of FRFP. We focus on explicit/tacit structure, boundary design, safe horizons, and institutional governance in a large, complex financial institution.

### 14.1.1 Background

In early 2012, JPMorgan’s CIO, operating out of London, accumulated very large positions in credit default swap (CDS) indices and tranches as part of its Synthetic Credit Portfolio (SCP). The trading was led by Bruno Iksil, a trader soon dubbed the “London Whale” because his positions were so large they moved prices in the CDX index markets.

The SCP was described internally as a hedge for the bank’s overall credit exposure, but in practice it became a large, complex, and partially directional portfolio of credit derivatives. By April–May 2012, press reports and hedge funds on the other side of the trade highlighted mounting losses. JPMorgan initially disclosed an estimated \$2 billion loss in May 2012; this was later revised to about \$6.2 billion as positions were unwound.

Subsequent investigations by JPMorgan’s internal task force, the U.S. Senate’s Permanent Subcommittee on Investigations, the Federal Reserve, and other regulators found:

- poor judgment, weak risk management, and inadequate escalation in the CIO;
- risk limits that were too coarse and sometimes ignored;
- a new Value-at-Risk (VaR) model that halved reported risk but was flawed in design and implementation, including spreadsheet errors;
- mis-marking of positions and misleading information provided to senior management and regulators.

JPMorgan ultimately paid over \$900 million in fines to U.S. and U.K. regulators and significantly revised its risk governance. The incident became a canonical example of derivatives risk, model failure, and control breakdown in a major bank.

### 14.1.2 Explicit and Tacit Spaces in the CIO

#### Explicit space $E_{\text{CIO}}$

In FRFP terms, the explicit domain includes:

- the portfolio of CDS indices and tranches (e.g. CDX IG 9, high-yield indices, bespoke tranches);

- trade tickets, positions, and P&L records for the SCP;
- risk models and metrics:
  - VaR models (old and new) and their inputs;
  - stress scenarios, sensitivities, and limit reports;
- formal risk limits and escalation thresholds for the CIO;
- internal reports and presentations to senior management and the board;
- regulatory filings and disclosures regarding the bank’s risk profile.

The explicit pipelines in  $N \times E_{\text{CIO}}$  consist of:

- trading decisions that change the SCP positions;
- daily risk computation runs (VaR, stress, P&L);
- aggregation and reporting of risk and performance up the hierarchy;
- communications with regulators.

### **Tacit space $T_{\text{CIO}}$**

Tacit space comprises:

- traders’ and managers’ market intuition: views on credit spreads, correlation, and systemic risk;
- risk managers’ understanding of model limitations and data quality;
- senior executives’ beliefs about the CIO’s role (hedging vs. profit centre, “fortress balance sheet”);
- board and regulator expectations about JPMorgan’s risk appetite and control culture.

These tacit states determine:

- how seriously risk signals are taken;
- whether VaR and other metrics are treated as approximate guides or as quasi-authoritative measures of correctness;
- how rapidly concerns are escalated and constraints imposed.

### 14.1.3 Boundary Design: Risk Metrics vs. Capital Commitments

#### Intended boundary

In a well-functioning FRFP-style risk architecture:

- explicit models (VaR, stress tests, P&L) provide signals in  $E_{\text{CIO}}$ ;
- human decision-makers in  $T_{\text{CIO}}$  (traders, risk committees, senior management, board) interpret these signals and make capital allocation decisions;
- boundary events include:
  - approval of trading mandates and limits;
  - responses to limit breaches;
  - sign-off on model changes and risk reports;
  - major strategic decisions (e.g. growth of the SCP).

Correctness about risk—whether the portfolio is within appetite, appropriately hedged, and resilient—is supposed to live at these human boundaries.

#### De facto boundary: models and limits as surrogates

In the London Whale incident, several factors shifted the effective boundary toward the explicit layer:

- the CIO was granted broad discretion and large notional limits, framed under a “macro hedging” mandate, with limited granular scrutiny;
- a new VaR model, implemented in January 2012, immediately cut reported VaR for the SCP roughly in half, reducing apparent risk and helping the portfolio stay within limits, despite flawed methodology and manual data-handling errors;
- limit breaches and growing losses were not promptly escalated or were rationalised using the new model and other explicit metrics;
- senior management initially characterised concerns as a “tempest in a teapot,” signalling confidence in the explicit risk apparatus.

From an FRFP perspective:

- VaR and related explicit metrics became de facto bearers of correctness;
- tacit skepticism and local concerns were overridden or not adequately propagated;
- the boundary between signal and decision blurred: if VaR was within limits on paper, positions were allowed to grow.

### 14.1.4 Protocol-Level Failures in FRFP Terms

#### Z.8.1 Explicit-Space Mis-Specification: VaR Model and Limit Structure

**VaR model flaws.** The Senate report and internal JPMorgan task-force report highlight significant problems with the new SCP VaR model:

- complex, manual data handling and aggregation, including spreadsheet processes prone to error;
- inappropriate scaling and parameter choices that understated volatility and correlation risk;
- limited independent validation and rushed implementation, motivated in part by the desire to reduce reported VaR.

In FRFP language:

- the explicit mapping  $f_{\text{VaR}} : \text{positions, data} \rightarrow \text{risk metric}$  had defective semantics;
- it failed to preserve crucial invariants (e.g. higher concentration and complexity should not show lower risk);
- the explicit artefact (VaR number) ceased to be a reliable signal for tacit risk assessment.

**Risk limit granularity.** Reports also note that risk limits for the SCP and CIO were not sufficiently granular:

- limits were set at aggregated levels that allowed large, offsetting positions to mask concentrations;
- concentration limits by index/tenor or scenario were weak or absent;
- breaches were frequent but often waived or reinterpreted.

Within FRFP, this means that the explicit constraints on runs in  $N \propto E_{\text{CIO}}$  were too coarse: the admissible set of runs included structurally dangerous configurations that should have been excluded by design.

#### Z.8.2 Boundary Failure: From Hedge to Prop Book

**Mandate drift.** The SCP was intended as a hedge for JPMorgan’s broader credit exposure, but investigations concluded that:

- the portfolio grew far beyond what was needed for plausible hedging of underlying exposures;
- positions were often directional bets on credit indices and tranches;

- internal descriptions of the SCP as a “hedge” were, at best, partially accurate.

In FRFP terms:

- the explicit label “hedge” attached to the SCP no longer matched its tacit economic function;
- this misalignment distorted boundary decisions: risk-taking that should have been treated as proprietary trading was instead viewed as risk-reducing or neutral.

**Escalation and endorsement.** Limit breaches, growing P&L volatility, and market commentary about the “London Whale” should have triggered strong boundary events:

- reconsideration of the SCP mandate;
- immediate review and possible suspension of trading;
- board-level and regulatory engagement.

Instead, escalation was delayed and partial:

- some internal concerns were downplayed;
- senior management initial public statements sought to reassure markets rather than acknowledge structural issues;
- decisive actions (e.g. “phones down” order, unwinding trades) only came after large losses had already materialised.

FRFP reads this as boundary failure: tacit authorities did not reassert control when explicit signals and external information clearly indicated that admissibility conditions were being violated.

### Z.8.3 Safe Horizon Violations: Size, Complexity, and Time

**Long-horizon build-up.** The SCP positions grew over many months:

- notional exposures reached hundreds of billions of dollars;
- complexity increased as traders added offsetting positions, basis trades, and curve trades;
- risk metrics struggled to capture tail risks and liquidity effects.

FRFP would require:

- safe horizons on notional size, complexity, and model reliance;
- episodic re-grounding of the portfolio’s purpose and risk profile;
- periodic human re-evaluation of worst-case scenarios, independent of model output.

**Collapse.** Because such safe horizons were not structurally enforced:

- mispricing and model error accumulated;
- counterparties and hedge funds identified the CIO’s large positions and took the other side;
- when markets moved, the unwinding of positions generated realised losses far exceeding early estimates.

The London Whale incident is thus a prototypical safe-horizon violation in FRFP terms: a long run in  $N \propto E_{\text{CIO}}$  where explicit constraints and tacit oversight failed to cap risk accumulation.

### Z.8.4 Institutional Governance and Regulator Interaction

**Internal governance.** Internal reports and external commentary raise questions about JPMorgan’s governance:

- the CIO reported directly to the CEO, yet its evolving activities and risk profile were not matched by strengthened controls;
- the firm lacked a treasurer for a period, and risk leadership in the CIO was relatively inexperienced;
- model approval and risk-limit processes did not prevent the flawed VaR model from being adopted or used to justify risk-taking.

**Regulatory oversight.** Senate and Fed reports note that regulators received incomplete or misleading information about the size and risk of the SCP, and that regulatory frameworks (including the Volcker Rule debates) did not fully anticipate such complex hedging/prop hybrids.

In FRFP’s institutional semantics:

- internal governance and regulators together constitute institutional operators  $G$  acting on the tacit state of the firm and the financial system;
- these operators relied heavily on explicit artefacts (VaR numbers, risk dashboards, high-level descriptions of the SCP) without robust mechanisms to interrogate their semantics;
- non-automation of governance is violated if one imagines that such explicit metrics can replace tacit supervisory judgment.

The London Whale case illustrates the limits of explicit-only governance in complex financial institutions.

#### 14.1.5 Counterfactual: An FRFP-Aligned CIO Architecture

What might a CIO risk architecture look like if designed under FRFP?

**Z.8\* Clear Explicit/Tacit Role Separation**

- Explicit systems:
  - compute risk metrics (VaR, stress, greeks);
  - maintain detailed position and P&L records;
  - generate scenario analyses and what-if projections.
- Tacit decision-makers (traders, risk executives, board, regulators) retain responsibility for:
  - defining the economic purpose of each portfolio (hedge vs. prop vs. liquidity management);
  - accepting or rejecting risk levels indicated by metrics;
  - interpreting model limitations and external market signals.

**Z.8\* Boundary Design for Models and Limits**

- Model changes (e.g. new VaR models) are treated as major boundary events:
  - require independent model validation;
  - require parallel runs and backtesting;
  - require board-level awareness for portfolios of systemic importance.
- Risk limits are decomposed and granular:
  - limits by index, tenor, scenario, and concentration;
  - clear escalation steps for any breach, with documented justifications for temporary overrides.
- Limit waivers and policy changes are logged as explicit boundary events with responsible signatories.

**Z.8\* Safe Horizons for Portfolio Growth**

- Notional and risk exposures for complex portfolios grow only within explicitly defined safe horizons:
  - caps on exposure under a given model configuration;
  - episodic reviews after each incremental growth phase;
  - scenario-based drills of worst-case losses and liquidity.
- External signals (e.g. concentrated positioning visible to counterparties, press reports about unusual trades) trigger automatic re-grounding reviews and potential position caps.

### Z.8\* Institutional Governance with FRFP Operators

- A board-level risk committee acts as an FRFP institutional operator:
  - defines which risk decisions can be automated or delegated;
  - sets admissibility criteria for strategies like the SCP;
  - reviews and approves safe-horizon frameworks.
- Regulators adopt FRFP-aware expectations:
  - stress testing and model validation that explicitly consider boundary design and governance, not just quantitative metrics;
  - requirements for banks to document explicit/tacit roles, model limits, and escalation procedures for major portfolios.

Under such an architecture, the SCP might still suffer trading losses, but:

- mis-specified models would be more likely caught in validation;
- portfolio growth would be bounded by safe horizons;
- early signals (limit breaches, external market reactions) would have triggered corrective boundary events sooner.

#### 14.1.6 Lessons for FRFP and Financial Risk Management

The London Whale case reinforces several themes in FRFP’s treatment of finance:

1. **Model numbers are not correctness.** VaR and similar metrics are explicit artefacts whose semantics depend on modelling choices and data quality; treating them as definitive measures of risk leads to boundary capture.
2. **Labels (“hedge”) are not guarantees.** Calling a portfolio a hedge does not make it one. FRFP emphasises alignment between explicit descriptions and tacit economic reality.
3. **Safe horizons matter as much as limits.** Traditional limits are static constraints; FRFP highlights the need for dynamic safe horizons that govern how fast and how far complex positions can evolve before mandatory re-grounding.
4. **Institutional non-automation holds in finance.** Governance based largely on explicit risk metrics and dashboards, without strong tacit oversight and FRFP-style boundary design, cannot reliably prevent large, complex failures.

5. **Formal architectures clarify systemic patterns.** The London Whale is often framed as a story of a rogue trader, a bad model, or a management oversight. FRFP reveals a structural pattern: mis-specified explicit semantics, weak boundaries, absent safe horizons, and explicit-only governance together made the CIO’s trading run inadmissible at scale.

Accordingly, the London Whale episode serves as a detailed, well-documented example of how FRFP-style reasoning can illuminate the interplay between models, human judgment, and institutional structures in large financial institutions.

## Chapter 15

# FRFP Case Study: AlphaFold and Protein Structure Prediction

## 15.1 FRFP Case Study (Science / Discovery): AlphaFold

This appendix analyses DeepMind’s AlphaFold as an FRFP case study in long-horizon scientific discovery. We focus on how AlphaFold’s design and deployment interact with the FRFP architecture: explicit/tacit separation, boundary design, safe horizons, and institutional semantics in the life sciences.

### 15.1.1 Background

Predicting a protein’s three-dimensional structure from its amino-acid sequence was a central open problem in structural biology for roughly 50 years. Experimental techniques such as X-ray crystallography, NMR, and cryo-EM are precise but slow, expensive, and unevenly distributed across labs.

In 2020, at the CASP14 (Critical Assessment of protein Structure Prediction) competition, DeepMind’s AlphaFold2 demonstrated unprecedented accuracy: for many targets it predicted backbone atom positions with median error  $\lesssim 1$  Å, substantially outperforming all other methods and reaching a level CASP’s organisers described as effectively “solving” the protein folding problem for many single-chain proteins.

Subsequently:

- DeepMind and EMBL–EBI released the AlphaFold Protein Structure Database (AF-DB), which by 2024 covered over 214 million protein sequences across most known organisms, a  $\sim 500$ -fold expansion relative to the initial 2021 release.
- AF-DB increased structural coverage of the human proteome from roughly 48% to about 76% of residues, dramatically shrinking the set of proteins with no structural information.
- AlphaFold and related tools have been used in thousands of research projects, including studies of nuclear pore complexes, malaria and neglected disease targets, and antibiotic resistance, and are now integrated into mainstream structural biology workflows.
- AlphaFold 3, released in 2024, extends predictions to protein complexes and interactions with ligands, nucleic acids, and antibodies, targeting drug design more directly.

This makes AlphaFold an outstanding example of AI success that is tightly coupled to human scientific practice and data ecosystems, rather than a standalone agent.

### 15.1.2 Explicit and Tacit Spaces in Structural Biology

**Explicit space**  $E_{AF}$

For AlphaFold and AF-DB, explicit space comprises:

- amino-acid sequences from UniProt / UniRef and other databases;

- multiple sequence alignments (MSAs) and evolutionary couplings;
- experimentally determined structures in the Protein Data Bank (PDB);
- AlphaFold’s network parameters and prediction code;
- predicted 3D structures, per-residue confidence scores (pLDDT), and predicted aligned error (PAE) matrices;
- AF-DB, which serves predictions and metadata via web, APIs, and bulk download.

These are all artefacts in  $E_{AF}$ : bits and tensors that can be processed by machines, versioned, and shared.

### **Tacit space $T_{bio}$**

Tacit space is the community’s scientific understanding:

- structural biologists’ intuition about protein domains, flexibility, disorder, and function;
- medicinal chemists’ sense of ligandability, binding-site quality, and risk of false positives in virtual screening;
- experimentalists’ judgement about which predictions are good enough to guide experiments, and where empirical validation is essential;
- institutions’ and funders’ beliefs about the value and limitations of in silico structure prediction in drug discovery pipelines.

Correctness in science is ultimately judged in this tacit space: by whether findings replicate, support plausible mechanistic narratives, and fit into broader theoretical and empirical structures.

### **15.1.3 Boundary Design: Predictions vs. Scientific Claims**

#### **Formal interface: predictions with uncertainty**

AlphaFold’s design and release strategy incorporate explicit boundary elements between its outputs and scientific conclusions:

- Each structure prediction comes with per-residue confidence scores (pLDDT) and PAE maps, making uncertainty explicit and machine- and human-readable.
- AF-DB documentation clearly states that predictions are not a substitute for experiments, especially for low-confidence regions, disordered proteins, and conformationally dynamic systems.

- The system is framed as providing structural hypotheses and starting points for further work, not definitive answers about function or druggability.

In FRFP terms, the boundary functor  $J-K$  maps explicit predictions in  $E_{AF}$  into tacit states  $T_{bio}$  via:

- scientists’ interpretation of pLDDT/PAE as reliability signals;
- domain-specific heuristics about when to trust or discount AlphaFold models;
- peer-review and replication processes that convert structural hypotheses into accepted knowledge.

### **De facto use: high trust, but still mediated**

Empirically, the community has come to treat AlphaFold structures as:

- near-experimental quality for many well-behaved, single-chain, globular proteins with high-confidence scores;
- less reliable for intrinsically disordered regions, flexible loops, large complexes, and ligand-bound conformations;
- excellent starting models for phasing crystallographic data or building cryo-EM models, substantially accelerating standard experimental pipelines.

From an FRFP standpoint:

- the boundary remains with human scientists and experimentalists;
- AlphaFold outputs strongly inform tacit reasoning but are not, in general, treated as final claims about mechanism or drug efficacy;
- this arrangement—highly capable explicit system plus robust tacit interpretation—is relatively close to an FRFP-aligned design.

#### **15.1.4 Protocol-Level Properties in FRFP Terms**

##### **Z.9.1 Explicit-Space Strength: A Mature, Well-Specified Semantics**

AlphaFold’s explicit semantics are unusually strong for an AI system:

- The task is well-defined: map a sequence to a static 3D structure under standard physical conventions, up to known caveats (flexibility, alternate states).
- Ground truth exists: experimentally-determined structures provide a rich, high-quality, globally shared reference database (PDB).

- Evaluation protocols (CASP) offer strict, blind metrics for accuracy, with community acceptance of what counts as “good enough”.

From an FRFP perspective:

- the mapping  $f_{\text{AF}} : \text{sequence, MSAs} \rightarrow \text{structure}$  has a clear semantics and strong external validation;
- $E_{\text{AF}}$  is unusually close to a self-contained domain of explicit artefacts whose correctness can be assessed without extensive recourse to tacit local context;
- this is precisely the kind of problem where FRFP expects explicit AI to be able to “go far” safely before requiring tacit correction.

### Z.9.2 Explicit Transparency: Confidence and Coverage

Key design choices that support FRFP-style use:

- Per-residue pLDDT scores and PAE maps provide structured, explicit uncertainty, allowing downstream users to modulate trust.
- AF-DB exposes coverage and confidence statistics across proteomes, clarifying where the system is strong vs. weak.
- Documentation emphasises limitations: dynamics, complexes, and ligand binding are only partially captured, especially for AlphaFold2-era models.

This explicit communication of uncertainty and domain of competence is an instance of FRFP’s recommendation that explicit pipelines carry their own self-descriptions and limits, making boundary decisions in  $T_{\text{bio}}$  better informed.

### Z.9.3 Safe Horizons in Scientific Use

Although AlphaFold is powerful, its safe usage horizons are still bounded:

- For many basic structural biology tasks (e.g. annotating domain architectures, suggesting mutational impacts on stability), predictions can be used relatively directly, within the guardrails of pLDDT/PAE.
- For mechanistic inferences and drug discovery (binding energetics, allosteric regulation, conformational ensembles), AlphaFold outputs are starting points that must be combined with other modelling and experimental data.
- Structural biology remains an iterative loop: predictions suggest experiments; experiments confirm, refine, or refute predictions; tacit understanding evolves accordingly.

FRFP’s safe-horizon lens thus suggests:

- AlphaFold enables very long runs in explicit structure space, but tacit scientific correctness is safeguarded by experimental validation and peer review;
- attempts to push AlphaFold beyond these horizons—e.g. to fully automate target validation and drug design without adequate data—are precisely where current AI drug discovery efforts have struggled.

### 15.1.5 Institutional and Ecosystem-Level Semantics

#### Open Data and Shared Infrastructure

AlphaFold’s impact is amplified by institutional choices:

- AF-DB is freely accessible and integrated with UniProt, InterPro, and other resources, functioning as a public infrastructure layer for structural biology.
- This enables a large population of scientists to build tacit understanding of when and how to use predictions, via thousands of papers and lab practices.
- OpenFold and related community projects extend and reimplement AlphaFold-like models, allowing independent scrutiny and adaptation.

In FRFP terms, these institutions act as operators on the global tacit state of the structural biology community, spreading knowledge of AlphaFold’s capabilities and limits and helping to stabilise the explicit/tacit interface.

#### Drug Discovery and Commercial Ecosystem

AlphaFold has also shaped commercial AI-for-drug-discovery:

- companies like Isomorphic Labs and startups such as Latent Labs build on AlphaFold-style models to design drugs and synthetic proteins;
- large pharma consortia are now pooling structure data (e.g. OpenFold3 collaborations) to train next-generation structural models, explicitly citing AlphaFold as a foundation.
- industry analysts highlight AlphaFold as a key bright spot in an otherwise slower-than-hyped AI drug discovery landscape.

Here, FRFP’s institutional semantics emphasise:

- firms are treating AlphaFold outputs as high-quality explicit artefacts that feed into broader, still-challenging tacit tasks (biology, chemistry, clinical development);

- where companies attempt to over-automate tacit scientific judgment (e.g. fully AI-driven target selection with limited wet-lab work), failures and delays mirror the non-automation claims in the main text.

### 15.1.6 Counterfactual: FRFP Pitfalls AlphaFold Mostly Avoided

It is instructive to note which FRFP failure modes AlphaFold largely *avoided*, compared to some failures in other case studies:

- **Misplaced boundaries.** AlphaFold was framed, and is mostly used, as a prediction tool feeding into human-led science, not as an autonomous decision-maker about patient care or investment. The boundary stays with scientists.
- **Opaque or uncalibrated output.** Confidence measures and evaluation benchmarks are built into the explicit interface, reducing the risk of uncalibrated over-trust.
- **Weak explicit semantics.** The problem definition (static structure prediction) and evaluation are unusually clean, unlike in domains such as legal reasoning or clinical decision-making, where semantics are more entangled with tacit norms.
- **Closed, proprietary governance.** By partnering with EMBL–EBI and releasing AF-DB openly, DeepMind effectively created a shared institutional operator that diffuses control and supports community governance and critique.

These design and deployment choices align AlphaFold more closely with FRFP’s principles than many other high-profile AI systems.

### 15.1.7 Lessons for FRFP and Scientific AI

AlphaFold provides a constructive counterpart to failure case studies:

1. **Explicit problems with strong ground truth are fertile ground.** FRFP predicts that domains with clear explicit semantics and robust evaluation (like structure prediction) are particularly suitable for powerful explicit AI systems.
2. **Uncertainty-aware interfaces support healthy boundaries.** By surfacing confidence and error explicitly, AlphaFold helps scientists keep correctness in tacit space, rather than blindly trusting predictions.
3. **Open infrastructure shapes population-level tacit states.** AF-DB and community reimplementations act as institutional operators that propagate good usage norms and collective understanding of limitations.

4. **Non-automation still applies downstream.** Even with AlphaFold’s success, drug discovery and mechanistic biology remain only partially automatable; tacit scientific judgment and experimentation remain structurally indispensable.
5. **Formal FRFP analysis helps generalise from success.** By analysing why AlphaFold works as a socio-technical system—not just as a model—FRFP provides guidance for designing future scientific AI tools that integrate explicit strength with tacit, institutional oversight.

In this sense, AlphaFold is both a triumph of explicit AI in a well-chosen domain and a demonstration of how FRFP-compatible infrastructure can amplify an AI breakthrough into a durable, community-wide scientific advance.

## Chapter 16

# FRFP Case Study: Autonomous Diabetic Retinopathy Screening AI

## 16.1 FRFP Case Study (Clinical / Public Health): Autonomous AI for Diabetic Retinopathy Screening

This appendix analyses autonomous AI systems for diabetic retinopathy (DR) screening as a positive FRFP case study. We focus primarily on FDA-cleared systems such as LumineticsCore<sup>®</sup> (formerly IDx-DR), EyeArt<sup>®</sup>, and AEYE-DS<sup>®</sup>. These systems are among the first examples of autonomous AI diagnostics deployed at scale in routine care, particularly in primary care clinics and underserved settings.

### 16.1.1 Background

Diabetic retinopathy is a leading cause of preventable blindness among working-age adults worldwide. Regular retinal screening and timely treatment can prevent most vision loss, but screening rates are chronically low, especially in primary care and underserved populations.

To address this gap, several autonomous AI systems have been developed and cleared:

- **LumineticsCore<sup>®</sup> (IDx-DR)**: in 2018, the first autonomous AI diagnostic system authorized by the U.S. FDA in any field of medicine. It automatically detects more-than-mild DR (mtmDR) from retinal images taken on a specific fundus camera in primary-care settings, without requiring an eye-care specialist to interpret each image.
- **EyeArt<sup>®</sup> (Eyenuk)**: FDA-cleared in 2020 as an autonomous AI system that detects both mtmDR and vision-threatening DR (vtDR) and can work across multiple fundus camera models.
- **AEYE-DS<sup>®</sup> (AEYE Health)**: FDA-cleared as an autonomous AI for DR screening, designed to operate with certain non-mydratic fundus cameras and, in newer configurations, with a single image per eye and no dilation, targeting sub-minute screening workflows.

Pivotal trials for IDx-DR showed sensitivity around 87% and specificity around 90% for mtmDR in primary care populations, meeting pre-specified performance endpoints for autonomous use. Subsequent implementation studies (e.g. at Johns Hopkins and in federally qualified health centers) indicate that autonomous AI screening can significantly increase screening rates and care-gap closure when integrated into primary care workflows.

From an FRFP perspective, these systems are interesting because:

- they *are* autonomous diagnostic systems (they issue a clinical-level decision: refer vs. no-refer);
- yet they are deployed in ways that keep ultimate correctness at human and institutional boundaries (treatment, follow-up, programme design);
- their success depends heavily on careful explicit problem design, uncertainty labelling, and regulatory/institutional governance.

### 16.1.2 Explicit and Tacit Spaces in DR Screening AI

#### Explicit space $E_{\text{DR}}$

For systems such as LumineticsCore, EyeArt, and AEYE-DS, explicit space includes:

- input images:
  - colour fundus photographs from specified camera models;
  - typically macula- and disc-centered views per eye;
- model internals:
  - trained deep convolutional or related networks;
  - pre-processing and image-quality assessment modules.
- output artefacts:
  - a categorical diagnostic conclusion (e.g. “mtmDR present” vs. “mtmDR absent”);
  - image-quality verdict (gradable vs. ungradable);
  - sometimes a DR severity grade or referral recommendation.
- deployment infrastructure:
  - cloud or local servers receiving images and returning decisions;
  - user interfaces in primary care clinics and retail settings.

All these artefacts live in  $E_{\text{DR}}$ : they are explicit bits and decisions produced and consumed by machines.

#### Tacit space $T_{\text{DR}}$

Tacit space contains the epistemic states of:

- primary-care clinicians: their understanding of diabetes care pathways, risk factors, and when to emphasize follow-up;
- eye-care specialists (ophthalmologists, retina specialists): nuanced knowledge of DR progression, treatment thresholds, and patient-specific considerations;
- patients: their understanding of vision risk, willingness to attend referrals, and trust in the healthcare system;
- health-system administrators and payers: beliefs about programme design, reimbursement, and equity.

These tacit states determine how explicit AI outputs are interpreted and acted upon: whether a “refer” result leads to timely specialist care, or whether a “no-refer” result alters behaviour around future screening.

### 16.1.3 Boundary Design: From Image Classification to Clinical Workflow

#### Regulatory definition of the AI boundary

Regulatory documents for IDx-DR and similar systems define a very specific boundary:

- The AI system is authorised to *autonomously* detect more-than- mild DR (and for EyeArt/AEYE-DS, in some cases, vtDR) from images acquired under specified conditions.
- If image quality is insufficient, the system returns an “ungradable” result and the patient must be referred or re-imaged.
- For negative results (no mtmDR detected), guidelines and labelling still require repeat screening at defined intervals (often yearly), not discharge from care.
- For positive results, the system issues a referral recommendation, but an ophthalmologist ultimately determines diagnosis and treatment.

In FRFP terms:

- the AI’s explicit decision boundary is:  $\text{classify image} \rightarrow \{\text{mtmDR present, mtmDR absent, ungradable}\}$ ;
- the clinical boundary—where correctness is endorsed—remains:
  - in primary care: deciding whether a patient has satisfied the screening requirement or needs referral;
  - in specialist care: diagnosing DR stage and choosing treatment (laser, anti-VEGF, observation, etc.).

The FDA explicitly recognises this structure by authorising the AI to make a narrow, well-defined diagnostic decision while leaving broader care decisions to human clinicians.

#### Workflow integration as boundary implementation

Implementation studies describe how DR AI is embedded in primary-care workflows:

- a nurse or medical assistant captures fundus images during a routine diabetes visit;
- the AI system returns a result in minutes;
- if positive or ungradable, staff schedule an eye-care referral and provide counselling;
- if negative, the result is documented in the EHR, closing the DR screening care gap for that year.

From an FRFP perspective:

- AI operates fully within explicit space (image  $\rightarrow$  label), but boundary events (referral decisions, treatment) are explicitly structured in clinical workflows and guidelines;
- the architecture intentionally shifts some correctness responsibility (for *screening* classification) into the AI pipeline, but within a tightly circumscribed domain and with oversight via quality metrics and regulatory conditions.

#### 16.1.4 Protocol-Level Properties in FRFP Terms

##### Z.10.1 Explicit Semantics: A Narrow, Well-Specified Task

DR screening AI enjoys strong explicit semantics:

- **Task definition:** detect the presence of more-than-mild DR (and sometimes vtDR) from retinal photographs, according to accepted grading scales.
- **Gold standard:** independent, masked grading of images by retina specialists serves as a clear ground truth in pivotal trials.
- **Performance targets:** pre-specified sensitivity/specificity thresholds for autonomous use (e.g. rule-out sensitivity, rule-in specificity) are agreed with regulators.

In FRFP language:

- the mapping  $f_{\text{DR}} : \text{fundus image} \rightarrow \{\text{mtmDR}, \text{no mtmDR}, \text{ungradable}\}$  has a precise semantics under trial conditions;
- $E_{\text{DR}}$  has reliable, externally validated metrics to assess correctness of the AI’s explicit outputs;
- this is exactly the kind of narrow medical task where explicit AI can legitimately assume part of the correctness burden.

##### Z.10.2 Uncertainty and “Ungradable” as Structural Safeguards

An important design feature is the explicit handling of uncertainty:

- systems classify some images as ungradable, prompting referral to an eye-care provider instead of issuing a false negative;
- image-quality checks and fail-safe logic are part of the cleared indication for use;
- in practice, this means that when the explicit pipeline cannot produce a reliable decision, it falls back to tacit human judgement.

In FRFP terms:

- “ungradable” is a designed escape hatch back into tacit space  $T_{\text{DR}}$ ;

- the explicit system is not forced to speak when its semantics would be unreliable;
- this is a concrete instantiation of FRFP’s recommendation that explicit pipelines carry their own limitations to the boundary.

### Z.10.3 Safe Horizons in Screening Programs

DR screening is a repeated, population-level process:

- patients with diabetes should be screened annually or every few years, depending on guidelines;
- autonomous AI is intended for long-horizon use across large populations (health systems, national programmes).

FRFP’s safe-horizon lens highlights several structural properties:

- **Temporal horizons:** even after a negative AI screen, guidelines insist on re-screening after a fixed interval; no sequence of negative outputs permanently discharges the patient from surveillance.
- **Data shift monitoring:** camera changes, population differences, and disease prevalence shifts can degrade performance; responsible deployments monitor quality metrics and periodically revalidate the AI.
- **Programme-level review:** implementation studies treat DR AI roll-outs as quality-improvement projects, with ongoing measurement of screening uptake, time-to-referral, and adherence.

Thus, even though the AI is autonomous at the single-visit level, the overall programme is embedded in human and institutional cycles that limit how far the system can run without tacit correction.

## 16.1.5 Institutional Semantics: Regulation, Reimbursement, and Equity

### Regulatory governance as FRFP operator

Regulators act as institutional operators in FRFP terms:

- The FDA’s De Novo authorization of IDx-DR set a precedent for autonomous AI in medicine, including requirements on performance, indications, and image-acquisition conditions.
- Subsequent 510(k) clearances for EyeArt and AEYE-DS followed, broadening the set of devices and cameras but within a shared regulatory framework.
- CPT codes and reimbursement policies have been developed to support billing for autonomous AI DR screening, reinforcing its role in routine care.

These institutional choices:

- formalise the boundary between AI output and clinical use;
- define safe deployment domains (primary care, camera types, age ranges);
- require ongoing post-market surveillance and device updates.

### Health-system deployments and equity goals

Large health systems and public programmes increasingly deploy DR AI with explicit equity aims:

- Johns Hopkins Medicine’s deployment across primary-care sites showed increased closure of screening care gaps, especially in some underserved groups.
- Multicomponent trials in federally qualified health centres (FQHCs) evaluate whether AI screening plus EHR integration and patient education improves adherence to screening and standards of diabetes care.
- In India, AI-based DR screening platforms such as MadhuNetrAI / MadhuNETrAI are being piloted or rolled out in government programmes to reach rural and military populations with limited ophthalmology access.

From an FRFP perspective, these programmes:

- treat autonomous AI as part of an institutional operator that adjusts population-level tacit states (e.g. awareness, screening norms, referral patterns);
- rely on explicit AI outputs to trigger care, but leave diagnosis, treatment, and policy adaptation in human hands;
- illustrate how explicit systems can be embedded in equity-oriented institutional dynamics, rather than only in revenue-maximising workflows.

#### 16.1.6 Counterfactual Pitfalls and FRFP Safeguards

DR screening AI also highlights potential FRFP failure modes that are being actively managed:

- **Overextension of indications.** Using an AI system beyond its cleared population (e.g. different ethnic mix, camera, or disease spectrum) without revalidation could degrade performance. FRFP would insist on explicit safe horizons and re-grounding when context shifts.
- **Silent false negatives.** If negative AI results led clinicians or patients to skip future exams, long-horizon risk would increase. Current guidelines and labelling prevent this by mandating periodic re-screening regardless of AI output.

- **Complacency in specialist care.** There is a risk that specialists might over-trust AI-based triage and pay less attention to borderline or atypical cases. FRFP would recommend clear communication that AI covers only the screening threshold, not full disease management.

So far, regulatory frameworks, trial designs, and deployment practices have kept many of these issues under control, making DR screening AI a relatively successful instance of autonomy within an FRFP-compatible architecture.

### 16.1.7 Lessons for FRFP and Clinical AI

Autonomous DR screening systems illustrate several of FRFP’s central points in a positive light:

1. **Narrow, well-specified tasks can bear autonomous decisions.** When explicit semantics are strong (image  $\rightarrow$  DR threshold classification) and ground truth is available, AI can legitimately take on a clearly-bounded diagnostic role.
2. **Explicit uncertainty and “ungradable” outputs are crucial.** Structured fallbacks to human care are built into the model’s explicit interface, embodying FRFP’s recommendation that pipelines carry their own limits back to the boundary.
3. **Safe horizons can be encoded as clinical intervals.** Annual (or guideline-based) re-screening and ongoing performance monitoring function as safe-horizon mechanisms in a public-health setting.
4. **Institutional governance makes autonomy possible.** Regulatory authorisation, reimbursement structures, and programme design act as FRFP-style institutional operators that shape how AI is used and prevent over-automation of tacit clinical judgment.
5. **Explicit AI can advance equity when embedded correctly.** By bringing DR screening to primary care, FQHCs, and low-resource settings, DR AI demonstrates how explicit systems can extend access while keeping correctness at human and institutional boundaries.

As such, autonomous DR screening is a valuable counterpoint to high-profile failures: it shows that, under FRFP-compatible conditions—narrow task, strong semantics, explicit uncertainty, and robust institutional governance—AI can safely and measurably improve population health outcomes.

## Chapter 17

# FRFP Case Study: Waymo Driverless Ride-Hailing

## 17.1 FRFP Case Study (Autonomy / Mobility): Waymo Driverless Ride-Hailing

This appendix analyses Waymo’s driverless ride-hailing service as an FRFP case study in long-horizon autonomy. We focus on how its technology, safety processes, and deployment model map onto FRFP’s explicit/tacit split, boundary design, safe horizons, and institutional governance.

### 17.1.1 Background

Waymo, originally the Google self-driving car project (launched in 2009), has evolved into a commercial provider of robotaxi services. As of 2025, Waymo operates fully driverless ride-hailing (no human safety driver in the car) in multiple U.S. cities, including Phoenix, San Francisco, and parts of Los Angeles and other regions.<sup>1</sup>

Key milestones (very briefly):

- Early closed-course and supervised public-road testing (2010s);
- Limited public pilot in Chandler, Arizona, with safety drivers;
- Transition to fully driverless operations (no safety driver) in parts of the Phoenix metro area, expanding to 24/7 service;
- Gradual expansion to San Francisco and Los Angeles, with driverless service in designated operating domains;
- Public safety reports and disengagement statistics through California and Arizona regulators, and Waymo’s own safety cases.

Compared to many other high-profile autonomy efforts, Waymo has:

- accumulated very large numbers of driverless miles without a correspondingly large number of serious crashes;
- maintained a comparatively cautious, constrained operational domain;
- invested heavily in explicit safety cases and institutional relationships with regulators and local authorities.

From an FRFP perspective, Waymo is a prominent example of an AI-intensive, agentic system that has reached real-world, at-scale deployment in a domain with stringent safety demands.

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<sup>1</sup>For an overview of Waymo’s deployments and miles driven, see public documentation and safety reports on the Waymo and Alphabet websites, as well as regulatory filings and independent analyses.

### 17.1.2 Explicit and Tacit Spaces in Waymo’s System

#### Explicit space $E_{\text{Waymo}}$

Waymo’s explicit domain consists of the artefacts processed by its autonomous driving stack and operations:

- **Perception data and models:** sensor inputs (LiDAR, radar, cameras), object detections, semantic segmentations, and tracking.
- **HD maps and localisation:** high-definition road maps, lane-level geometry, traffic control information, static infrastructure features.
- **Prediction and planning:** models of other agents’ behaviour; trajectory plans; control commands (steering, throttle, braking).
- **Operational data:**
  - logs of each trip (telemetry, interventions, incidents);
  - simulation scenarios and replay environments;
  - safety metrics (e.g. collision rates, policy violations).
- **Policies and procedures:** formal documentation of safety protocols, fallback behaviours, disengagement rules, and operational design domains (ODDs).

These are all explicit artefacts in  $E_{\text{Waymo}}$ , manipulated by Waymo’s software systems and interpreted by engineers and regulators.

#### Tacit space $T_{\text{mobility}}$

Tacit space contains the epistemic and normative states of:

- safety drivers (in earlier phases) and remote operations staff: their sense of when the system is behaving oddly or unsafely;
- Waymo’s safety engineers and leadership: judgement about acceptable risk, release readiness, and go/no-go decisions for new features and cities;
- regulators and city officials: beliefs about safety, public acceptability, and regulatory obligations;
- passengers and the public: evolving trust, expectations, and perceptions of risk vs. benefit.

Correctness in this domain is not just “did the car follow the rules of the road?” but “is this system, in this city, under these conditions, safe enough to operate driverlessly?”, which is a tacit institutional judgement.

### 17.1.3 Boundary Design: From Onboard Autonomy to Human Oversight

#### Onboard boundary: the vehicle as an explicit agent

Within the vehicle, the autonomy stack operates as an explicit agent:

- It perceives the environment, predicts others' behaviour, and plans and executes a trajectory in real-time.
- In driverless service, there is no human in the loop for routine control; the system is authorised to actuate steering, acceleration, and braking directly.
- Safety fallbacks (e.g. minimal-risk conditions, safe stop) are encoded explicitly and triggered when the system detects uncertainties or failures.

In FRFP terms:

- many low-level and mid-level decisions are fully automated within  $N \times E_{\text{Waymo}}$ ;
- correctness for these decisions is delegated to explicit semantics (control algorithms, safety monitors, redundancy mechanisms).

#### System-level boundary: fleet operations and governance

Above the vehicle level, there are stronger boundaries:

- **Operational Design Domain (ODD)**: driverless operation is restricted to predefined regions, weather conditions, and scenarios where the system has been validated.
- **Fleet operations centre**: remote staff can monitor vehicles, respond to unusual events, and provide high-level guidance or issue commands (e.g. instructing the vehicle to pull over and stop).
- **Safety release process**: major expansions (e.g. new cities, night driving, highway segments) go through internal review and, in some jurisdictions, explicit regulatory approval.

These structures place the key correctness boundary at:

- decisions about where and when the system may operate driverlessly;
- how incidents and anomalies are handled at the fleet and governance level.

This matches FRFP's principle that high-level correctness for long-horizon, high-stakes systems must remain with human and institutional actors in  $T_{\text{mobility}}$ , even if many sub-decisions are fully automated.

### 17.1.4 Protocol-Level Properties in FRFP Terms

#### Z.11.1 Explicit Semantics and Strong Evaluation

Compared to many AI systems, autonomous driving has relatively strong, structured explicit semantics:

- Rules of the road, traffic regulations, and physical constraints define a clear set of admissible behaviours at the vehicle level.
- Safety metrics (e.g. crashes per million miles, policy-violation rates, near-miss statistics) can be defined and tracked at scale.
- Simulation and replay allow systematic stress-testing of the autonomy stack against rare or adversarial scenarios.

For Waymo:

- large-scale data collection and years of testing have created rich explicit datasets;
- incident and disengagement logs support post-hoc evaluation and refinement of explicit pipelines;
- safety reports and external analyses provide some transparency on performance and limitations.

In FRFP language, the mapping from sensor data and maps to control actions has a well-defined semantics constrained by physics and law, and there is an ongoing, explicit evaluation apparatus to test it.

#### Z.11.2 ODD as an Explicit Safe Horizon

Waymo's use of Operational Design Domains (ODDs) is a concrete instance of FRFP's safe-horizon concept:

- The system is only allowed to operate driverlessly in regions, times, and conditions where it has been validated (e.g. specific parts of Phoenix and SF, typical weather, certain speed limits).
- Expansion of the ODD is gradual and gated: new neighbourhoods, nighttime operation, or more complex intersections are added only after extensive testing.
- When conditions fall outside the ODD (e.g. extreme weather, severe construction changes), the system may restrict service, request remote assistance, or execute safe stops.

From an FRFP perspective:

- the ODD defines an explicit horizon in state-space: outside this region, the autonomy stack is not authorised to operate without additional tacit oversight;
- this enforces a structural limitation on how far explicit autonomy can run before requiring institutional re-grounding (via testing, recalibration, and regulatory review).

### Z.11.3 Incident Handling and Feedback Loops

Waymo’s operational model incorporates explicit feedback mechanisms:

- Any incident or unusual behaviour (e.g. road blockage, confused negotiation with other vehicles, passenger complaints) is logged.
- Engineers can replay sensor data in simulation, examine why the behaviour occurred, and adjust perception, prediction, or planning modules.
- Policy changes (e.g. more conservative behaviour at certain unprotected left turns) can be rolled out fleet-wide.

In FRFP terms:

- each incident creates a run segment in  $N \times E_{\text{Waymo}}$  that is then subjected to tacit evaluation by safety teams;
- this evaluation updates both explicit pipelines (software) and tacit states (risk understanding, operating procedures);
- correctness is thus enforced at the institutional boundary, with explicit artefacts feeding into a dynamic governance loop.

## 17.1.5 Institutional and Societal Semantics

### Regulatory and city-level governance

Waymo operates within a complex regulatory landscape:

- State motor-vehicle regulators and public-utilities commissions grant permits for testing and driverless deployment.
- City authorities influence where vehicles may pick up and drop off, and how they interact with emergency services and road closures.
- Incidents are scrutinised by regulators, city officials, and the public, shaping tacit perceptions of safety.

FRFP views these institutions as operators on the population-level tacit state  $T_{\text{mobility}}^{\text{POP}}$ :

- approvals, suspensions, and rule changes update beliefs about acceptable risk and control;
- public hearings and media coverage feed back into Waymo’s own safety posture and expansion plans;
- over time, a shared tacit understanding of what “safe enough” autonomy means in different cities is gradually constructed.

### Public trust and behavioural adaptation

Passengers and other road users form tacit beliefs about Waymo vehicles:

- riders experience the cars as cautious but competent; many report high trust after a few trips, while others remain sceptical;
- human drivers learn to anticipate Waymo behaviour (e.g. its caution at narrow passes or merges);
- incidents—even rare ones—can cause sharp updates in public opinion, prompting calls for stricter oversight or pauses.

These dynamics matter for FRFP because they shape:

- how often and where people choose to ride driverless vehicles;
- how regulators balance innovation vs. safety;
- how institutions design future autonomy policies and infrastructures.

#### 17.1.6 Counterfactual Pitfalls and FRFP Alignment

Waymo’s trajectory can be contrasted with hypothetical or real failure modes:

- **Overly broad ODD.** A less cautious system might claim nationwide operation in all conditions, leading to frequent boundary violations and unsafe behaviours. Waymo’s constrained ODDs are a partial FRFP-aligned safeguard.
- **Weak incident feedback.** Failing to systematically log and analyse incidents would erode the link between explicit artefacts and tacit safety judgements. Waymo invests heavily in these loops.
- **Purely metric-driven governance.** Over-reliance on disengagement statistics or proprietary safety metrics, without robust external scrutiny, would be an instance of explicit-only governance. Waymo’s external reporting and regulatory engagement, while not perfect, move toward a more FRFP-consistent institutional layer.

Nonetheless, FRFP would highlight ongoing open questions:

- How well do current metrics capture tail risks and rare scenarios?
- Are city-level ODD boundaries and fleet behaviours transparent and contestable enough for the public and regulators?
- How should responsibility be allocated between Waymo, regulators, and riders in edge cases and accidents?

These questions sit squarely in tacit and institutional semantics, beyond what explicit models can resolve.

### 17.1.7 Lessons for FRFP and Long-Horizon Autonomy

Waymo's driverless service illustrates several key FRFP themes:

1. **Explicit autonomy can work within strong structural constraints.** Carefully designed explicit semantics (perception, planning), validated in limited ODDs, can support safe autonomous operation for many trips and miles.
2. **ODDs are practical safe horizons.** Geographic and situational constraints act as explicit safe horizons on where the autonomy stack is authorised to operate, aligning with FRFP's call for bounded runs in high-stakes domains.
3. **Incidents must feed into tacit governance.** Logging, simulation, and human safety reviews turn each anomaly into an opportunity to update both models and institutional understanding, rather than trusting explicit systems blindly.
4. **Institutional non-automation remains.** Decisions about expansion to new cities, policies for interacting with emergency services, and responses to public concern cannot be automated; they require human and institutional judgement.
5. **Formal FRFP perspective helps generalise autonomy design.** By seeing Waymo not just as a technical stack but as an FRFP system with explicit/tacit and institutional layers, we gain a template for evaluating and designing other long-horizon autonomous services (delivery robots, industrial fleets, etc.).

In this sense, Waymo is a high-profile example of explicit AI autonomy that has achieved non-trivial real-world deployment, while still depending essentially on FRFP-style boundaries, safe horizons, and institutional governance to remain acceptable and safe over long horizons.

## Chapter 18

# FRFP Case Study: Recommenders, Machine Translation, and ChatGPT-like Systems

## 18.1 FRFP Case Study (Platforms / Everyday Use): Recommenders, Machine Translation, and ChatGPT

This appendix treats three families of widely deployed AI systems as a single FRFP-style case study:

1. large-scale recommender systems (e.g. YouTube, Netflix, Amazon);
2. neural machine translation (NMT), exemplified by Google Translate;
3. large language models (LLMs) in general-purpose tools like ChatGPT.

These systems are striking AI successes:

- Recommenders and ranking models drive much of the engagement and revenue of modern platforms, and increasingly mediate what people watch, read, and buy.
- NMT (e.g. Google Neural Machine Translation, GNMT) has moved web translation from brittle phrase-based systems to fluent, sentence-level neural models across 100+ languages.
- ChatGPT and similar LLM tools now have hundreds of millions of weekly users and process billions of prompts per day, becoming a general “language layer” for everyday tasks.

Unlike AlphaFold or DR screening AI, these systems live directly in the everyday cognitive environment of billions of people, which makes FRFP’s explicit/tacit and population-level lenses especially important.

### 18.1.1 Explicit and Tacit Spaces for Everyday AI Platforms

#### Explicit space ( $E_{\text{platform}}$ )

Across these systems, explicit artefacts include:

- user interaction logs: clicks, watch time, purchases, queries, prompts, translations;
- content and item representations: videos, products, web pages, sentences, embeddings, token sequences;
- model parameters: recommender networks, translation models, LLM weights;
- outputs: ranked lists of items, translated text, generated responses;
- metrics: CTR, dwell time, retention, BLEU/COMET scores, engagement, satisfaction proxies.

These are all explicit structures in ( $E_{\text{platform}}$ ), *manipulated by training and serving pipelines in* ( $N \times E$ ).

## **Tacit space ( $T_{\text{platform}}$ )**

Tacit space contains:

- users’ beliefs, preferences, habits, and values;
- creators’ and merchants’ strategies in responding to platform incentives;
- engineers’ and product managers’ intuitions about trade-offs (e.g. diversity vs. engagement, safety vs. expressivity);
- regulators’ and the public’s evolving norms about fairness, safety, and acceptable influence;
- workers’ practices when using tools like MT and ChatGPT in their jobs.

Platform correctness—whether these systems are beneficial, safe, fair, and aligned with human purposes—ultimately resides in this tacit and institutional space, not in any single metric inside ( $E_{\text{platform}}$ ).

## **18.1.2 Large-Scale Recommender Systems**

### **Z.12.1 Overview**

Modern recommender systems drive:

- video and content discovery (YouTube, TikTok, Netflix);
- product discovery and advertising (Amazon, e-commerce, app stores);
- social feeds and news ranking (various platforms).

They have evolved from nearest-neighbour and matrix factorisation methods to deep, sequence-based and graph-based models that learn rich user/item representations.

Explicit objectives typically blend:

- short-term engagement (clicks, watch time);
- long-term retention and revenue;
- sometimes additional constraints (diversity, safety rules, content policies).

Field experiments show that recommender changes can significantly affect what people consume, with ambiguous effects on diversity and welfare.

### Z.12.2 FRFP view: explicit pipelines and population-level tacit states

From an FRFP standpoint:

- The explicit mapping ( $f_{\text{rec}} : \text{context, history} \rightarrow \text{ranked items}$ ) *is optimised for platform metrics, not directly for user welfare*.
- Users' tacit states are updated as they consume recommended content: beliefs change, habits crystallise, preferences drift; creators and advertisers adapt their behaviour in response to what the system rewards.
- The platform's institutional operators (product teams, policy teams, regulators) shape objective functions and constraints, but often through explicit metrics rather than direct control of tacit outcomes.

This is a classic FRFP configuration where:

- explicit systems operate at massive scale and high frequency;
- tacit and institutional effects (on public discourse, attention, markets) accumulate over long horizons;
- governance is largely metric-driven, risking explicit-only control over complex population-level phenomena.

### Z.12.3 Boundaries and safe horizons

Key boundary questions:

- How much of a user's informational diet should the recommender effectively control?
- To what extent are editorial and civic responsibilities delegated to optimisation over engagement metrics?
- How are long-horizon externalities (polarisation, addiction, market power) fed back into objective design?

FRFP would recommend:

- explicit guardrails and quotas (e.g. diversity constraints, exploration budgets, limits on reinforcement of recent behaviour);
- safe horizons in user modelling (e.g. bounded memory, reset mechanisms, opportunities for users to re-declare goals and interests);

- institutional oversight bodies that treat recommender optimisation as a socio-technical process, not purely a metric-maximisation problem.

Current practice partially reflects this (e.g. “responsible recommender” research on diversity and sustainability), but FRFP suggests a more formal architectural treatment.

### 18.1.3 Neural Machine Translation (NMT)

#### Z.12.4 Overview

Google Neural Machine Translation (GNMT) and successors replaced phrase-based MT with deep neural sequence-to-sequence models using attention, and later Transformer-based models, enabling more fluent, context-aware translations.

By the late 2010s:

- Google Translate supported 100+ languages, many via a single multi-lingual NMT model;
- NMT reduced error rates substantially (e.g. claims of  $\sim 60\%$  error reduction on some benchmarks when first deployed);
- quality has continued to improve, particularly for high-resource language pairs.

NMT is now woven into browsers, phones, and productivity apps, enabling cross-lingual communication at global scale.

#### Z.12.5 FRFP view: explicit semantics with clear but limited ground truth

NMT is closer to AlphaFold than to recommenders in one sense:

- the task—map a source sentence to an adequate target translation—has relatively clear explicit semantics, even though nuance and pragmatics complicate evaluation;
- there is abundant training data (parallel corpora) and usable automatic metrics (BLEU, COMET, etc.) as proxies for human quality;
- human evaluation remains the ultimate tacit arbiter, especially in professional or high-stakes translation.

FRFP interpretation:

- the mapping ( $f_{\text{NMT}} : \text{source text} \rightarrow \text{target text}$ ) *has strong but not perfect explicit semantics; some meaning, possibly*
- everyday use (tourism, casual chat, rough reading) often places correctness de facto in the model, with users accepting outputs without deep scrutiny;
- professional translators treat NMT as a drafting tool, keeping final correctness in tacit judgement—an FRFP-aligned boundary.

### Z.12.6 Boundaries by use case

FRFP would distinguish:

- **Low-stakes, everyday use:** using NMT to get the gist of a foreign-language web page; explicit semantics are good enough; tacit oversight is light but acceptable.
- **High-stakes or formal contexts:** legal contracts, medical instructions, diplomatic communication; here, NMT outputs should be treated as drafts needing human verification, not final artefacts.

Misuse arises when the boundary is blurred—for example, when institutions rely on raw MT output for official communications without adequate review. FRFP suggests organisational policies that encode these boundaries: which uses can be automated, which require human sign-off, and how safe horizons are maintained (e.g. when domain drift or new jargon makes previous evaluation obsolete).

## 18.1.4 ChatGPT and General-Purpose LLMs

### Z.12.7 Scale and role

ChatGPT-style LLMs are now general-purpose language tools:

- OpenAI reports ~700–800 million weekly active users in 2025 and 2–2.5 billion prompts per day.
- Usage spans everyday personal tasks, education, writing, coding, and work support; a recent study finds ~75% practical guidance, information seeking, and writing.

Unlike recommenders or MT, LLMs are:

- open-domain;
- generative rather than purely translational or ranking-based;
- increasingly integrated into tools that can act (agents, office suites, browsers).

### Z.12.8 FRFP view: extremely broad explicit semantics, strong tacit coupling

From an FRFP standpoint:

- the explicit mapping ( $f_{\text{LLM}} : \text{prompt, context} \rightarrow \text{text (and actions)}$ ) *is learned from huge corpora and reflects*
- unlike NMT or DR screening, there is no single task or metric: correctness is highly context-dependent and often normative;

- hallucination, bias, and value misalignment are structural, not incidental, features of this explicit mapping.

This makes the explicit/tacit split especially delicate:

- users can easily over-assign correctness authority to the system because it is fluent and broadly knowledgeable;
- tacit skills (search, critical reading, drafting) may atrophy if large portions of cognitive labour are repeatedly outsourced;
- organisational processes may start to treat LLM outputs as de facto truth or policy guidance, even where tacit judgement is essential.

### Z.12.9 Boundaries and safe horizons for LLM use

FRFP suggests several structural design choices:

- **Role labelling:** treat LLMs as assistants (drafting, summarisation, brainstorming), not as oracles or autonomous decision-makers, except in very narrow, well-specified domains.
- **Human-in-the-loop boundaries:** require human endorsement for filings, public statements, and high-stakes decisions drafted with LLM assistance (cf. the *Mata v. Avianca* case study).
- **Safe horizons in workflows:** limit how far LLMs can drive multi-step processes without human checkpoints (e.g. code deployment, legal filings, medical advice); design agent architectures where each major boundary event is explicitly approved.
- **Institutional oversight:** treat adoption of LLMs in organisations as an FRFP-level change, requiring policies, training, and governance, not just tool rollout.

At the population level, FRFP also worries about:

- epistemic effects (LLMs as a default interface to knowledge);
- labour and skill effects (who gains or loses tacit capabilities);
- cultural and linguistic homogenisation (if models favour high- resource language norms).

These require institutional operators (educational systems, media, policy) to treat LLM deployment as a long-horizon socio-technical transformation, not just a productivity boost.

### 18.1.5 Cross-Cutting FRFP Lessons from Platform AI

Taken together, recommender systems, NMT, and ChatGPT / LLMs illustrate:

1. **Explicit strength is uneven across tasks.** NMT has relatively clear semantics and ground truth; recommenders and LLMs operate in domains where correctness is entangled with tacit and normative judgements.
2. **Population-level effects dominate.** These systems act not just on individual users but on entire populations’ attention, beliefs, and workflows; FRFP’s population- level semantics and institutional operators are essential to reason about them.
3. **Boundaries are often informal or implicit.** Everyday users rarely have clear cues about when to trust or distrust MT and LLM outputs, or about the objectives behind recommenders. FRFP argues for explicit boundary design and user-facing semantics.
4. **Safe horizons are rarely explicit but deeply needed.** Long-horizon reliance on recommenders and LLMs without reset or re-grounding mechanisms risks drift in tacit skills, norms, and epistemics. NMT is somewhat safer here because tasks are local and evaluation is easier, but high-stakes use still needs boundaries.
5. **Institutional non-automation is central.** Platform governance, media regulation, and organisational policy cannot be replaced by optimisation over engagement or proxy metrics; they require FRFP-style tacit judgement and multi-stakeholder oversight.

These platform-scale successes thus function in the FRFP monograph as counterpoints to purely “failure” case studies: they show that explicit AI can be enormously powerful and useful, but also how much of its real-world impact is governed—for better or worse—by the tacit and institutional semantics in which it is embedded.

## Part II

# Patterns, Playbooks, and Metrics

## Chapter 19

# Design Patterns and Anti-Patterns in FRFP Systems

This chapter distils the case studies of Volume II into a small set of *design patterns* and *anti-patterns* for FRFP systems. The goal is a reusable pattern language: a way to recognise structural moves that tend to go well or badly across domains, and to apply those moves when designing new systems.

We write patterns in the same vocabulary as Volume I:

- explicit space  $E$  and explicit pipelines in  $N \times E$ ;
- tacit space  $T$  and tacit states of humans and institutions;
- boundaries where explicit artefacts cross into tacit endorsement;
- safe horizons that bound how far runs can proceed without re-grounding;
- institutional operators that shape which runs are admissible.

Each pattern is illustrated (implicitly or explicitly) by one or more case studies in this volume: AlphaFold, DR screening AI, Waymo, Watson, Zillow, COMPAS, Project Vend, Mata v. Avianca, the London Whale, and the platform-scale systems.

### 19.1 Pattern P1: Narrow, Strong Explicit Semantics

**Name.** Narrow, Strong Explicit Semantics.

**Intent.** Restrict the AI system to tasks where the mapping it implements in explicit space is well-defined, externally evaluable, and closely aligned with the tacit notion of correctness for that task.

## Structure

- Choose a task with a clear, domain-accepted definition:
  - e.g. “predict protein 3D structure from sequence”, “detect more-than-mild diabetic retinopathy from a fundus image”, “classify a stop sign”.
- Ensure ground truth is available at scale: experimental structures, specialist-graded images, labelled scenarios, etc.
- Define explicit metrics that are good proxies for tacit correctness: RMSD, sensitivity/specificity, error rates on held-out data.
- Train and evaluate models *only* within this well-defined semantic scope.

## Examples

- AlphaFold: sequence  $\rightarrow$  static structure, evaluated in CASP.
- DR screening AI: fundus image  $\rightarrow$  mtmDR/no-mtmDR/ungradable, evaluated against masked specialist grading.
- Core perception modules in Waymo: sensor data  $\rightarrow$  object detections and lane geometry.

## Consequences

- Explicit metrics track the domain’s tacit notion of correctness.
- Failures are mainly statistical (noise, distribution shift), not semantic (wrong task).
- It is easier to reason formally about safe horizons, because the task semantics do not mutate as the system scales.

## When to use

- In high-stakes domains where a narrow, well-defined subtask is available.
- As the foundation on top of which more complex, tacit-heavy workflows are built.

## 19.2 Pattern P2: Explicit Uncertainty and “I Don’t Know”

**Name.** Explicit Uncertainty and “I Don’t Know”.

**Intent.** Design explicit pipelines so that they can express when they are unreliable, and route those cases back into tacit space instead of forcing a brittle decision.

## Structure

- Augment outputs with explicit uncertainty or quality measures:
  - per-residue confidence scores (pLDDT, PAE);
  - image-quality flags and “ungradable” outcomes;
  - confidence intervals, anomaly flags, or out-of-domain detection.
- Treat “I don’t know” or “ungradable” as first-class outputs, *not* as errors.
- Define routing rules: when uncertainty exceeds a threshold, control passes to human tacit judgement or to a different, more conservative pipeline.

## Examples

- AlphaFold’s confidence scores and PAE maps: users interpret low-confidence regions differently from high-confidence cores.
- DR screening AI’s “ungradable” outcome: poor images trigger referral rather than a false negative.
- Waymo’s safe-stop behaviours when perception is degraded or conditions leave the ODD.

## Consequences

- Reduces catastrophic errors in regions where explicit semantics are fragile.
- Forces boundary events back into  $T$  when  $E$  loses grip.
- Makes it easier for institutional operators to specify when AI may or may not be used autonomously.

## When to use

- Whenever models will unavoidably face out-of-distribution cases or ambiguous inputs.
- In any domain where false confidence is more dangerous than equivocation (law, medicine, finance, safety-critical control).

## 19.3 Pattern P3: Boundary-Centred Interfaces

**Name.** Boundary-Centred Interfaces.

**Intent.** Design systems so that boundaries—points where humans endorse or commit—are explicit, instrumented, and supported by the right information, instead of being implicit in a UI or process.

## Structure

- Identify boundary events: filings, orders, diagnoses, trade approvals, deployments, policy adoptions.
- For each boundary, specify:
  - who is authorised to endorse;
  - what artefacts they see (raw data, model outputs, uncertainty, alternatives);
  - what gets logged (identity, time, context, rationale).
- Design UIs and workflows around boundaries: the interface foregrounds the boundary decision, not just model outputs.
- Treat signatures and approvals as semantic operations: transitions from “proposal” to “endorsed”, visible in the system state.

## Examples

- DR screening workflows: AI result → documented screening status / referral.
- Waymo: system-level decisions about where and when to operate driverlessly, with explicit ODD declarations and permits.
- The legal case-study counterfactuals: AI-generated drafts, but human-verified citations at filing.

## Consequences

- Makes it harder for AI systems to silently capture correctness.
- Creates a clear audit trail of tacit endorsements.
- Supports institutional oversight: committees can review boundary events, not just raw logs.

## When to use

- For any system affecting rights, money, safety, or scientific claims.
- Wherever existing professional norms already treat certain actions (signing, prescribing, executing a trade) as serious commitments.

## 19.4 Pattern P4: Engineered Safe Horizons

**Name.** Engineered Safe Horizons.

**Intent.** Make the limits of a system’s operation in time, scale, and domain explicit and enforce them with mechanisms that force re-grounding in tacit judgement.

## Structure

- Define horizons along three axes:
  - temporal (how long between thorough reviews);
  - scale (how many cases, trades, patients, rides, or items affected);
  - domain (how far from the context where the system was validated).
- Associate each horizon with triggers: caps, trial endpoints, or conditions under which operation must pause or shift mode.
- Define re-grounding procedures: audits, revalidation, redesign, or human-only periods.
- Use phase-gated deployment: pilot → limited expansion → broader roll-out, with explicit go/no-go decisions between phases.

## Examples

- DR screening AI: yearly re-screening intervals; post-market surveillance; limits on indications and camera types.
- Waymo: operational design domains (cities, weather, road types); gradual expansion; service suspension in out-of-ODD conditions.
- Scientific workflows with AlphaFold: predictions feed into experiments; no deployment of predictions as clinical facts without separate validation.

## Consequences

- Reduces the risk of slow, unnoticed drift.
- Forces institutions to periodically reflect and adjust.
- Provides natural points to change objectives, models, and governance in light of new evidence.

## When to use

- For any long-horizon, high-impact system (clinical programmes, trading desks, platform-wide algorithms).
- Especially when scaling pilots to full production.

# 19.5 Pattern P5: Governance as a First-Class Operator

**Name.** Governance as a First-Class Operator.

**Intent.** Treat governance as a semantic layer with its own operators on tacit states and admissible runs, not as an afterthought described only in prose or policy PDF files.

## Structure

- Identify governance bodies: internal committees, external regulators, boards, professional societies, ethics panels.
- For each, specify:
  - what information they see (metrics, incidents, case studies, whistleblower reports);
  - what operations they can perform: approve, suspend, restrict, mandate changes, redefine objectives;
  - how frequently they act and under what triggers.
- Ensure governance is structurally distinct from the teams that build and operate the system, even if they share information.
- Model governance decisions as changes in:
  - boundaries (what AI can/cannot decide);
  - horizons (caps, trial lengths, ODDs);
  - acceptable explicit tasks and objectives.

## Examples

- FDA and similar regulators for DR screening AI: explicit indications-for-use, performance targets, and post-market requirements.
- EMBL–EBI and the structural biology community around AF-DB: shared norms on how AlphaFold predictions should be used.
- City and state regulators for Waymo: permits, geographic constraints, and operational restrictions.

## Consequences

- Makes institutional non-automation visible and analysable.
- Gives a formal place to value judgements and political choices.
- Allows comparison of governance designs across domains.

## When to use

- Always, for systems that affect large populations or high-stakes domains.
- Even for internal tools, where a “governance board” might be a cross-functional risk committee rather than a regulator.

We now contrast these positive patterns with recurrent anti-patterns from the failure case studies.

## 19.6 Anti-Pattern A1: Advisory-in-Name-Only

**Name.** Advisory-in-Name-Only.

**Smell.** The system is officially “decision support”, but in practice its outputs are rarely overruled and often determine outcomes.

### Structure

- Model outputs are integrated into workflows as default decisions: risk scores that drive bail decisions, hiring scores that determine interview lists, recommendations that become offers.
- Human “overrides” are possible but rare, costly, or discouraged.
- Interfaces emphasise model scores, not uncertainty or alternative reasoning.

### Consequences

- De facto boundary capture: correctness migrates from tacit space  $T$  to the explicit pipeline  $N \times E$ , contradicting stated governance.
- Formal accountability remains with humans, but their practical control is diminished.
- Failures are attributed to “human error” despite being driven structurally by model design and integration.

### Examples

- COMPAS scores in courts which, while officially advisory, often significantly influence bail and sentencing.
- Algorithmic ranking in hiring or admissions that effectively determines the candidate pool.

### Refactoring toward patterns

- Introduce boundary-centred interfaces: require structured justification for following or overriding the tool, logged for review.
- Narrow the explicit task and ensure strong explicit semantics (Pattern P1) and uncertainty signalling (Pattern P2).

## 19.7 Anti-Pattern A2: Metric Worship and Explicit-Only Governance

**Name.** Metric Worship and Explicit-Only Governance.

**Smell.** Decision-makers treat dashboards and metrics as authoritative, with little attention to their semantics, limits, or unmodelled effects.

### Structure

- A small set of key metrics (VaR, engagement, accuracy, NPS, ROC-AUC) dominates governance conversations.
- Decision-makers focus on whether metrics are within target bands, not on whether the metrics still measure what matters.
- Qualitative and tacit signals (frontline feedback, user discomfort, external concerns) are dismissed as “anecdotal” or noisy.

### Consequences

- Model mis-specification and distribution shift go unnoticed until large failures occur.
- Governance is effectively automated: institutional operators become operators on metrics rather than on real-world states.
- Incentives skew toward metric gaming and cosmetic fixes.

### Examples

- Zillow Offers scaling based on backtested pricing metrics while ignoring selection effects and regime shifts.
- London Whale relying on a flawed VaR model that understated risk but was accepted because the numbers looked good.
- Watson Health emphasising demo performance and marketing narratives over prospective clinical evidence.

### Refactoring toward patterns

- Make governance a first-class operator (Pattern P5) with independent tacit input.
- Use metrics as inputs, not replacements, for boundary decisions.
- Introduce safe horizons (Pattern P4) tied to independent evaluations, not just internal metrics.

## 19.8 Anti-Pattern A3: One-Shot Validation, Infinite Deployment

**Name.** One-Shot Validation, Infinite Deployment.

**Smell.** The system passes an initial validation (pilot, test, audit) and then is deployed at scale indefinitely, with no explicit plan for revalidation or sunset.

### Structure

- Initial testing on a fixed dataset or limited pilot population.
- A one-time “go” decision leads to wide or permanent deployment.
- Monitoring reduces to passive metrics; there are no hard endpoints or caps that *force* a strategic review.

### Consequences

- The system gradually drifts away from the conditions under which it was validated.
- Accumulated effects (capital exposure, reliance, behaviour change) are not revisited until visible failure.
- It becomes politically and economically difficult to pause or decommission the system, even when evidence of misalignment appears.

### Examples

- Zillow Offers scaling rapidly after initial model success, without clear phase gates tied to external conditions.
- Long-lived use of risk scores or heuristic tools in law, finance, or medicine, long after their original evidence base has aged or the context has shifted.

### Refactoring toward patterns

- Replace one-shot validation with engineered safe horizons (Pattern P4): pilots with explicit endpoints, caps, and revalidation rules.
- Tie horizon extensions to independent evidence, not just internal enthusiasm.

## 19.9 Anti-Pattern A4: Workflow-Led Design Without Semantics

**Name.** Workflow-Led Design Without Semantics.

**Smell.** The system is specified in terms of phases, swimlanes, and diagrams, but there is no clear account of what the AI components in fact compute, or what states and boundaries mean.

## Structure

- Design documents emphasise:
  - who talks to whom;
  - which tickets are created;
  - which committees review what, when.
- There is little or no:
  - formal description of explicit tasks and mappings;
  - explicit modelling of tacit states;
  - representation of boundaries as state transitions.

## Consequences

- It is difficult to reason about correctness or failure modes: everything is in narrative form.
- Workflows become brittle when context changes; there is no semantic core to adapt.
- Governance devolves into checking boxes rather than evaluating runs in  $N \times E$ .

## Examples

- The initial V01 framework for long-horizon collaboration (the meta-case study): rich in work-flow concepts, poor in semantic structure.
- Many contemporary “AI governance” and “responsible AI” documents that specify processes but not semantics.

## Refactoring toward patterns

- Start from explicit and tacit spaces (Steps 2 and 3 in Chapter 6), not from swimlanes.
- Introduce explicit semantics (Pattern P1) and boundaries (Pattern P3) before designing work-flows.
- Treat governance as a semantic layer (Pattern P5), not just a flowchart box.

## 19.10 Anti-Pattern A5: Tacit Erosion by Over-Automation

**Name.** Tacit Erosion by Over-Automation.

**Smell.** Over time, humans become less able or willing to exercise independent judgement, because AI systems handle more of the cognitive work.

## Structure

- AI tools provide increasingly complete drafts, diagnoses, recommendations, or decisions.
- Human users are nominally “in the loop” but:
  - rarely override the system;
  - lose practice in underlying skills;
  - face time pressure that incentivises acceptance.
- Training and evaluation assume stable human capabilities, even as those capabilities change under AI influence.

## Consequences

- When AI systems fail or encounter novel cases, human backstops are weaker than assumed.
- Organisational claims that “humans are ultimately responsible” become less realistic over time.
- Tacit norms (e.g. about verification, scepticism) decay.

## Examples

- Lawyers relying on LLMs for research and drafting, leading to hallucinated citations and weakened verification norms (*Mata v. Avianca* and follow-ons).
- Potential long-run effects of heavy LLM usage on students’ writing, critical reading, or coding skills.

## Refactoring toward patterns

- Design boundaries and interfaces (Pattern P3) that require substantive human engagement at key points.
- Use safe horizons (Pattern P4) to limit how far AI-driven workflows can proceed without deeper human involvement.
- Explicitly model tacit skills and norms as part of  $T$  and treat their preservation as a design objective.

## 19.11 Using the Pattern Language

This chapter offers a compact language for describing FRFP systems in practice:

- Patterns P1–P5 describe structural moves that appear again and again in successful or robust systems.
- Anti-patterns A1–A5 capture recurrent failure modes across very different domains.

When analysing or designing a system, you can use this language in two ways:

1. **Diagnostic.** Walk through the anti-patterns and ask:

- Do we claim to be “advisory” while behaving otherwise (A1)?
- Are we governing primarily by metrics (A2)?
- Are we treating validation as a one-time gate (A3)?
- Are our documents workflow-led rather than semantic (A4)?
- Are we assuming tacit skills that may erode (A5)?

2. **Design.** For each pattern, ask how to instantiate it:

- What is the narrow, strong task we can safely automate (P1)?
- How do we express uncertainty and “I don’t know” states (P2)?
- Where are our boundaries, and how do we surface them in UI and logs (P3)?
- What are our safe horizons and re-grounding procedures (P4)?
- What does governance look like as an operator, not just a checklist (P5)?

## Chapter 20

# Domain-Specific FRFP Implementation Playbooks

This chapter gives concrete, domain-specific playbooks for implementing FRFP in three high-impact areas:

- (a) scientific and engineering research;
- (b) finance and risk management;
- (c) clinical practice and healthcare delivery.

Each playbook is structured as a short, practical guide:

- typical explicit and tacit spaces in the domain;
- recommended boundary design;
- safe-horizon choices;
- governance patterns;
- a compact implementation checklist.

The aim is not to exhaust domain detail, but to show how the general FRFP architecture can be concretely instantiated.

## 20.1 Playbook I: Scientific and Engineering Research

### 20.1.1 Typical Explicit and Tacit Spaces

**Explicit space**  $E_{\text{res}}$ .

- datasets, experimental logs, simulation outputs;
- code, models, pipelines, configuration files;
- manuscripts, preprints, lab notebooks, design docs;
- metrics and benchmarks (loss curves, accuracy, error bars).

**Tacit space  $T_{\text{res}}$ .**

- researchers’ intuitions about what questions are important or tractable;
- judgement about experimental design, artefact quality, and whether results “make sense”;
- field-wide norms about evidence, novelty, and acceptable risk;
- institutional incentives (career, funding, publication venues).

**20.1.2 FRFP Boundary Design in Research****Key boundaries.**

- **Research claims:** when a result moves from “internal exploration” to “external claim” (preprint, paper, patent).
- **Code and model releases:** when an internal tool becomes a public artefact or shared infrastructure.
- **Project commitments:** when a lab or organisation commits resources to a long-horizon programme (e.g. a multi-year line of work or experiment).

**Design moves.**

- Require that any *claim-bearing document* (paper, report, safety case) be explicitly marked as such and go through a human review pipeline distinct from AI-assisted drafting.
- Maintain explicit provenance: AI-generated analyses and text are tagged, so that reviewers know when they are endorsing machine-produced content.
- For code/model releases, implement a release gate where technical leads and domain experts jointly sign off on:
  - \* domain of validity;
  - \* known failure modes;
  - \* recommended and prohibited uses.

**20.1.3 Safe Horizons in Research Programmes****Temporal horizons.**

- Set explicit review intervals for large programmes: e.g. every 6–12 months, with an external or cross-team review of assumptions, goals, and methods.
- For high-risk directions (e.g. dual-use capabilities), enforce shorter horizons and require renewal of approval.

**Scale horizons.**

- Limit the number of downstream projects that can depend on a single unvalidated model or dataset before more thorough validation is performed.
- For AI tools used across a lab or organisation, cap adoption (e.g. number of users or critical workflows) until an initial evaluation phase is complete.

**Domain horizons.**

- Require explicit justification before using a tool outside the domain where it was validated (e.g. reusing a model trained for one regime in a qualitatively different one).
- Trigger re-grounding when key assumptions (hardware, scale, data sources, institutional context) change.

**20.1.4 Governance Patterns for Research**

- Establish an internal **FRFP board** (or adapt an existing research committee) with responsibility for:
  - \* approving high-impact projects and releases;
  - \* setting safe-horizon policies;
  - \* reviewing incident reports and near-misses.
- Integrate FRFP concepts into:
  - \* lab meeting culture (explicit discussion of boundaries and tacit judgement);
  - \* mentoring and training (how to use AI tools without eroding core skills);
  - \* collaboration agreements (roles and responsibilities for human and AI components).
- Align with external institutions: IRBs, safety review boards, professional societies and journals (e.g. disclosure of AI-generated analyses and code).

**20.1.5 Implementation Checklist (Research)**

For a research group or engineering team:

- (a) **Write down**  $E_{\text{res}}$  and  $T_{\text{res}}$  for your main projects.
- (b) **Mark boundaries:** which artefacts count as claims, releases, or commitments?
- (c) **Define safe horizons** for:
  - programme duration before review;
  - scale of deployment of internal tools;
  - reuse of models outside their training domain.

- (d) **Set up a lightweight FRFP board** (even 2–3 people) to approve boundary events and horizon extensions.
- (e) **Instrument provenance:** tag AI-generated analyses and text; log endorsements at boundaries.

## 20.2 Playbook II: Finance and Risk Management

### 20.2.1 Typical Explicit and Tacit Spaces

**Explicit space  $E_{\text{fin}}$ .**

- trades, orders, positions, and P&L records;
- risk metrics: VaR, stress tests, greeks, liquidity measures;
- client data, credit histories, transaction logs;
- models and algorithms: pricing models, risk engines, execution algos, ML forecasters;
- policies and limit tables implemented in code or configuration.

**Tacit space  $T_{\text{fin}}$ .**

- traders’ and portfolio managers’ market intuition;
- risk officers’ judgement about model limitations and stress scenarios;
- executives’ and boards’ beliefs about risk appetite and strategy;
- regulators’ evolving expectations;
- client and market participants’ perceptions of stability and fairness.

### 20.2.2 FRFP Boundary Design in Finance

**Key boundaries.**

- **Capital allocation and limits:** approvals for large positions, new strategies, or structural hedges.
- **Model changes:** implementation or retirement of risk/pricing models.
- **Regulatory and public disclosures:** risk reports, investor communications.

**Design moves.**

- Treat *model approval* as a major boundary event:
  - \* independent model validation;
  - \* parallel runs with legacy models;
  - \* sign-off by risk and business leadership.

- Make limit changes and waivers explicit boundary events: who approved, under what rationale, with what horizon.
- Distinguish between *hedging* and *directional* strategies in both code and governance, rather than relying on labels alone.

### 20.2.3 Safe Horizons in Financial Systems

#### Temporal horizons.

- Set review cycles for strategies (e.g. quarterly or annual deep dives) independent of short-term P&L.
- For complex or illiquid portfolios, require more frequent stress tests and scenario reviews.

#### Scale horizons.

- Cap notional exposure and risk measures per strategy or book until performance and model fit are demonstrated.
- Tie horizon extensions to both quantitative evidence and tacit review of market structure and counterparty behaviour.

#### Domain horizons.

- Restrict model use to markets, instruments, and regimes where assumptions hold; treat entry into new asset classes or liquidity regimes as new domains requiring fresh validation.
- For ML models, monitor for data drift and structural breaks (e.g. regime shifts, new regulations); trigger re-grounding.

### 20.2.4 Governance Patterns for Finance

- Define a **FRFP-aware risk committee** with authority over:
  - \* model approvals and retirements;
  - \* safe-horizon definitions for strategies;
  - \* limit frameworks and escalation procedures.
- Ensure clear separation of roles:
  - \* traders and quants design strategies and models;
  - \* independent risk validates and challenges;
  - \* the committee arbitrates and sets boundaries.
- Align with external regulators: treat regulatory stress tests and model-risk rules as institutional operators in FRFP terms, and design internal processes to be legible to them.

### 20.2.5 Implementation Checklist (Finance)

For a trading business, risk group, or fintech:

- (a) **Catalogue key models and metrics** in  $E_{\text{fin}}$ , including their intended semantics and data sources.
- (b) **Map tacit risk decisions** in  $T_{\text{fin}}$ : who actually decides on exposures, hedges, and new products.
- (c) **Define boundaries**:
  - model approval / retirement;
  - strategy launch / scale-up;
  - limit changes and waivers.
- (d) **Set safe horizons** for:
  - maximum exposure pre-review;
  - maximum time between independent stress-testing;
  - valid domain of each model.
- (e) **Establish or adapt a risk committee** to act as a governance operator with real authority over the above.

## 20.3 Playbook III: Clinical Practice and Healthcare Delivery

### 20.3.1 Typical Explicit and Tacit Spaces

**Explicit space**  $E_{\text{clin}}$ .

- electronic health records (EHRs), lab results, imaging data;
- clinical guidelines and order sets;
- AI models for:
  - \* diagnosis and screening (e.g. DR detection, imaging triage);
  - \* risk prediction (readmission, deterioration, sepsis);
  - \* workflow optimisation and resource allocation.
- decision-support outputs: alerts, risk scores, recommendations, triage flags, auto-draft notes.

**Tacit space**  $T_{\text{clin}}$ .

- clinicians’ tacit knowledge of patients, disease courses, local practice norms;
- nurses’ and allied staff’s understanding of workflows and practical constraints;
- institutional culture around safety, liability, and autonomy;

- patient trust, preferences, and understanding of care;
- regulator and payer expectations.

### 20.3.2 FRFP Boundary Design in Clinical Settings

#### Key boundaries.

- **Diagnostic and treatment decisions:** diagnoses entered, treatments ordered, prescriptions signed.
- **Screening and triage decisions:** who is referred, prioritised, or discharged.
- **Documentation and billing:** what goes into the official record and claims.

#### Design moves.

- Treat AI outputs as structured inputs to clinical reasoning, not as replacements for it: UIs should present recommendations alongside underlying data and uncertainty.
- For autonomous systems (like DR screening AI), make the AI's *scope* explicit: the specific question it answers, and what it does *not* decide (e.g. treatment).
- Require explicit clinician endorsement at key boundaries: entering a diagnosis, confirming a treatment plan, signing a discharge.

### 20.3.3 Safe Horizons in Clinical Programmes

#### Temporal horizons.

- For screening and risk-prediction programmes, define re-evaluation intervals (e.g. annually) for both:
  - \* model performance;
  - \* workflow and outcome impact (over- and under-treatment).
- For AI tools used at the bedside, schedule periodic clinician-survey cycles and qualitative feedback rounds.

#### Scale horizons.

- Roll out new AI tools in phases: pilot units → hospital-wide → system-wide, with clear criteria for progression.
- Limit their use in high-acuity or vulnerable populations until sufficient evidence is available.

#### Domain horizons.

- Do not deploy models outside the populations in which they were validated (e.g. different hospitals, countries, or demographic mixes) without adaptation and revalidation.

- Explicitly track changes in clinical practice patterns, technology (e.g. new imaging devices), and epidemiology that may invalidate prior validations.

### 20.3.4 Governance Patterns for Clinical AI

- Use or extend existing structures:
  - \* clinical governance and safety committees;
  - \* quality-improvement (QI) and patient-safety teams;
  - \* IRBs and ethics boards for research deployments.
- Give these bodies explicit FRFP responsibilities:
  - \* approve indications-for-use and deployment domains;
  - \* set safe horizons and monitoring plans;
  - \* review incidents and near-misses involving AI tools.
- Align with external operators: regulators (e.g. device approvals), payers (reimbursement), and professional societies (guidelines) as external FRFP operators that can change admissible runs.

### 20.3.5 Implementation Checklist (Clinical)

For a hospital, clinic, or health system:

- (a) **Inventory AI tools** in  $E_{\text{clin}}$ : what models exist, what they predict, and where they are used in workflows.
- (b) **Map tacit actors** in  $T_{\text{clin}}$ : clinicians, nurses, administrators, patients, regulators.
- (c) **Define boundaries**:
  - which decisions AI may take autonomously (if any);
  - which require clinician endorsement;
  - what gets written into the medical record.
- (d) **Set safe horizons**:
  - pilot sizes and durations;
  - review intervals for performance and impact;
  - domain limits (sites, populations, devices).
- (e) **Empower a clinical AI committee** to:
  - approve tools and indications;
  - oversee monitoring and incident review;
  - communicate with external regulators and payers.

## 20.4 Using the Playbooks

These playbooks are deliberately schematic. For each domain, they provide:

- a first pass at identifying  $E$ ,  $T$ , boundaries, horizons, and governance operators;
- concrete steps an organisation can take in months, not years, to bring existing or planned systems closer to FRFP alignment.

In practice, implementing FRFP will involve adapting these templates to local context and constraints. The case studies in Volume II provide detailed illustrations of how similar patterns have failed or succeeded in related systems. By combining the playbooks with the pattern language in Chapter II and the methodology in Chapter 6, practitioners can build domain-specific FRFP architectures that are both principled and operational.

## Chapter 21

# Measuring FRFP Health: Metrics, Indicators, and Checklists

This chapter offers a practical toolkit for assessing how well a system (or organisation) aligns with FRFP principles. It does *not* define a single scalar “FRFP score”. Instead, it gathers:

- structural indicators for explicit semantics, boundaries, safe horizons, and governance;
- simple maturity levels;
- concrete checklists you can apply to an existing or proposed system.

The intent is diagnostic and comparative: to help you see where a system is strong or weak in the FRFP sense, and where design or governance changes are most urgent.

### 21.1 Why FRFP Metrics Are Different

Traditional evaluation often asks: *How accurate is the model? How fast is the system?* FRFP asks additional questions:

- Is the explicit mapping the model implements well-defined and appropriate for the tacit function we care about?
- Where do humans or institutions actually endorse decisions, and are those boundaries meaningful?
- How far can the system run (in time, scale, and domain) before someone must re-ground assumptions?
- Which parts of governance are de facto automated, and which remain genuinely tacit?

FRFP “metrics” are therefore mostly *structural indicators* and *maturity assessments*, not just numbers you can compute from logs. They are intended to be used alongside conventional measures of performance and safety, not instead of them.

## 21.2 Core FRFP Indicator Families

We group indicators into five families:

- (a) explicit-task semantics;
- (b) boundary design and implementation;
- (c) safe horizons and dynamics;
- (d) governance and institutional operators;
- (e) tacit-skill and norm preservation.

For each family, we provide:

- key questions;
- example indicators;
- rough maturity levels.

### 21.2.1 Explicit Semantics Indicators

#### Key questions.

- Is there a clear, written description of what the system is trying to compute?
- Is there a plausible ground truth or evaluation procedure?
- Are failure modes understood as *semantic* (wrong question) or *statistical* (noise, data limits)?

#### Indicative signals.

- Presence of a concise “task definition” document that could be given to a sceptical domain expert.
- Availability of representative test sets or benchmarks that reflect real use, not just training data splits.
- Explicit list of assumptions and known *out-of-scope* cases.

#### Maturity levels.

**Level 0 (unclear):** Task definition is vague (e.g. “rank the best content”, “optimise risk-adjusted returns”), with no precise mapping or ground truth.

**Level 1 (local clarity):** Task is clearly defined for a narrow slice of cases, but the system is already used more broadly.

**Level 2 (aligned core):** Task is well-defined and most usage stays inside that scope; there is a known set of out-of-scope cases.

**Level 3 (strong semantics):** Task is sharply defined; there is robust evaluation and continuous checking that usage remains within validated semantics.

### 21.2.2 Boundary Indicators

#### Key questions.

- Can we name the moments where humans or institutions endorse or commit?
- Are those moments technically and procedurally real, or merely rhetorical?
- Does the system log and support those boundary events?

#### Indicative signals.

- Clear documentation of which decisions AI systems may take autonomously and which require human sign-off.
- User interfaces where boundary actions (approve, reject, refer, sign, file, deploy) are explicit and foregrounded.
- Logging that ties each boundary event to:
  - \* responsible agent;
  - \* information available at decision time;
  - \* relevant model outputs and uncertainties.

#### Maturity levels.

**Level 0 (implicit):** Humans are nominally “in the loop”, but in practice no clear boundary points exist; model outputs are simply followed.

**Level 1 (documented):** Policies describe boundaries (e.g. “doctor is responsible”), but systems and logs do not enforce or reveal them.

**Level 2 (implemented):** Boundary events are explicit in UI and logs; humans do endorse, but information quality and incentives may still be weak.

**Level 3 (robust):** Boundaries are explicit, instrumented, and aligned with incentives; external reviewers can reconstruct who endorsed what and on what basis.

### 21.2.3 Safe-Horizon Indicators

#### Key questions.

- How long can the system run without being deeply re-examined?
- How many people, assets, or decisions can it affect before someone is required to reconsider design assumptions?
- Are there explicit triggers to pause, review, or redesign?

**Indicative signals.**

- Existence of phase-gated deployment plans: pilots, limited roll-outs, full deployment.
- Formal review intervals for models and programmes (e.g. yearly clinical revalidation, quarterly strategy deep dives).
- Predefined caps on scale (exposure, user count, jurisdiction) before further governance action is required.

**Maturity levels.**

**Level 0 (unbounded):** Once deployed, the system runs indefinitely and scales freely; no planned re-grounding.

**Level 1 (reactive):** Reviews occur only after problems or complaints; no explicit horizons.

**Level 2 (scheduled):** Regular reviews and audits exist, but may not be tied to explicit caps or triggers.

**Level 3 (engineered horizons):** Safe horizons are explicit; hitting a horizon or trigger forces structured re-grounding and possible redesign.

**21.2.4 Governance and Institutional Indicators****Key questions.**

- Who can change the system’s objectives, boundaries, and horizons?
- What information do they see?
- Are they independent of the teams building and operating the system?

**Indicative signals.**

- Existence of a named governance body (committee, board, panel) with scope over the system.
- Published charters or terms of reference for that body.
- Evidence that governance decisions affect:
  - \* what the system is allowed to decide;
  - \* where and for whom it may operate;
  - \* when it must be paused or redesigned.
- Use of case studies and tacit evidence (frontline feedback, near-miss reports), not just aggregate metrics.

**Maturity levels.**

**Level 0 (absent):** No dedicated governance; product or line managers implicitly play this role.

**Level 1 (nominal):** A committee exists on paper but has limited information or authority; decisions remain driven by metrics and local teams.

**Level 2 (active):** Governance has real influence; it can halt, modify, or constrain deployments, but may lack a structured FRFP vocabulary.

**Level 3 (FRFP-aware):** Governance bodies explicitly reason in terms of  $E$ ,  $T$ , boundaries, and horizons; they are structurally independent and use both metrics and tacit input.

### 21.2.5 Tacit Skill and Norm Preservation Indicators

#### Key questions.

- Are human skills and norms being preserved, or eroded, by AI use?
- Do users retain the ability and habit of questioning the system?
- Is the organisation tracking these changes?

#### Indicative signals.

- Training and onboarding explicitly address:
  - \* when to trust and when to doubt AI outputs;
  - \* how to verify or override them;
  - \* examples of failures and near-misses.
- Periodic assessment of human skills (e.g. sample cases where staff must solve tasks without AI support).
- Monitoring of override rates and rationales: are humans still actually exercising judgement?

#### Maturity levels.

**Level 0 (ignored):** No attention to tacit skill or norm drift; AI is treated purely as a productivity boost.

**Level 1 (acknowledged):** Informal awareness and discussion, but no systematic measurement or design.

**Level 2 (monitored):** Some indicators (override rates, training, skill assessments) are tracked and inform local adjustments.

**Level 3 (managed):** Tacit preservation is a design goal; systems and workflows are explicitly shaped to maintain critical human capabilities over time.

## 21.3 An FRFP Maturity Snapshot

For quick diagnosis, you can summarise a system's FRFP posture in a table:

| Dimension          | Level 0  | Level 1–2           | Level 3           |
|--------------------|----------|---------------------|-------------------|
| Explicit semantics | unclear  | partial / local     | strong, validated |
| Boundaries         | implicit | documented / UI     | explicit, logged  |
| Safe horizons      | none     | ad hoc / scheduled  | engineered        |
| Governance         | absent   | nominal / active    | FRFP-aware        |
| Tacit preservation | ignored  | acknowledged / mon. | actively managed  |

The goal is not perfection, but awareness. Many real systems will be at Level 1 or 2 on most dimensions. The value of this snapshot is in:

- making gaps visible (e.g. strong semantics but weak horizons);
- helping prioritise interventions;
- providing a common language across technical and governance stakeholders.

## 21.4 Checklists

We now give three concrete checklists you can use directly:

- (a) a system design checklist (pre-deployment);
- (b) an operational monitoring checklist (post-deployment);
- (c) an institutional governance checklist.

Each question is essentially Yes/No (or “not applicable”); the aim is to trigger discussion and action, not to produce a numeric score.

### 21.4.1 System Design Checklist (Pre-Deployment)

#### (a) Task semantics

- i. Have we written down, in one paragraph, what function the system is supposed to approximate?
- ii. Is there any plausible ground truth or external evaluation procedure for that function?
- iii. Have we listed the main types of cases that are *out of scope*?

#### (b) Data and evaluation

- i. Do we have test data that reflects real deployment conditions, not just training distributions?
- ii. Are evaluation metrics aligned with what domain experts care about?

#### (c) Boundaries

- i. Have we identified the specific decisions the AI will make vs. those reserved for humans?

- ii. Are boundary events (approvals, sign-offs, filings, orders) explicitly represented in the design?
- iii. Will our logs capture who endorsed what, and when?

(d) **Uncertainty and abstention**

- i. Can the system say “I don’t know” or defer to humans when inputs are low-quality or out-of-domain?
- ii. Are those deferrals routed to appropriate tacit decision-makers?

(e) **Safe horizons**

- i. Have we defined initial limits on:
  - time (how long the system can run before a deep review)?
  - scale (how many users, patients, trades, etc.)?
  - domain (which segments or contexts)?
- ii. Are there explicit triggers for pausing, rolling back, or redesigning (performance drops, incident rates, context changes)?

(f) **Governance**

- i. Is there a named body or role with authority over this system’s deployment and evolution?
- ii. Have we agreed what information they will see and how often they will act?

(g) **Tacit preservation**

- i. Have we identified which human skills and norms are critical to preserve (e.g. verification, domain intuition)?
- ii. Does the design explicitly support those skills (e.g. by requiring review, offering raw data, or including training)?

## 21.4.2 Operational Monitoring Checklist (Post-Deployment)

(a) **Usage**

- i. Is actual use consistent with the intended task and domain?
- ii. Are there emerging use cases we did not anticipate?

(b) **Performance and drift**

- i. Are key performance metrics stable in the deployed setting?
- ii. Have we checked for distribution shift in inputs or outcomes?

(c) **Boundaries in practice**

- i. Are humans still making the decisions they are supposed to make, or has the AI effectively taken over?
- ii. Do override rates and patterns suggest genuine human judgement, or rubber-stamping?

- (d) **Horizon adherence**
  - i. Are we still inside the originally defined safe horizons (time, scale, domain)?
  - ii. If not, has any explicit decision been taken to extend them?
- (e) **Incidents and near-misses**
  - i. Are incidents and near-misses being logged and investigated?
  - ii. Do we see repeated patterns that suggest deeper design issues?
- (f) **Human feedback**
  - i. Are frontline users reporting concerns or workarounds?
  - ii. Have we incorporated their tacit knowledge back into design updates?
- (g) **External context**
  - i. Have regulations, norms, or competitive conditions changed in ways that affect admissible runs?
  - ii. Have we adjusted the system or governance in response?

### 21.4.3 Institutional Governance Checklist

- (a) **Mandate**
  - i. Does a governance body have an explicit mandate over the system (scope, powers, responsibilities)?
  - ii. Is that mandate documented and communicated to stakeholders?
- (b) **Information**
  - i. Does the body see both quantitative metrics and qualitative case reports?
  - ii. Are FRFP-relevant artefacts (boundaries, horizons, incident logs) part of its regular agenda?
- (c) **Authority**
  - i. Can the body actually change:
    - where and how the system is used?
    - what tasks it may perform?
    - when it must be paused or withdrawn?
  - ii. Are its decisions binding on operational teams?
- (d) **Independence and composition**
  - i. Is membership cross-functional (technical, domain experts, legal, ethics, affected stakeholders)?
  - ii. Is it sufficiently independent from the teams whose incentives are tied to system growth or metrics?
- (e) **Processes**

- i. Are there scheduled reviews (horizon checkpoints) as well as incident-driven meetings?
  - ii. Are decisions minuted, with rationale and links to specific evidence?
- (f) **External linkage**
  - i. Does the body interact with external operators (regulators, professional societies, public bodies) as needed?
  - ii. Are external developments (new standards, regulations, prominent failures elsewhere) fed back into governance?

## 21.5 How to Use This Chapter

In practice, the recommended workflow is:

- (a) Pick a concrete system (existing or proposed).
- (b) Use the *System Design Checklist* to assess its FRFP structure.
- (c) Assign rough maturity levels on each dimension (semantics, boundaries, horizons, governance, tacit preservation).
- (d) Identify 2–3 most concerning gaps.
- (e) Consult the pattern and anti-pattern chapter to choose design moves that address those gaps.
- (f) Revisit periodically with the *Operational Monitoring* and *Governance* checklists.

The metrics and indicators here are deliberately simple. They are meant to be used quickly in real organisations, as conversation starters and design aids, not as precise scientific instruments. Their value lies in making FRFP questions unavoidable: for any serious system, someone should be able to say, concretely, what the task is, where the boundaries are, how far it may run, who governs it, and how we will know if tacit control is being lost.

## Chapter 22

# Research Agenda and Open Problems

This chapter sketches a research agenda emerging from FRFP and the case studies in Volume II. It is intentionally short and selective: its goal is to identify a few tractable directions where theory, empirics, and institutional work can reinforce each other, rather than to catalogue every possible question.

We group open problems into four clusters:

- (a) foundations and theory;
- (b) empirical characterisation of long-horizon systems;
- (c) institutional and governance design;
- (d) tools and practice for FRFP-aligned development.

### 22.1 Foundations and Theory

#### 22.1.1 T1. Formal conditions for safe horizons

FRFP treats safe horizons as structural bounds on how far a run in  $N \times E$  may proceed without re-grounding in tacit judgement. Volume I defines this notion; Volume II shows qualitatively where horizon failures occur. We lack sharp theorems of the form:

Given a class of systems with properties  $A, B, C$ , any run of depth  $> D$  without re-grounding has probability at least  $p$  of entering an inadmissible region.

Open problems:

- characterise safe horizons in simplified models: bandit processes, Markov decision processes, or repeated games with partial monitoring;
- prove sufficient conditions under which periodic intervention by tacit operators  $T$  guarantees bounds on error or drift;
- analyse trade-offs between horizon length, intervention frequency, and performance.

### 22.1.2 T2. Population-level FRFP dynamics

Volume II informally treats populations of users, institutions, and systems. A more formal treatment would:

- model distributions over tacit states  $T$  (e.g. beliefs, skills, norms) for populations;
- define institutional operators acting on these distributions (platform policies, regulations, scientific norms);
- study fixed points, cycles, and instability modes when AI systems and institutions co-evolve.

Concrete questions:

- Under what conditions do recommender or LLM-mediated environments converge to narrow attractors in belief or preference space?
- How do different governance operators (e.g. content policies, audit regimes) alter these dynamics?

### 22.1.3 T3. Non-automation theorems at the institutional level

Volume I states non-automation results for certain classes of problems; Volume II extends this intuition to governance:

There is no purely explicit procedure that can fully replace tacit institutional judgement about acceptable socio-technical systems.

Open problems:

- formalise classes of governance tasks (e.g. acceptable-risk assessment, fairness adjudication) and show limits on their automation;
- connect these limits to impossibility results in social choice, mechanism design, or learning theory;
- identify precise boundaries where partial automation is safe or beneficial.

### 22.1.4 T4. Semantics for large foundation models

Foundation models and general-purpose LLMs have diffuse explicit semantics. We need better theoretical tools to:

- describe what these models are actually computing, beyond generic sequence prediction;
- relate this to task-specific semantics when they are fine-tuned or prompted for particular roles;
- characterise when and how their behaviour can be constrained by FRFP-style boundaries and safe horizons.

Open problems include:

- compositional semantics for prompted or tool-augmented models;
- formal models of hallucination and its inevitability under certain training regimes;
- conditions under which foundation models can safely serve as components inside FRFP architectures.

## 22.2 Empirical Characterisation of Long-Horizon Systems

### 22.2.1 E1. Measuring tacit-skill erosion and preservation

Volume II highlights tacit erosion as a recurring risk. Empirical work is needed to:

- measure how prolonged AI use affects human skills in law, medicine, engineering, and everyday tasks;
- disentangle short-term productivity gains from long-term dependence;
- evaluate design interventions (e.g. mandatory verification, occasional AI-free tasks) that preserve key skills.

This includes controlled experiments, longitudinal field studies, and instrumentation of real systems.

### 22.2.2 E2. Boundary capture in practice

Many case studies feature systems that are “advisory” on paper but decisive in practice. We currently lack good empirical measures of boundary capture.

Open questions:

- How often do humans override AI recommendations in different domains, and why?
- How do override rates evolve over time as trust and habit set in?
- What interface and incentive designs help maintain meaningful boundaries?

### 22.2.3 E3. Horizon creep and drift

Horizon creep—the gradual extension of time, scale, or domain beyond initial intentions—is visible in several failures, but rarely measured.

Research directions:

- define empirical indicators of horizon creep (expansion of user base, jurisdictions, asset classes, patient populations);
- correlate these with incident rates, near-misses, and qualitative signs of misalignment;
- study organisational dynamics that accelerate or resist horizon creep (competition, regulation, internal culture).

### 22.2.4 E4. Comparative case-study programmes

Volume II provides a handful of detailed case studies. A larger research programme could:

- build a curated library of FRFP-analysed systems across domains and scales;
- develop standardised coding schemes for patterns and anti-patterns;
- quantitatively test hypotheses such as: “systems with engineered safe horizons have lower incidence of catastrophic failure, controlling for domain and scale”.

This would turn FRFP from a primarily conceptual framework into a cumulative empirical research area.

## 22.3 Institutional and Governance Design

### 22.3.1 G1. FRFP-aware regulatory regimes

Regulators in finance, healthcare, transport, and digital platforms increasingly face AI systems with long-horizon effects. Open questions:

- How can regulatory frameworks be rewritten to explicitly address:
  - \* task semantics and domain of validity;
  - \* boundaries (who decides what);
  - \* engineered safe horizons;
  - \* institutional non-automation?
- What templates for permits, approvals, and ongoing reporting best capture these elements?
- How can regulators collaborate across domains using a shared FRFP vocabulary?

### 22.3.2 G2. Organisational FRFP architectures

Within organisations, there is a design space of governance architectures:

- centralised vs. federated AI risk committees;
- domain-specific vs. cross-cutting boards;
- internal vs. external oversight mechanisms.

Research questions:

- Which architectures are most effective at preventing anti-patterns (metric worship, advisory-in-name-only, horizon creep)?
- How can FRFP concepts be embedded in standard corporate processes (risk registers, audits, product reviews)?
- How do different incentive structures affect the success of FRFP-style governance?

### 22.3.3 G3. Professional norms and FRFP

Many domains (medicine, law, science, engineering) already have strong professional norms around judgement, evidence, and responsibility.

Open problems:

- How can FRFP concepts be translated into domain-native language and integrated into existing codes of practice?
- What training and certification regimes are needed for “FRFP-literate” professionals and leaders?
- How can professional societies act as FRFP operators, updating norms as AI capabilities change?

## 22.4 Tools and Practice

### 22.4.1 P1. FRFP-aware development and operations tools

Today’s MLOps and product tools are largely agnostic to FRFP structure.

Opportunities:

- development tools that:
  - \* encourage explicit task-definition and scope;
  - \* annotate model outputs with uncertainty and provenance;
  - \* make boundary events first-class objects.
- operations tools that:
  - \* track safe horizons and trigger reviews;
  - \* surface FRFP metrics to governance bodies;
  - \* log institutional decisions as part of the system state.

### 22.4.2 P2. Simulation environments for FRFP systems

Several case studies depend on long-horizon, population-level effects. We need simulation tools that:

- model interactions between AI systems, users, and institutions;
- allow testing of governance and boundary designs before deployment;
- support “what if” analysis: what happens if we change the ODD, screening interval, or content policy?

This suggests a line of work on FRFP-aware agent-based simulations and testbeds.

### 22.4.3 P3. Education, documentation, and patterns

Finally, there is practical work to do in:

- creating FRFP teaching materials for engineers, policy-makers, and students;
- collecting and publishing patterns and anti-patterns (as in Chapter II) in a form usable by practitioners;
- developing documentation standards that include FRFP elements (semantics, boundaries, horizons, governance) alongside conventional API or model docs.

## 22.5 Closing Remarks

FRFP was motivated by a gap: the absence of a formal language for reasoning about long-horizon human–AI systems that are simultaneously technical, social, and institutional. Volume I filled part of that gap at the theoretical level; Volume II tested the language against concrete cases.

The research agenda outlined here points in both directions:

- inward, toward deeper formal results on safe horizons, population dynamics, and non-automation;
- outward, toward empirical measurement, institutional design, and practical tools.

The hope is that FRFP can function as a common scaffold across these efforts: a way for theorists, engineers, clinicians, financiers, regulators, and others to talk about the same objects—explicit spaces, tacit spaces, boundaries, horizons, and operators—even as they work at very different levels of abstraction and practice.

# Conclusion: What FRFP Learns from Its Case Studies

Volume II has treated a diverse set of systems as FRFP objects: trading desks and oncology clinics; hiring engines and risk scores; vending machines and robotaxis; protein-structure predictors and legal chatbots. Despite the diversity, a small number of structural themes recur with striking regularity. This conclusion distils those themes into a set of lessons for designers, institutions, and theorists.

## Lesson 1: Explicit Success Does Not Guarantee Tacit Safety

Many of our failure case studies looked superficially impressive in explicit metrics:

- Watson for Oncology passed internal benchmarks and demos;
- Zillow’s pricing models performed well on backtests;
- Amazon’s recruiting engine optimised predictive accuracy on historical data;
- COMPAS satisfied certain calibration metrics;
- the CIO’s VaR model produced numbers within nominal limits.

Yet in each case, the explicit semantics were misaligned with tacit notions of correctness:

- Watson’s training on synthetic cases and idiosyncratic expert judgements did not capture the real distribution of oncology cases;
- Zillow’s models ignored selection effects and regime shifts that local experts understood;
- COMPAS could not simultaneously meet all fairness desiderata in the presence of base-rate differences;
- VaR metrics failed to encode key tail risks and liquidity constraints.

By contrast, AlphaFold and DR screening AI succeed precisely where explicit semantics are strong and well-matched to tacit needs. The mapping from sequence to structure, or from retinal image to a clinically defined threshold, is compact enough to be captured by explicit models, and ground truth is abundant.

FRFP’s first conclusion is therefore:

*Explicit performance metrics are not proxies for tacit safety or correctness. They are informative only when the explicit task is well-specified, the data and model semantics match the tacit function we care about, and the limits of that match are understood.*

## Lesson 2: Boundaries are Designed, Not Assumed

A central claim of Volume I is that correctness lives at boundaries: at the points where explicit artefacts are endorsed, rejected, or transformed by agents in tacit space. Volume II shows how often these boundaries are treated as rhetorical rather than structural. In many failures:

- systems are labelled “decision support” while functioning as de facto decision-makers (COMPAS, recruiting engines, recommenders);
- lawyers or clinicians treat AI outputs as if they were vetted results (Watson, *Mata*) rather than hypotheses;
- risk metrics and model outputs replace human assessment at the point of commitment (Zillow offers, London Whale limits).

In successful or more robust systems:

- DR AI explicitly reverts to human care when images are ungradable;
- AlphaFold outputs are clearly presented as predictions with confidence, not as experimental facts;
- Waymo draws a sharp operational boundary via its ODD, with human and institutional sign-off governing any expansion.

Our second conclusion:

*Boundaries between explicit systems and tacit endorsement must be deliberately engineered and institutionally enforced. Naming a system “supportive” or “advisory” does not make it so; interfaces, logging, and governance must make boundary events concrete and auditable.*

The first attempt at long-horizon collaboration failed partly because it took boundaries for granted: it assumed that humans would “stay in the loop” without describing the mechanisms that would force this to remain true under pressure.

## Lesson 3: Safe Horizons are Structural, Not Optional

The notion of a safe horizon—a bound on how far a run may proceed without re-grounding in tacit judgement—is perhaps the most distinctive FRFP contribution. Volume II confirms that, across domains, trouble arises where runs are effectively unbounded:

- trading strategies accumulate risk over months without strong episodic review (London Whale, Zillow);
- long-horizon judicial use of risk scores gradually reshapes practice (COMPAS);
- repeated use of LLMs and legal-research surrogates erodes verification norms (*Mata* and its successors);
- recommender systems adjust millions of people’s content diets continuously without clear reset mechanisms.

Successful systems encode safe horizons in their very design:

- DR programmes enforce periodic re-screening and post-market validation;
- Waymo restricts autonomy to defined ODDs and phases in expansions;
- scientific practice around AlphaFold, by tradition, requires experimental replication and peer review.

Thus:

*Any long-horizon deployment of AI—clinical, financial, legal, or epistemic—needs explicit safe horizons: limits in time, scale, and scope, together with procedures for systematic re-grounding. Without them, even well-behaved systems will drift into regimes where their explicit semantics no longer match tacit reality.*

The initial, informal human–AI collaboration framework lacked such a notion. It described workflows but not their depth limits. FRFP arose when it became clear that we needed not just good workflows but principled bounds on their iteration.

## Lesson 4: Governance Itself Cannot Be Automated

Maybe the deepest pattern in Volume II is the failure of explicit-only governance.

In case after case:

- boards and executives rely on risk dashboards, backtests, and demos to justify scaling (Watson Health, Zillow, CIO);
- courts and regulators accept black-box scores under the rubric of “data-driven” decision-making (COMPAS, early risk tools);
- firms treat AI tools as internal experiments without firm-level policies (Amazon recruiting, early generative-AI use in law);
- platforms rely on engagement and safety metrics to manage recommender systems without fully grappling with their societal externalities.

Where governance is more successful:

- regulators explicitly define indications, performance thresholds, and post-market surveillance for DR AI;

- scientific communities, through peer review and replication, institutionalise scepticism toward explicit models, even AlphaFold;
- autonomous-vehicle regulators and cities negotiate permits, ODD constraints, and incident-reporting frameworks with operators like Waymo.

This supports FRFP’s non-automation claims at the institutional level:

*There is no purely explicit procedure that, given model outputs and metrics, can reliably determine whether a socio-technical system is acceptable. Governance and oversight are themselves tacit functions carried out by humans and institutions, and must be modelled as such. Good governance uses explicit tools and metrics, but does not reduce itself to them.*

The first collaboration framework implicitly treated governance as a design parameter: write down the right rules and processes, and the system will behave. FRFP, informed by the case studies here, insists that governance is dynamic, contested, and structurally non-automatable, and therefore must be part of the formal semantics.

## Lesson 5: FRFP is Both Descriptive and Prescriptive

A final lesson concerns the role of FRFP itself.

On the descriptive side, Volume II demonstrates that a single vocabulary—explicit/tacit spaces, runs, boundaries, safe horizons, institutional operators—is capable of reorganising a wide range of case studies into a small number of recurring structural patterns. This is evidence that FRFP is not merely an arbitrary formalism but is tracking real invariants in human–AI systems.

On the prescriptive side, the case studies suggest concrete design moves:

- **Narrow and stabilise explicit tasks** wherever possible, especially in high-stakes domains, and be honest about where semantics become diffuse (e.g. general-purpose LLMs).
- **Make uncertainty explicit** in interfaces, including “I don’t know” and “ungradable” states that revert to human judgement.
- **Elevate boundaries to first-class objects**: log them, design UI around them, and give institutional structures authority at those points.
- **Encode safe horizons** in policies: caps, phases, revalidation triggers, and reset protocols, rather than relying on ad hoc audits after failures.
- **Build governance as an explicit layer** with its own semantics and operators: cross-functional committees, regulatory regimes, professional norms, and public deliberation.

FRFP does not guarantee good outcomes, any more than having a theory of mechanics guarantees a bridge will not fall. But just as mechanical engineering makes certain failure

modes legible (buckling, resonance, fatigue), FRFP makes certain socio-technical failure modes legible and therefore, in principle, avoidable.

## Looking Forward: From Volume II Back to Practice

The story of this monograph is itself a small FRFP case study. An initial, mostly narrative attempt to design long-horizon human–AI collaboration proved inadequate. That failure, when reflected on, led to a more formal architecture (Volume I). The architecture, in turn, has been tested against real systems (Volume II) and refined by their successes and failures.

The next step is to close the loop:

- to use FRFP prospectively in the design of new systems and institutions;
- to co-develop mathematical results, engineering practices, and governance patterns;
- to treat FRFP not as a static theory of “AI safety” but as an evolving language shared across technical, social-science, and institutional actors.

The case studies in this volume show that powerful AI can be embedded in human institutions in ways that are disastrous, benign, or genuinely transformative. They also suggest that the difference is not mysterious: it lies in the structure of the explicit pipelines, the design of boundaries and safe horizons, and the seriousness with which we treat institutional non-automation.

If Volume I argued that such structure is *necessary*, Volume II suggests that it is *already* present, implicitly, in systems that work reasonably well, and conspicuously absent in those that fail. The task ahead is to make that structure explicit—in our tools, our organisations, and our laws—before the next generation of long-horizon AI systems arrives by default rather than by design.

# Bibliography